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LIST OF SYMBOLES

A.C	:	Alternating current
D.C	:	Direct current
M_{CT}	:	Mutual couplings for current transformer
M_{VT}	:	Mutual couplings for voltage transformer
M_{CS}	:	Mutual couplings between conductor C and sheath S
emf	:	Electric motive force
E_t	:	emf induced in the ground loop from the transformer
E_c	:	emf induced in the ground loop from the conductor current
CTs	:	Current transformers
VTs	:	Voltage transformers
I_{Sr}	:	Sheath circulating current in phase R
I_{Ss}	:	Sheath circulating current in phase S
I_{St}	:	Sheath circulating current in phase T
XLPE	:	Cross linked polyethylene
PVC	:	Polyvinyl Chloride
PE	:	Polyethylene
L3	:	Minor section length no. 3
U_t	:	Residual voltage at sheath terminal
IEEE	:	Institute of Electrical and Electronic Engineers.
SVLs	:	sheath voltage limiters
ecc	:	Earth continuity conductor
I_p	:	Sheath circulating current
e_p	:	Sheath induced voltage
I_c	:	Conductor current
I_1, I_2, I_3	:	The line current in phases (1), (2) and (3) respectively
V_{S1}, V_{S2}, V_{S3}	:	Induced voltage in sheaths (1), (2) and (3) respectively
I_{CS1}, I_{CS2}, I_{CS3}	:	The circulating currents in sheaths of phases (1), (2) and (3) respectively
R_S	:	The resistance of sheath at its maximum operating temperature
$M_{1,2}$	:	The mutual inductance between core (1) and sheath (2)
$M_{1,3}$	:	The mutual inductance between core (1) and sheath (3)

$M_{2,3}$	:	The mutual inductance between core (2) and sheath (3)
W_{CS}	:	The circulating sheath loss per meter
I	:	The line currents in phases (1), (2) and (3) with balance condition
S	:	Spacing between axes of adjacent conductors
r_{sh}	:	Mean of outer and inner radii of sheath
X	:	The reactance per unit length of sheath
R	:	The resistance of conductor at its maximum operating temperature
X_m	:	Mutual reactance per unit length of cable between the sheath of an outer cable and the conductors of the other two, when cables are in flat formation
V_0	:	Residual voltage along the cable sheath
IEC	:	International Electro-technical Commission
I_{SE1}, I_{SE2}, I_{SE3}	:	Sheath Eddy Current in phase no. 1,2 and 3 respectively
D_S	:	The external diameter of cable sheath
t_S	:	The thickness of sheath
m	:	factor depends on power frequency and metallic sheath resistance
R_{dc}	:	The d.c. resistance of the conductor at 90 ^oC
R_{20}	:	The d.c. resistance of the conductor at 20 ^oC
y_s	:	The skin effect factor
y_p	:	The proximity effect factor
A_S	:	The sheath cross-sectional area
d_S	:	The mean diameter of the sheath
D_{Se}	:	The external diameter of the sheath
R_{strand}	:	Resistance of one strand
n	:	Number of strands
d_C	:	Diameter of conductor
D_e	:	External diameter of cable
I_{CS-R}, I_{CS-S}, I_{CS-T}	:	The sheath circulating currents in R, S and T phases respectively
h	:	an operator which rotates a phasor 120° counter clock-wise
I_{CSX},I_{CSY}, I_{CSZ}	:	The sheath circulating currents in sheath circuits X, Y and Z respectively
Z_X , Z_Y , Z_Z	:	The sheath impedances of the X, Y and Z circuits respectively

V_X , V_Y, V_Z	:	The induced voltages in sheaths of the X, Y and Z circuits respectively
R_e	:	The equivalent resistance of sheath and armour in parallel
R_A	:	The resistance of armour per unit length of cable at its maximum operating temperature
d	:	The mean diameter of sheath and armour
d_S	:	The mean diameter of sheath
d_A	:	The mean diameter of armour
I_S	:	Sheath current (circulating or eddy)
I_A	:	Armour current (circulating or eddy)
I_{SA}	:	Sheath-armour combination current (circulating or eddy)
I_{AE1}, I_{AE2}, I_{AE3}	:	Armour Eddy Current in phase no. 1,2 and 3 respectively
I_{AC1}, I_{AC2}, I_{AC3}	:	Armour Circulating Current in phase no. 1,2 and 3 respectively
E_{AE},E_{BE},E_{CE}	:	Voltages between sheaths of phases A,B and C respectively and the earth conductor
I_F	:	Short-circuit current in cable conductor
S_{AE},S_{BE},S_{CE}	:	The geometric mean spacing between cables A, B and C respectively and the earth conductor
R_C	:	Resistance of earth conductor
r_c	:	Geometric mean radius of earth conductor
E_{AB},E_{BC},E_{CA}	:	Voltages between sheaths of phases A&B, B&C and C&A respectively
CIGRE	:	International Council on Large Electric Systems
rms	:	Root mean square
f	:	power frequency (50 Hz)
ω	:	2π x frequency (in cycles per second)
λ_{CS}	:	The circulating sheath loss factor
λ_{CS1}, λ_{CS2}, λ_{CS3}	:	The circulating sheath loss factor for sheaths (1), (2) and (3) respectively
λ_{SE1},λ_{SE3},λ_{SE2}	:	Sheath Eddy loss factor in phase no. 1,2 and 3 respectively
ρ_S	:	The electrical resistivity of sheath material at operating temperature
Δ_1 ,Δ_2	:	factors depend on the types of cable layouts formation
g_S , β_1	:	factors depend on the cable parameters

θ_s	:	sheath temperature
ρ_{S20}	:	The electrical resistivity of sheath material at 20 ^oC
ℓ	:	The length of lay of the tape or wire
ρ_{C20}	:	The electrical resistivity of conductor material at 20 ^oC
α_{C20}	:	The constant mass temperature coefficient at 20 ^oC for conductor
$\theta_{C\ max}$	:	maximum operating temperature of conductor
$\theta_{S\ max}$	:	maximum operating temperature of sheath
ℓ_i	:	The length of section number i
$\lambda_{AE1}, \lambda_{AE1}, \lambda_{AE1}$	:	Armour Eddy Loss Factor in phase no. 1,2 and 3 respectively
$\lambda_{AC1}, \lambda_{AC2}, \lambda_{AC3}$	:	Armour Circulating Loss Factor in phase no. 1,2 and 3 respectively

ABSTRACT

Single-core underground power cables can induce voltages and currents in their metallic sheaths. The sheath induced currents are undesirable and generate power losses and reduce the cable ampacity whereas the induced voltages can generate electric shocks to the workers that keep the power line. This means that it is very important to know the values of sheath currents and induced voltages and the factors affecting them. So this thesis discussed the following:

- Calculations of the induced voltages in single-core cables with various voltages levels from 11 kV to 500 kV with briefly studying the factors affecting them.

- Studying the factors affecting the sheath losses in single-core cables by calculating the sheath currents (eddy-circulating) and their sheath losses in single-core cables with various metallic sheath materials and various voltages levels from 11 kV to 500 kV with taking into consideration the following factors:
Types of sheath bonding methods (single-point bonding, two-points bonding, cross bonding) and cable layouts (trefoil, flat), cable parameters, cable spacing, sheath resistance, phase rotation, conductor current, power frequency, the minor section length in cross bonding arrangement and cable armoring. This study is carried out depending mainly on IEC 60287 by a proposed computer program using MATLAB.

- Studying the overvoltages in the metallic sheaths of single-point bonding and cross bonding due to different types of external faults, which may cause the sheath multi-points break-down and result in a large sheath circulating losses.

CHAPTER (1)

INTRODUCTION

1.1 Introduction

With the rapid increase in demand for electric energy and the trend for large infra-structures and vast expansion of highly-populated metropolitan areas, the use of underground power cables has grown significantly over the years [1].

Three separate single-core cables are usually used instead of three-core cables. The principal reasons are [2, 3]:

1. To transmit large quantities of power, for which three-conductors cable would be unwieldy.
2. To obtain phase isolation.
3. To gain advantage of the inherently higher unit dielectric strength of the insulation in single-conductor cable.
4. The handling of large multi-conductors cable can be difficult, especially compared to the relative ease of handling of several smaller conductors.

In a single-core power transmission cable, normally a metallic sheath is coated outside the insulation layer to prevent the ingress of moisture, protect the core from possible mechanical damage, serves as an electrostatic shield (the electric field is enclosed in between the conductor and the sheath), and act as
a return path for fault current and capacitive charging currents [4, 5].

When an isolated single conductor cable carries alternating current, an alternating magnetic field is generated around it. If the cable has a metallic sheath, the sheath will be in the field, the sheath of a single-conductor cable for A.C service acts as a secondary of a transformer; the current in the conductor induces a voltage in the sheath. When the sheaths of single-conductor cables are bonded to each other, as is common practice for multi-conductor cables, the induced voltage causes current to flow in the completed circuit. This current causes losses in the sheath [6].

The problems of the induced voltages and currents associated with using single-core cables (for example, failure of sheath insulators, failure of cable jackets and sheath corrosion) have been recognized since metallic sheathed cables were first used, and

the fundamentals of calculating sheath voltages and currents have been defined for many years [6].

Much work has been done, for the purpose of minimizing sheath losses by introducing various methods of bonding.

Any sheath bonding or grounding method must perform the following functions [2, 6]:

1- Limit sheath voltages as required by the sheath section- alizing joint.
2- Reduce or eliminate the sheath losses.
3- Provide low impedance path for fault currents.
4- Maintain a continuous sheath circuit to permit adequate lightning and switching surge protection.
5- Limit abnormal sheath voltages during failure to the lowest possible values.

The above objects must be accomplished without causing the following objectionable features [2]:

1- Excessive losses in the sheath bonding devices.
2- Introduction of triple or other harmonic currents into the sheath circuit causing inductive interference with telephone circuits.
3- Interference with proper current drainage to prevent D.C electrolysis; also adverse effect on operation of the A.C sheath bonding method by flow of stray D.C currents.
4- Excessive size, weight, space, or cost of bonding devices.

Due to the importance of the sheath losses especially in single-core cables, the factors affecting them in single-core underground cables have been studied in this thesis.

1.2 Book Outline

The remaining chapters in this thesis are arranged as follows:

Chapter (2): This chapter discusses some necessary theories and background information that related to sheath losses in single-core cables such as the sheath phenomena, types of sheath bonding and types of losses in the metallic sheath.

Chapter (3): This chapter provides some of the methods used to reduce the sheath circulating currents and losses in single-core cables.

Chapter (4): This chapter discusses the different factors affecting the sheath losses in single-core underground power cables by using a suitable mathematical algorithm by MATLAB programming depending mainly on IEC 60287.

Chapter (5): In this chapter over voltages are calculated for single-point bonding and cross bonding under different types of external faults for systems having solidly earthed neutral.

Chapter (6): The conclusions obtained from this thesis are listed.

CHAPTER (2)

SHEATH BONDING AND GROUNDING

Before studying the factors affecting the sheath losses in single-core underground cables it is reasonable to understand how are the voltage and current induced in the metallic sheath which is known as sheath phenomena, also discussion of the various methods of sheath bonding are carried out. Finally the types of metallic sheath losses are discussed.

2.1 Sheath Phenomena

When single-core power cables are used in A.C systems, the presence of a metallic sheath around each conductor causes one or both the following two phenomena:

2.1.1 Sheath voltage

The sheath of a single conductor cable acts as a secondary of a transformer and the current in the conductor induces a voltage in the sheath. This voltage does not depend upon the sheath material [7].

The value of this induced sheath voltage depends on the flux interlinked with the metallic sheath, and it increases as the inter-axial spacing of the cables is increased.

This value is also higher if cables are placed in separate ducts.

First, it was not industry practice to insulate the sheaths of cables, hence under normal operating conditions it was necessary to limit the sheath voltage to an acceptable level (12 V to 25 V) in order to avoid electric shock to either operating personnel and also to avoid corrosion [4].

However, with the advent of the insulating polyethylene jacket both of these problems have been solved very largely since corrosion became no longer a problem and operating personnel are protected so it became the presently accepted value of sheath voltage to 100 to 400 volts for normal load conditions [4].

Since the fault currents are much higher than the load currents, it is usually considered that the shield voltage during fault conditions be kept to a few thousand volts. This is controlled by using sheath voltage limiters, which is a type of surge arrester [4].

Limitations remain on the upper value of permissible induced voltages but at much higher level, these limitations are [6]:

1. Flashover voltage of the insulating jacket under faulty conditions.
2. Flashover voltage of the insulating joints.

2.1.2 Sheath current

If the sheaths of single conductor cable are bonded to each other at more one point, as is the common practice for three conductor cable, the induced voltage causes current to flow in the completed circuit. The circulating current value may achieve the same order as wire-core current.

One other important concept regarding multiple grounds is that the distance between the grounds has no effect on the magnitude of the current [4].

The circulating current will lead to energy loss and the falling of transmission efficiency, on the other hand, the circulating current will cause the cable temperature to rise, influence the cable's life, and decrease the transmission capacity.

2.2 Sheath Bonding Arrangements

The IEEE Standard 575 [6] introduces guidelines into the various methods of sheath bonding. The most common types of bonding are single point, two-points or multiple points and cross bonding

2.2.1 Sheath bonded at two-points (solid bonding):

In a 3-phase circuit, with single-core cables, where the cables are solid bonded the sheaths of all 3 cables will be connected together at both ends of the run. For safety reasons one end of the sheaths must also be earthed. It is common practice to earth the sheaths at both ends of the run, as given in Fig.(2-1),

to allow them to be used as an earth return conductor to carry through fault currents.

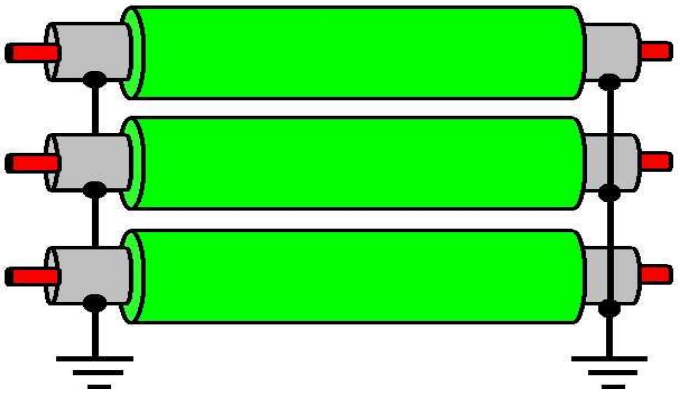


Fig. (2-1): Two-points bonding

In a solid bonded system, where the sheaths are bonded and earthed at each intermediate joint, the magnitude of the circulating current is independent of the circuit length [7, 8].

With modest loads sheath losses may be tolerated with each length being solidly bonded.

This method of bonding is the one way of eliminating the induced voltages. If the screen of a cable is bonded at both sides, the following effects will appear:

1. Due to the magnetic field of the main cable and the closed loop of the cable screen, a circulating current is flowing in the screen.
2. These currents can cause significant sheath losses and heating which can adversely affect the thermal rating of the cable's core conductor, hence reducing the current carrying capacity of the circuit.

This arrangement is most suitable for three-core cables and is not usually used at voltages above 66 kV [9] where there is a need to maximize the current carrying capacity of the circuits.

Also solid bonding would allow fault current to be transmitted along the sheath of a healthy cable in the event of an earth fault at one substation causing a rise in ground potential relative to that at another connected substation. Such a flow of fault currents is undesirable [8].

When load requirements reached higher level, other sheath bonding methods became necessary especially with the wider spacing of cables in ducts bank rather than in direct buried trefoil.

2.2.2 Sheath bonded at one end only:

The simplest form of bonding, for three-phase single-core cable, consists in arranging for the sheaths of the three cables to be connected together and earthed at one point only along their length, as given in Fig. (2-2-a), at the other end of the run the cable sheaths will be terminated at an insulated fitting.

If the cable screen is bonded at one side only, the following effects are appearing:

1. As the screen is open, there is no circulating current, hence, there are practically no losses in the Screen and the ampacity is higher compared with both sides bonding.
2. At all other points, a voltage will appear from sheath to ground that will be a maximum at the farthest point from the ground bond, as given in Fig.(2-2-b), so particular care must be taken to insulate and provide surge protection (using sheath voltage limiters SVLs) at the free end of the sheath to avoid danger from the induced transient voltages due to lighting and switching surges as well as limiting the voltage under fault current conditions, as given in Fig.(2-2-c).

The maximum sheath voltage permitted at full load varies considerably between different countries [6]; in most cases it precludes the use of single point bonding for anything other than cable circuits of a few hundred meters in length.

When the circuit length is such that sheath induced voltage limitation would be exceeded if the earth bond were connected at one end of the circuit, this bond may be connected at some other point, for example the centre of the length. In this situation, only half of the previous voltage appears on the sheath (as shown in Fig. (2-3-a,b)). If the circuit is too long to be dealt with by this means it may be

sectionalized by the use of sheath sectionalizing joints so that the sheath voltage for each elementary section is within the limitation imposed as shown in Fig. (2-3-c).

It is necessary to install an earth continuity conductor (ecc) to carry fault currents which would normally return via the cable sheaths. To maintain a low voltage between the cable sheaths and the ground under fault conditions the ecc is grounded at the cable terminals and possibly along the cable route and being sufficiently close from the cable circuit conductor.

To avoid circulating currents and losses in this conductor it is preferable, when the power cables are not transposed, to transpose the parallel ground continuity conductor (as shown in Fig. (2-3-d)).

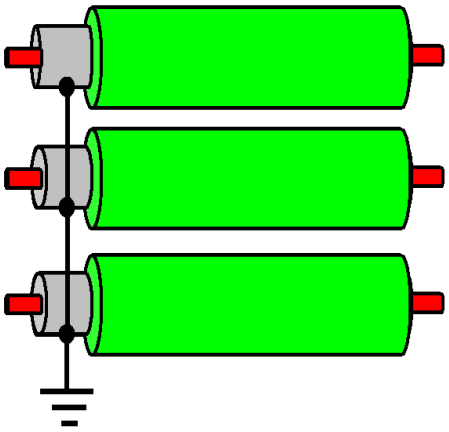


Fig.(2-2-a): Single-point bonding

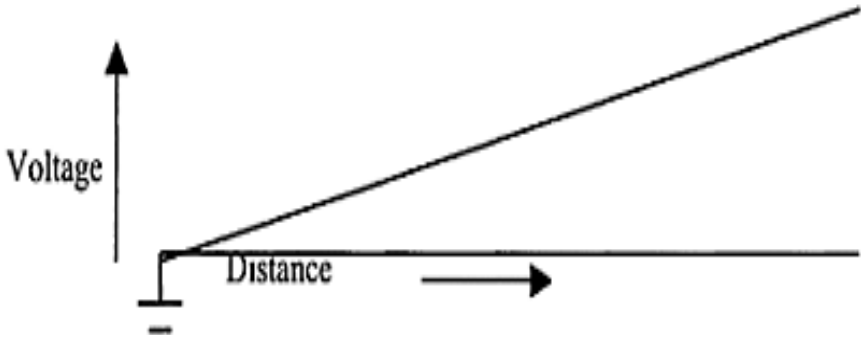


Fig.(2-2-b): Induced voltage in sheath with single-point bonding

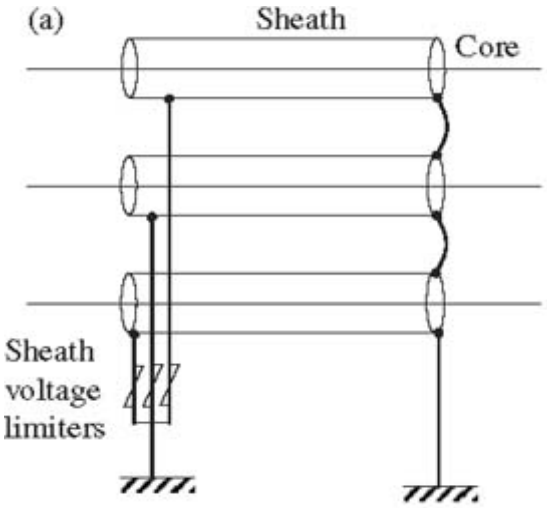


Fig. (2-2-c): Single-point bonding with sheath voltage limiter (SVL)

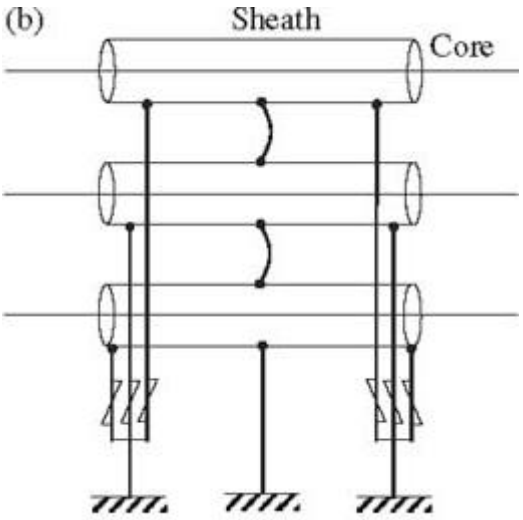


Fig.(2-3-a): Mid point bonding with sheath voltage limiter (SVL)

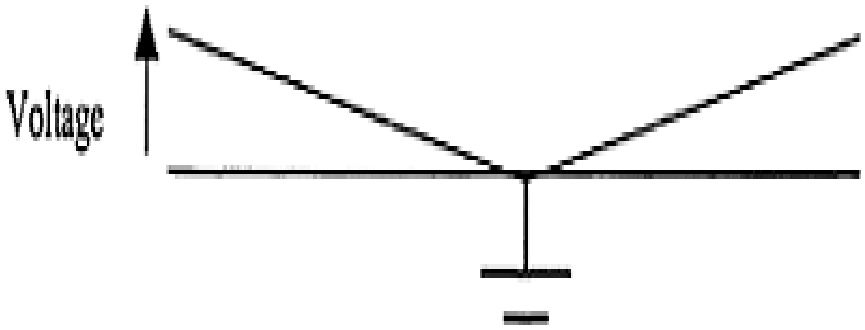


Fig.(2-3-b): Induced voltage in sheath with mid-point Bonding

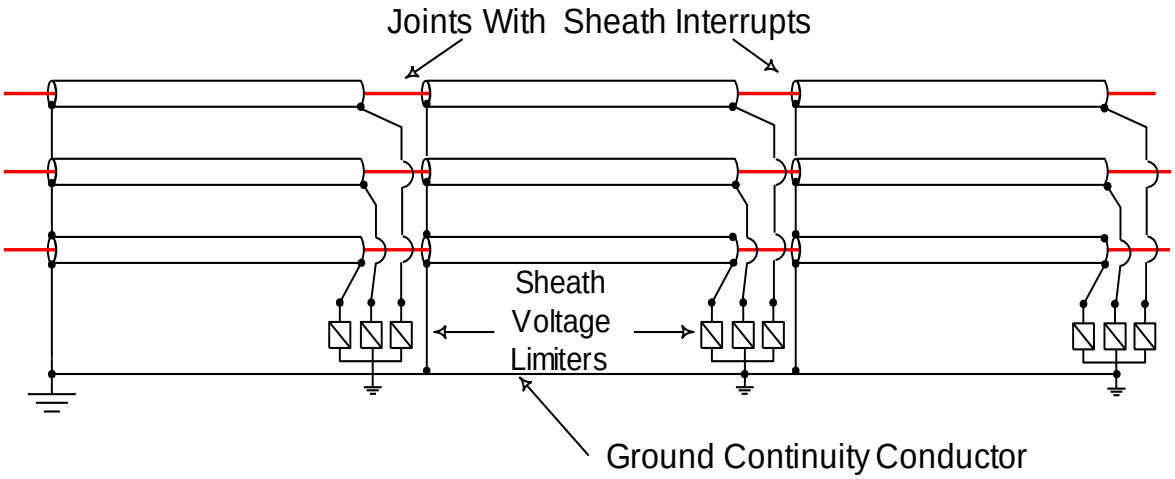


Fig. (2-3-c): Sectionalized run with single-point bonding

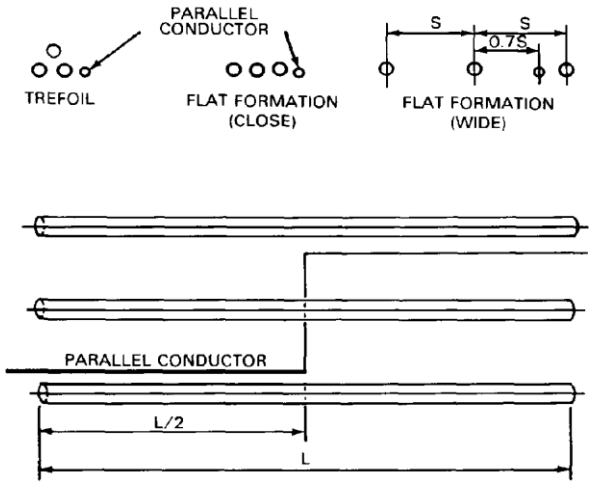


Fig. (2-3-d): Transposition of parallel conductor in flat formation or trefoil

2.2.3 Cross bonding system:

If the sheaths of three single core cables are not bonded electrically together, induction between conductors and each sheath can produce unacceptable voltages between sheaths. On the other hand, bonding at both ends will result in sheath currents following with associated losses, which is again not acceptable, especially for long cable routes [10]. Cross bonding of single core cable sheaths is a technique which has been common in different countries for many years. It has been

introduced in order to avoid circulating currents and excessive sheath voltages, hence, increases its current-carrying capacity.

It achieves that by dividing the cable route into three equal lengths (or six, or nine, etc.), and the sheath continuity is broken at each joint. The induced sheath voltages in each section of each phase are equal in magnitude and 120° out of phase. When the sheaths are cross connected each sheath circuit contains one section from each phase such that the total voltage in each sheath circuit sums to zero as shown in Fig. (2-4). If the sheaths are then bonded and earthed at the end of the run, the net voltage in the loop and the circulating currents will be zero and the only sheath losses will be those caused by eddy currents. This system provides a continuous earth path via the sheaths between the earth systems at the two ends of the cable, obviating the need for an auxiliary earth conductor.

Sheath voltage limiters (SVLs) are connected to earth at the intermediate cross bonding positions to dissipate any sheath voltage surges. This method of bonding allows the cables to be spaced to take advantage of improved heat dissipation without incurring the penalty of increased circulating current losses.

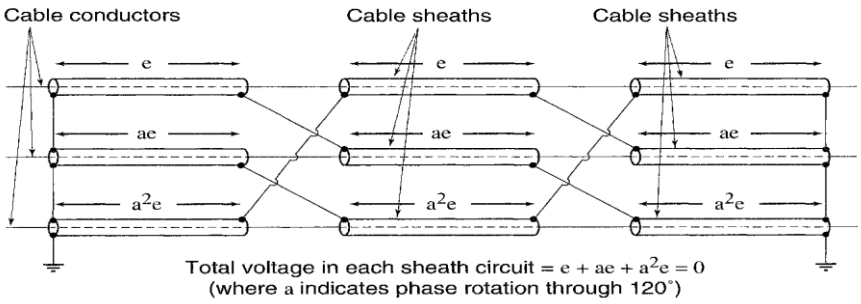


Fig. (2-4): Principle of cross-bonding

However, in practice it happens very often that the line is divided into unequal sections, which results in an unsymmetrical cross bonding and a residual voltage is measured at the end of the sheath, since the voltage triangle doesn't close [11].

Yet it is still useful to use this kind of bonding to at least reduce losses considerably, instead of canceling them completely.

Applying the method of cross bonding depends on the length of the cable and the length produced by the factory which is put on each drum

for transport, the length produced by the factory depends on many factors like weight, dimensions and transport facilities and limitations. Often cables produced in longer lengths than the average result in additional difficulties and are subjected to damage during transport or laying.

The length of each section of cable depends on the nature of the area in which the cable will be laid and any natural or man-made obstacles. Moreover, the costs of equipment necessary for cross bonding like junctions and special connections and junction protection against over voltages, etc., count for economical application of cross bonding and must be compared to the cost of the losses of sheath capitalized over the life time of the cable which can be estimated as an average of thirty years. It must be kept in mind that the cancellation or reduction of sheath losses results in a smaller conductor, since it increases the current carrying capacity and makes energy transmission more economical.

Generally, the higher the voltage applied, the power transmitted and the length of the cable line, the more is importance of the losses and the more cross bonding becomes a must for the cable designer.

Single-core cables of more than 500 mm^2 cross sectional conductor area and 3 km length will prove more economical with cross bonded sheaths in most cases [12].

In order to completely eliminate the sheath losses, the best arrangement is where the cores of the three minor sections within each major section are perfectly transposed but the sheaths are not, as shown in

Fig. (2-5).The voltages in the sheaths are now balanced and thereby there is no residual voltage which could circulate sheath currents and therefore they are absent [5, 9, and 11].

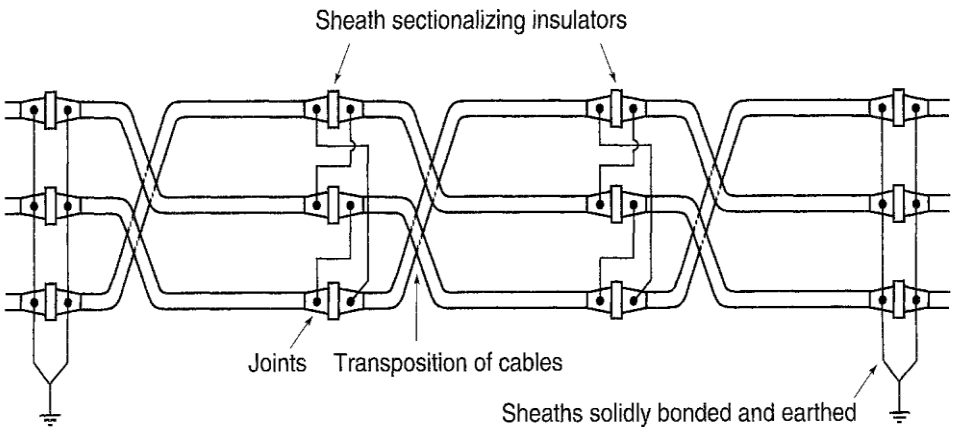


Fig. (2-5): Cross bonded cables with transposition

2.3 Types of Metallic Sheath Losses

Sheath losses are current dependent, and can be divided into two categories according to the type of bonding [5, 9, 10, and 11]:

1- Sheath eddy losses

2- Sheath circulating losses

2.3.1 Sheath eddy losses

The metallic sheath is immerged in the magnetic field generated by the conductor current (I_C). Therefore an induced voltage (e_P) appears in the sheath, which induces currents in the metallic sheath. These currents dissipate energy due to Joule effect.

The induced voltage is a maximum in the internal side of the sheath and minimum in its external side, this situation induces the circulation of eddy currents in the sheath as shown in Fig. (2-6). This is the origin of the eddy currents [13].

Eddy current losses occur in both 3-core and single-core cables, irrespective of the method of bonding [11].

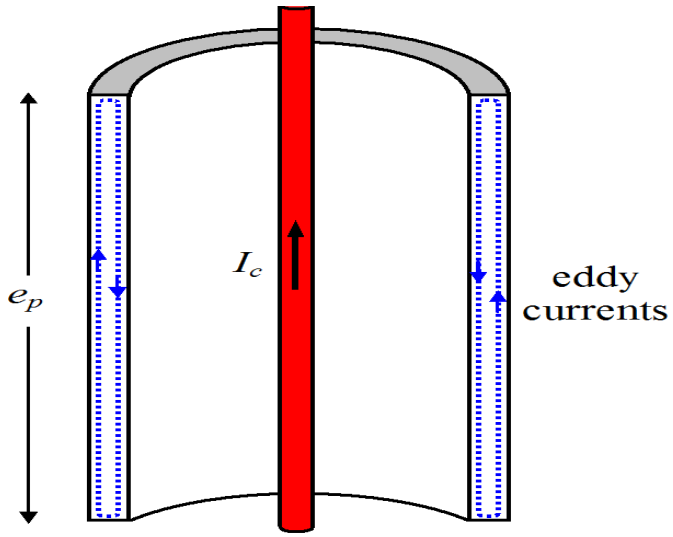


Fig. (2-6): Ungrounded metallic sheath

Sheath eddy currents and losses produced by them reach their maximum value when the cable conductors are situated as close as possible to one another.

2.3.2 Sheath circulating losses

When both ends of the sheath are grounded, the sheath voltage (e_p) induces a sheath circulating current (I_p) along the sheath, which returns through the ground circuit as shown in Fig.
(2-7).
The circulating currents I_p are usually much greater than the eddy currents. Therefore the eddy currents can be ignored when dealing with sheaths that have both ends grounded.
The sheath circulating loss occurs only in single-core cables systems [13].

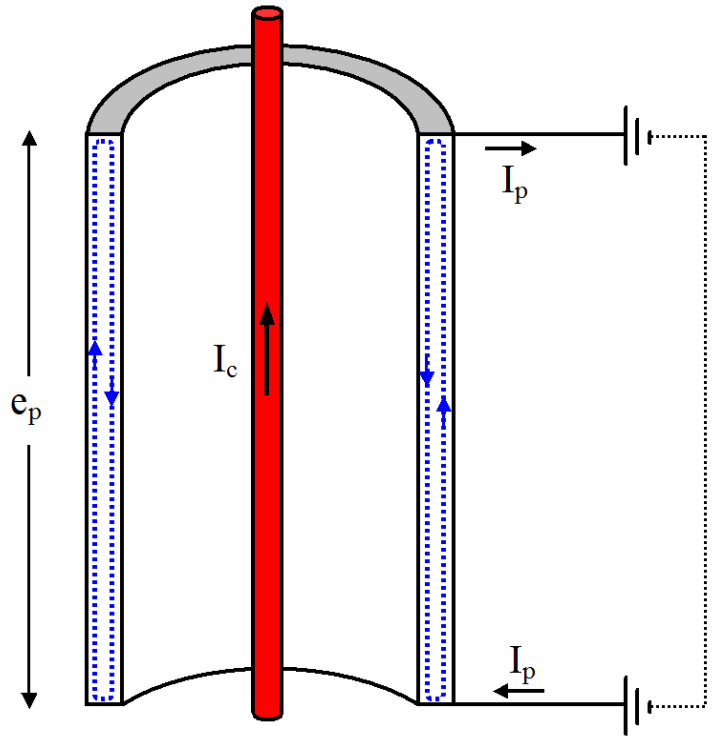


Fig. (2-7): Sheath grounded at both ends

CHAPTER (3)

METHODS TO REDUCE THE SHEATH CURRENTS AND LOSSES

3.1 Introduction

The sheath circulating current must be reduced in underground power cable systems to a safety level, as if the sheath circulating current rises, the loss caused by sheath circulating current will increase, and then the ratio of loss dissipated in sheath per unit length to loss in conductor per unit length will increase too. By such effect, the total thermal resistance of the cable is increasing, and the permissible current is reduced. Dry zone may be formed around the underground cable may lead to thermal failure of cable insulation [14]. So in this chapter the methods to reduce the sheath circulating currents and their losses will be discussed by classifying them into old and modern techniques.

3.2 Old Techniques to Reduce the Sheath Currents and Losses

Some of these methods are using up to date, while the others are not. So these methods will be discussed briefly.

3.2.1 Single-point and cross bonding methods

Prior to the development of outer coverings for cables that would provide reliable, long term, insulation of the metallic outer layer it was good practice to bond the metallic layers at both ends of the cable run. Although this practice effectively eliminated standing voltages on the metallic layer it allowed circulating currents to flow in the cable sheaths.

The development of extruded outer coverings for cables allowed single point bonded and cross bonded systems to be used in practice for either

eliminated or greatly reduced sheath circulating currents. These are single point bonded and cross bonded systems. Such special bonding systems were introduced into the UK in the late 1950s and early 1960s [8]. For more details about them refer to clauses (2.2.2) and (2.2.3).

3.2.2 Continuous cross bonding method

In which the cable sheaths were cross-bonded continuously along the complete line and the three sheaths are bonded and grounded at the two ends of the route only [2, 6].

3.2.3 Impedance bonding methods

The cable sheath sections are bonded together in some manner through impedance. The impedance of the devices is made considerably higher than the impedance of the sheaths, with the result that very little current flows and the voltage drop is almost entirely in the device. This impedance may consist of simple reactors or of devices such as saturable reactors and bonding transformers. To provide ground connections, the impedance devices are normally designed with center taps or grounding points [2, 6].

3.2.4 Resistance bonding method

The flow of sheath currents may be reduced by the installation of resistance in series with the cable sheaths. In general, resistance bonding is not practical, since the resistors have to be sized to take the fault currents and they are considered very large for high fault currents [2, 6].

3.3 Modern Techniques to Reduce the Sheath Currents and Losses

These methods are not famous, so they will be discussed in details.

3.3.1 Sheath current canceling device

A patent is introduced [15] based on the principle of electro-magnetic induction to reduce the circulating currents and the losses in the

metallic sheath loops of single-phase and three-phase system using single-core high voltage transmission cables, where the sheaths are grounded or bonded together at both ends of the cable run.

This invention consists of a current transformer at a sealing end of each single-phase cable, connected in series with a voltage transformer in the grounding or bonding connection of each sheath at the same cable end. The primary winding of each current transformer is the phase conductor, and the secondary winding of each voltage transformer is a sheath loop. The method involves inducing locally an emf into each sheath loop, essentially equal and opposite to that induced by the flux of the load current in each conductor acting along the whole cable length. The circulating sheath loop current and the losses are then nominally zero.

The principle of this method for a single-phase cable where the sheath ground loop (a-b-c-d) includes the ground returns path (a-d) is illustrated in Fig. (3-1). The dot notation (•) indicates the sense of the windings, and the mutual couplings, M_{CT} for transformer 1, M_{VT} for transformer 2, and M_{CS} between conductor C and sheath S [15].

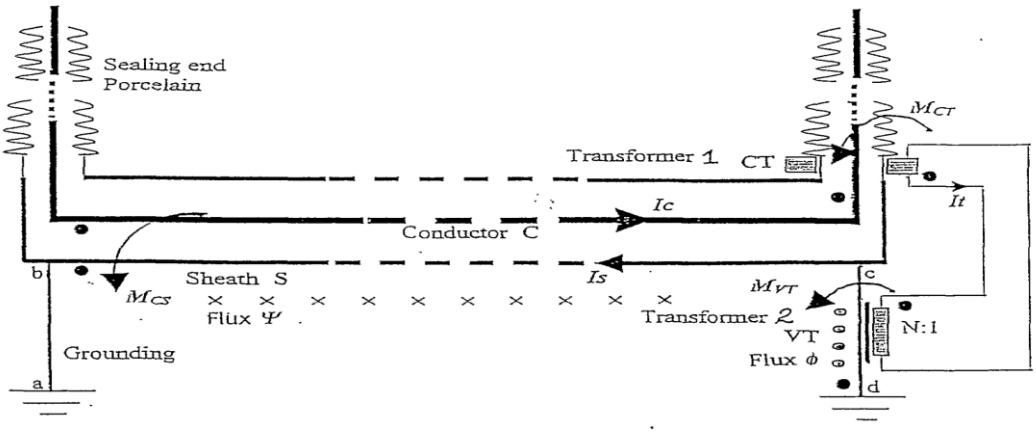


Fig. (3-1): Sheath current canceling device in single phase [15]

By a suitable choice of the windings of transformer 1, and transformer 2, the flux is arranged to be essentially equal and opposite with the flux from the conductor linking

the loop. Thus both the driving emf ($E_t + E_c$), E_t emf induced in the ground loop from the transformer 2 and E_c emf induced in the ground loop from the conductor current, and the circulating current I_S in the sheath ground loop (a-b-c-d) are essentially zero.

Fig. (3-2), illustrates the three-phase system with three sets of CTs and VTs set up for cancelling the normally circulating sheath currents I_{Sr}, I_{Ss} and I_{St}. The three current transformers are clearly not connected in series, as the device is designed to operate continuously in the steady state at power frequency, on high voltage single-phase cables with a metal sheath. Each cable conductor load current is used to introduce a continuous power frequency emf into its own sheath circuit via the VT, such that the normal circulating sheath current in a sheath ground loop, or sheath loop between phases is neutralized.

Exact equality between the opposing emfs is not necessary for the method to be effective, as the sheath losses are proportional to I_S^2 (where I_S^2 is the circulating sheath current). Even with I_S reduced by only 50 %, the losses are reduced by 75 %.

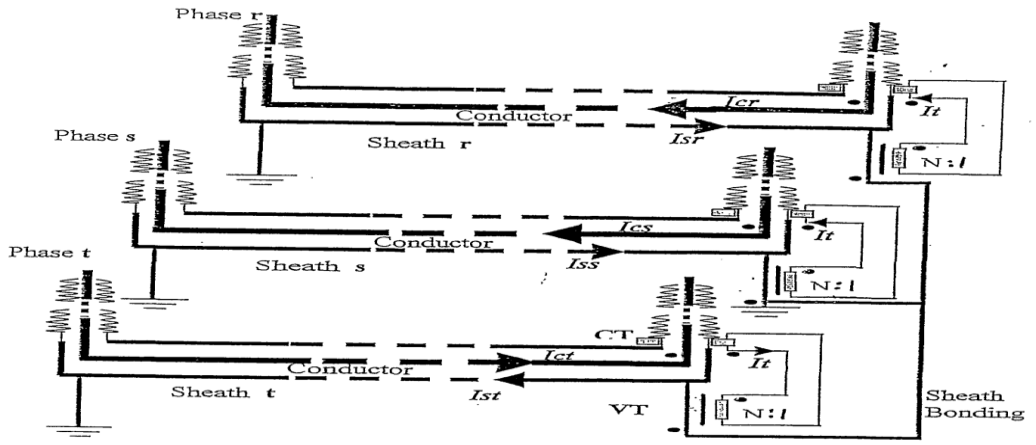


Fig. (3-2): Sheath current canceling device for three single-core cable [15]

This invention characterized by:

It can be applied to cables which are already laid, circulating sheath currents arising due to sheath insulation failure at any location on the sheath can be readily detected, as the secondary current in the current transformer is otherwise nominally zero and the method is passive and adjusts automatically to the prevailing load current on the cable.

3.3.2 Inductance compensation device

When laying down the cables asymmetrically or the length of three sections of the sheath is not equal due to the development of city constructions, there is will be a residual voltage appearing at the end of the sheath, since the voltage triangle does not close (Fig.(3-3)), the circulating current is generated in metal shield.

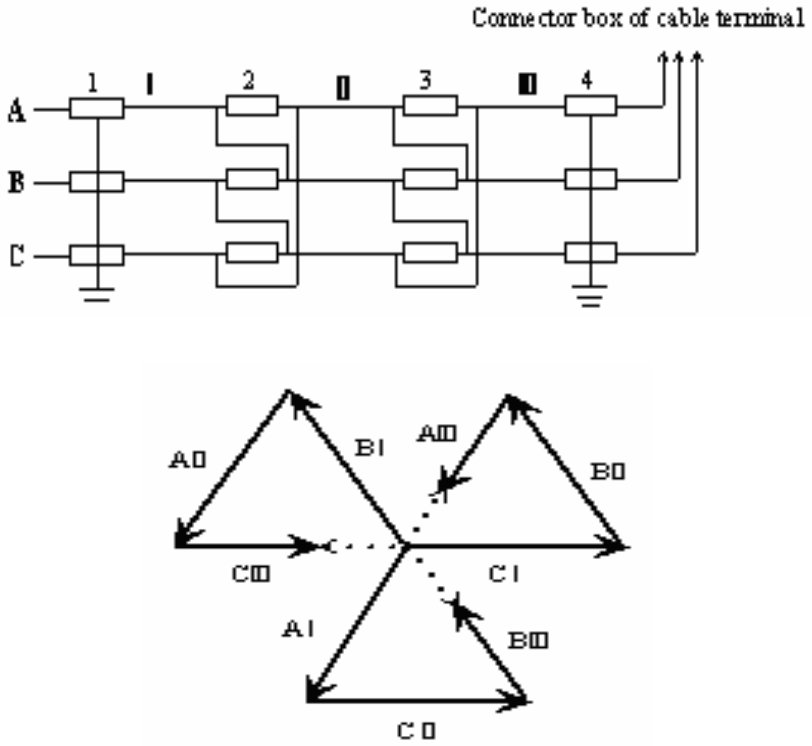


Fig. (3-3): Residual voltage at the end of the sheath

These factors affecting the sheath losses lead to development a new method to compensate the residual voltage by using an inductance compensation device [16, 17].

Compensating the inductance in the cable terminal enwinding coil around the iron core is used. One end of the winding connects to the end of the metal shield (a short one of two ends), and the other end connects to ground (Fig. (3-4)).

When there is alternating current in the single-core cable, the alternating magnetic field is generated around the single-core cable, which links the compensating coil, then the induced electromotive force is generated in the coil which can counteract the end voltage in

metal shield, hence the sheath current leads to zero, as shown in Fig.(3-5), the voltage in L3 is U_t, in the Fig.(3-5-a), and in the Fig.(3-5-b), the current in L3 is zero because of compensation.

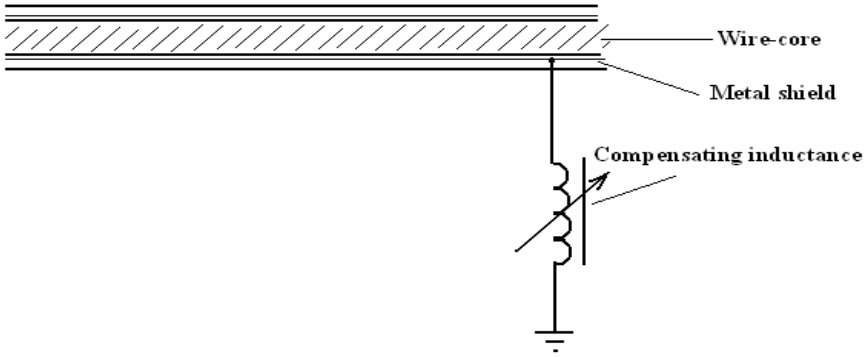


Fig. (3-4): Diagrammatic sketch of compensating inductance connect

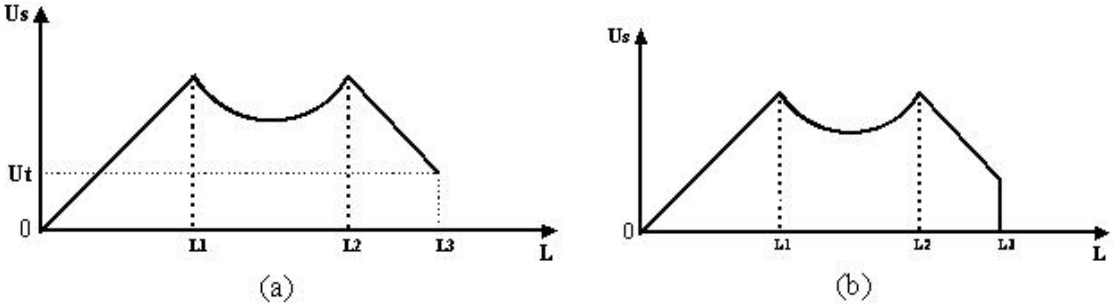


Fig. (3-5): Distribution diagram of voltage in metal shield before and after compensating inductance

To protect a compensating device against overvoltage which induced in the metal shield due to short circuit earth fault of one phase or thunder influences, the compensating device is made parallel to protection gap of overvoltage (Fig. (3-6)).

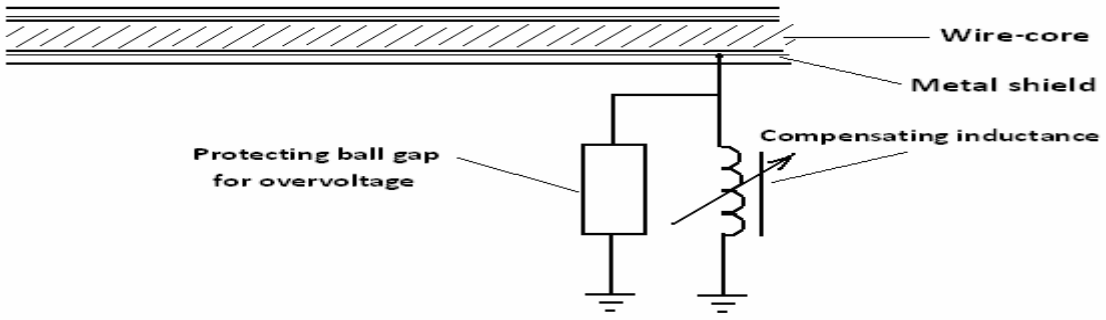


Fig. (3-6): Compensating device and overvoltage protector.

This method characterized by its easy installation, can be used for the system of which two ends earthed directly and for the system of which one end earthed with enhancing its length.

CHAPTER (4)

FACTORS AFFECTING THE SHEATH LOSSES IN SINGLE-CORE UNDERGROUND POWER CABLES

4.1 Introduction

Power losses in underground cables cause temperature rise of the cables during their operation, there are tow types of a power losses generated in the cables: current dependent powers and voltage dependent powers. Current dependent powers refer to the heat generated in metallic cable components (conductors, sheaths etc.); voltage dependent powers refer to the powers in cable insulation [18]. Sheath losses are current dependent and their values in single-core underground power cables can not be disregarded as they, in some cases, could be greater than power losses in the conductors. Sheath losses in single-core cables depend on a number of factors, these factors are:

1- Sheath bonding and cable layout formation
2- Cable parameters (conductor resistivity & conductor size)
3- Cable spacing
4- Sheath resistance
5- Phase rotation
6- Conductor current
7- Power frequency
8- The minor section length in cross-bonding arrangement
9- Cable armoring

In this chapter these factors are investigated depending mainly on IEC 60287.

4.2 Cable Layouts Formation

Two types of cable layouts formation usually used in practice are studied in this book:

1- A trefoil arrangement of three single-core cables, where the cables are laid as at the corners of an equilateral triangle. In this formation two single-core

cables are laid close together with one cable forming an upward apex, Fig. (4-1-a).

2- A flat arrangement of three single-core cables, where the three cables are laid in the same horizontal plane with the middle cable equidistant from two outer cables, Fig. (4-1-b).



(a) Trefoil formation (b) Flat formation

Fig. (4-1): Single-core cable layouts

4.3 Mathematical Algorithm

The single-core cables components are shown in Fig. (4-2).

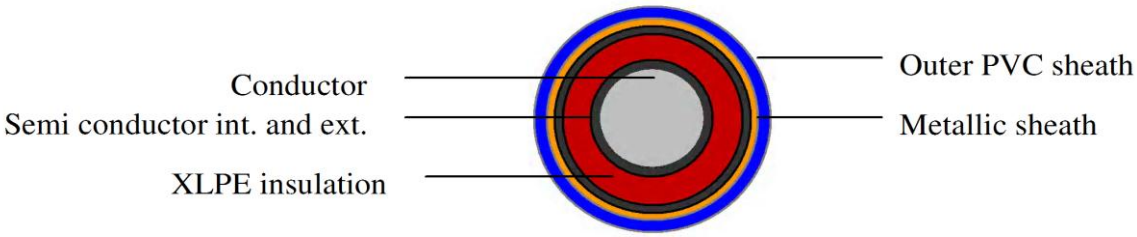


Fig. (4-2): Unarmored single-core cable components

4.3.1 Induced sheath voltages, sheath circulating currents and losses

The following assumptions are introduced in order to simplify the calculations of sheath losses in three phase power systems:

1- The sheath may be considered as a thin tube, of radius equal

to the mean of outer and inner radii of the sheath.

2- The capacitive currents returning along the cable sheaths will not appreciably affect the sheath losses.

At balance, every cable in the three-phase circuit, comprising phases 1, 2 and 3 can be regarded as a return line of the two others, i.e.

$I_1 + I_2 + I_3 = 0$ and $I_{cs1} + I_{cs2} + I_{cs3} = 0$

$$\left.\begin{aligned} I_{CS1} &= -(I_{CS2} + I_{CS3}) \\ I_{CS2} &= -(I_{CS1} + I_{CS3}) \\ I_{CS3} &= -(I_{CS1} + I_{CS2}) \end{aligned}\right\} \qquad \left.\begin{aligned} I_1 &= -(I_2 + I_3) \\ I_2 &= -(I_1 + I_3) \\ I_3 &= -(I_1 + I_2) \end{aligned}\right\} \text{ [10]} \qquad (4\text{-}1)$$

In general, the following equations for the phasors of the voltage drop per meter in the sheaths of each cable can be written as [10]

$$\left.\begin{aligned} V_{S1} &= I_{CS1}R_S - j\omega M_{1,2}(I_2 + I_{CS2}) - j\omega M_{1,3}(I_3 + I_{CS3}) \\ V_{S2} &= I_{CS2}R_S - j\omega M_{1,2}(I_1 + I_{CS1}) - j\omega M_{2,3}(I_3 + I_{CS3}) \\ V_{S3} &= I_{CS3}R_S - j\omega M_{1,3}(I_1 + I_{CS1}) - j\omega M_{2,3}(I_2 + I_{CS2}) \end{aligned}\right\} \qquad (4\text{-}2)$$

Where,

I_1, I_2, I_3 : The line current in phases (1), (2) and (3) respectively in A.

V_{S1}, V_{S2}, V_{S3} : Induced voltage in sheaths (1), (2) and (3) respectively Vm^{-1}.

I_{CS1}, I_{CS2}, I_{CS3} : The circulating currents in sheaths of phases (1), (2) and (3) respectively in A.

R_S : The resistance of sheath at its maximum operating temperature Ωm^{-1}.

$M_{1,2}$: The mutual inductance between core (1) and sheath (2) in Hm^{-1} .

$M_{1,3}$: The mutual inductance between core (1) and sheath (3) in Hm^{-1}.

$M_{2,3}$: The mutual inductance between core (2) and sheath (3) in Hm^{-1}.

ω : 2π x frequency (in cycles per second).

4.3.1.1 Three phase trefoil arrangement of cables

Due to the symmetrical disposition of cables [10]

$$M_{1,2} = M_{2,3} = M_{1,3} = M = 2x10^{-7} \ln\left(\frac{S}{r_{sh}}\right) \text{ H m-1} \qquad (4\text{-}3)$$

Balanced currents only are considered here. Consequently, the induced voltages and circulating currents in the sheaths will be respectively equal to each other for this system.

From equations (4-2) and (4-3), for the first cable,

$$V_{S1} = I_{CS1} R_S - j\,\omega\, M\,(I_2 + I_{CS2}) - j\,\omega\, M\,(I_3 + I_{CS3}) = I_{CS1} R_S + j\,\omega\, M\,(I_1 + I_{CS1}) \quad (4\text{-}4)$$

All cable sheaths are bonded at one end only, then

$I_{CS1} = 0$ and $V_{s1} = j\,\omega\, M\, I_1$

As a result, the induced sheath voltage per meter length will be numerically equals to

$$|V_{S1}| = \omega M I_1 = 2x10^{-7}\,\omega I \ln\left(\frac{S}{r_{sh}}\right)$$

Or $$|V_{S1}| = |V_{S2}| = |V_{S3}| = V_S = \omega M I = 2x10^{-7}\,\omega I \ln\left(\frac{S}{r_{sh}}\right) \qquad (4\text{-}5)$$

When all cable sheaths are bonded at each end of this circuit, then

$V_{S3} = 0 = V_{S1} = V_{S2}$

From equation (3-4) it follows that

$I_{S1} R_S + j\,\omega M\,(I_2 + I_{S1}) = 0$

and $$I_{CS1} = I_1 \frac{-j\omega M}{R_S + j\omega M}$$

Or numerically in general form:

$$|I_{CS}| = |I| \frac{\omega\ M}{\sqrt{R_S^2 + \omega^2 M^2}} = \frac{V_S}{\sqrt{R_S^2 + \omega^2 M^2}} \qquad (6\text{-}4)$$

The sheath loss per meter is

$$W_{CS} = |I_{CS}|^2 R_S = |I|^2 R_S \left(\frac{\omega^2 M^2}{R_S^2 + \omega^2 M^2} \right) \text{Wm-1} \quad (4\text{-}7)$$

From equation (4-7) as this loss is proportional to the square of the power current, it is most conveniently expressed as a ratio to the copper loss in the power conductor. This ratio then represents the amount by which the apparent resistance of the copper conductors is increased by the sheath losses.

The circulating sheath loss factor will be [20]:

$$\lambda_{CS} = \frac{R_S}{R} \left(\frac{\omega^2 M^2}{R_S^2 + \omega^2 M^2} \right) = \frac{R_S}{R} \left(\frac{1}{\left(\frac{R_S}{\omega M} \right)^2 + 1} \right) (4\text{-}8)$$

Let X = ω M

$$\lambda_{CS} == \frac{R_S}{R} \left(\frac{1}{\left(\frac{R_S}{X} \right)^2 + 1} \right) \qquad (4\text{-}9)$$

Where

I : The line currents in phases (1), (2) and (3) with balance condition

S : Spacing between axes of adjacent conductors in m

r_{sh} : Mean of outer and inner radii of sheath in m

W_{CS} : The circulating sheath loss in Wm^{-1}

λ_{CS} : The circulating sheath loss factor

X : The reactance per unit length of sheath Ω/m

R : The resistance of conductor at its maximum operating temperature Ωm^{-1}.

4.3.1.2 Three phase flat arrangement of cables

It is assumed that the phase rotation is such that

$$\left.\begin{aligned} I_1 &= I_2\left(-\frac{1}{2}+j\frac{\sqrt{3}}{2}\right) \\ I_3 &= I_2\left(-\frac{1}{2}-j\frac{\sqrt{3}}{2}\right) \end{aligned}\right\} \qquad (4\text{-}10)$$

When the cables are laid in a horizontal plane, with the middle cable equidistant from the two others, then [10]

$$M_{1,2} = M_{2,3} = 2x10^{-7}\ln\left(\frac{S}{r_{sh}}\right) \qquad \text{H m-1}$$

$$M_{1,3} = 2x10^{-7}\ln\left(\frac{2S}{r_{sh}}\right) = 2x10^{-7}\ln 2 + 2x10^{-7}\ln\left(\frac{S}{r_{sh}}\right) = M_m + M$$

Where

$$M_m = 2x10^{-7}\ln 2 = 1.389x10^{-7} \qquad \text{H m-1}$$

$$M = 2x10^{-7}\ln\left(\frac{S}{r_{sh}}\right) \qquad \text{H m-1}$$

When all cable sheaths are bonded at one end only, then

$I_{CS1} = I_{CS2} = I_{CS3} = 0$

The induced voltages in the cable sheaths per meter length, which can be found from equations (4-1) and (4-2) are

$$\left.\begin{aligned} V_{S1} &= -j\omega M_{1,2}I_2 - j\omega M_{1,3}I_3 \\ V_{S2} &= -j\omega M_{1,2}I_1 - j\omega M_{1,2}I_3 = j\omega M_{1,2}I_2 \\ V_{S3} &= -j\omega M_{1,3}I_1 - j\omega M_{1,2}I_2 \end{aligned}\right\} \qquad (4\text{-}11)$$

From equation (4-11) the numerical value of induced voltage in the sheath of the middle cable (V_{S2}), is equal to that of the trefoil layout.

The numerical values of V_{S1} and V_{S3} can be found from equations (4-11)

Let $X = \omega M$ and $X + X_m = \omega (M + M_m)$

The sheath voltages, V_{S1}, V_{S2}, V_{S3}, can be expressed by the following equations:

$$\left.\begin{aligned} V_{S1} &= -\frac{I_2}{2}\left[\sqrt{3}(X + X_m) + j(X - X_m)\right] \\ V_{S2} &= jI_2X \\ V_{S3} &= -\frac{I_2}{2}\left[-\sqrt{3}(X + X_m) + j(X - X_m)\right] \end{aligned}\right\} \qquad (4\text{-}12)$$

The numerical values of these voltages will be, for balanced three phase currents, as follows:

$$\left.\begin{aligned} |V_{S1}| &= |V_{S3}| = |I|\sqrt{X^2 + XX_m + X_m^2} \\ |V_{S2}| &= |I|X \\ |I_1| &= |I_2| = |I_3| = |I| \end{aligned}\right\} \qquad (4\text{-}13)$$

When all cable sheaths are bonded at each end of this circuit, then the circulating currents will flow and there may be a residual voltage a long the cable sheaths equal to V_0 Vm^{-1}.

V_0 could be zero when both ends of the cables are earthed.

Let

$$\left.\begin{aligned} V_{S1} &= V_{S2} = V_{S3} = V_0 \\ I_{CS1} + I_{CS2} + I_{CS3} &= 0 \end{aligned}\right\} \qquad (4\text{-}14)$$

From general equations (4-2) the sheath circulating currents could be found and therefore the sheath losses for the condition of balanced power currents.

The following equations are deduced from equations (4-2) and (4-10):

$$\left.\begin{aligned} V_{S1} = V_0 &= I_{CS1}(R_S + jX) - \frac{1}{2} jI_2(X - X_m) - \frac{\sqrt{3}}{2} I_2(X + X_m) - jI_{CS3}X_m \\ V_{S2} = V_0 &= I_{CS2}(R_S + jX) + jI_2X \\ V_{S3} = V_0 &= I_{CS3}(R_S + jX) - \frac{1}{2} jI_2(X - X_m) + \frac{\sqrt{3}}{2} I_2(X + X_m) - jI_{CS1}X_m \end{aligned}\right\} \quad (4\text{-}15)$$

From equations (4-15), the following equations can be obtained:

$$3V_0 = V_{S1} + V_{S2} + V_{S3} = j(I_2 + I_{CS2})X_m \quad (4\text{-}16)$$

$$3V_0 = 3V_{S2} = 3I_{CS2}(R_S + jX) + 3jI_2X \quad (4\text{-}17)$$

Or $$I_{CS2} = -I_2 \frac{j\left(X - \frac{X_m}{3}\right)}{R_S + j\left(X - \frac{X_m}{3}\right)} \quad (4\text{-}18)$$

$$I_{CS1} - I_{CS3} = I_2 \frac{\sqrt{3}(X + X_m)}{R_S + j(X + X_m)} \quad (4\text{-}19)$$

Let $$Q = X - \frac{X_m}{3}$$
$$P = X + X_m$$

Equations (4-18) and (4-19) can then be written respectively as

$$-I_{CS2} = (I_{CS1} + I_{CS3}) = I_2 \frac{jQ}{R_S + jQ} = I_2 \frac{(Q^2 + jR_SQ)}{R_S^2 + Q^2} \quad (4\text{-}20)$$

And $$(I_{CS1} - I_{CS3}) = I_2 \frac{\sqrt{3}P}{R_S + jP} = I_2 \frac{\sqrt{3}(R_SP - jP^2)}{R_S^2 + P^2} \quad (4\text{-}21)$$

From equations (4-20) and (4-21), I_{CS1}, I_{CS2}, I_{CS3} can be found

$$I_{CS1} = \frac{I_2}{2}\left[\frac{Q^2}{R_S^2+Q^2} + \frac{\sqrt{3}R_S P}{R_S^2+P^2} + j\left(\frac{R_S Q}{R_S^2+Q^2} - \frac{\sqrt{3}P^2}{R_S^2+P^2}\right)\right] \qquad (4\text{-}22)$$

$$I_{CS2} = -I_2\left(\frac{Q^2}{R_S^2+Q^2} + j\frac{R_S Q}{R_S^2+Q^2}\right) \qquad (4\text{-}23)$$

$$I_{CS3} = \frac{I_2}{2}\left[\frac{Q^2}{R_S^2+Q^2} - \frac{\sqrt{3}R_S P}{R_S^2+P^2}\right] + j\left(\frac{R_S Q}{R_S^2+Q^2} + \frac{\sqrt{3}P^2}{R_S^2+P^2}\right) \qquad (4\text{-}24)$$

From equations (4-22) and (4-24), it is interesting to note that the sheath currents as well as the sheath losses in the two outer cables are unequal. The un-equality is caused partly by the residual voltage along the sheaths and partly by the reactive effect of the sheath circulating currents.

Equations (4-22), (4-23) and (4-24) can be written as

$$\left.\begin{aligned} |I_{CS1}| &= |I|\sqrt{\frac{Q^2}{4\left(R_S^2+Q^2\right)} + \frac{3P^2}{4\left(R_S^2+P^2\right)} + \frac{\sqrt{3}PQR_S(Q-P)}{2\left(R_S^2+Q^2\right)\left(R_S^2+P^2\right)}} \\ |I_{CS2}| &= |I|\frac{Q}{\sqrt{R_S^2+Q^2}} \\ |I_{CS3}| &= |I|\sqrt{\frac{Q^2}{4\left(R_S^2+Q^2\right)} + \frac{3P^2}{4\left(R_S^2+P^2\right)} - \frac{\sqrt{3}PQR_S(Q-P)}{2\left(R_S^2+Q^2\right)\left(R_S^2+P^2\right)}} \end{aligned}\right\} \qquad (25\text{-}4)$$

The sheath losses per meter in each sheath are

$W_{CS1} = I_{CS1}^2 R_S$, $W_{CS2} = I_{CS2}^2 R_S$ and $W_{CS3} = I_{CS3}^2 R_S$

The sheath loss factor in each sheath is:

$$\lambda_{CS1} = \frac{R_S}{R}\left[\frac{\frac{1}{4}Q^2}{R_S^2+Q^2} + \frac{\frac{3}{4}P^2}{R_S^2+P^2} + \frac{\sqrt{3}R_S PQ(Q-P)}{2\left(R_S^2+Q^2\right)\left(R_S^2+P^2\right)}\right] \quad (4\text{-}26)$$

$$\lambda_{CS2} = \frac{R_S}{R}\frac{Q^2}{R_S^2+Q^2} \quad (4\text{-}27)$$

$$\lambda_{CS3} = \frac{R_S}{R}\left[\frac{\frac{1}{4}Q^2}{R_S^2+Q^2} + \frac{\frac{3}{4}P^2}{R_S^2+P^2} - \frac{\sqrt{3}R_S PQ(Q-P)}{2\left(R_S^2+Q^2\right)\left(R_S^2+P^2\right)}\right] \quad (4\text{-}28)$$

The three later equations can be written as:

$$\lambda_{CS1} = \frac{R_S}{R}\left[\frac{\frac{1}{4}Q^2}{R_S^2+Q^2} + \frac{\frac{3}{4}P^2}{R_S^2+P^2} - \frac{2R_S PQX_m}{\sqrt{3}\left(R_S^2+Q^2\right)\left(R_S^2+P^2\right)}\right] (4\text{-}29)$$

$$\lambda_{CS2} = \frac{R_S}{R}\frac{Q^2}{R_S^2+Q^2} (4\text{-}30)$$

$$\lambda_{CS3} = \frac{R_S}{R}\left[\frac{\frac{1}{4}Q^2}{R_S^2+Q^2} + \frac{\frac{3}{4}P^2}{R_S^2+P^2} + \frac{2R_S PQX_m}{\sqrt{3}\left(R_S^2+Q^2\right)\left(R_S^2+P^2\right)}\right] \quad (4\text{-}31)$$

Equations (4-9), (4-29), (4-30) and (4-31) are the same which have been listed in IEC-287 [19] for unarmored single-core cable in trefoil and flat formations.

Where

I_1 ,I_2 , I_3 : The vector current of cables 1, 2 and 3 respectively in A

X : The reactance of sheath per unit length of cable for two adjacent single-core cables Ωm^{-1}

X_m : Mutual reactance per unit length of cable between the sheath of an outer cable and the conductors of the other two, when cables are in flat

Formation Ωm^{-1}

V_0 : Residual voltage a long the cable sheath Vm^{-1}

$\lambda_{CS1}, \lambda_{CS2}, \lambda_{CS3}$: The circulating sheath loss factor for sheaths (1), (2) and (3) respectively.

4.3.1.3 Three phase arrangement with sheaths cross-bonded

According to IEC-287 [19], the circulating current loss is zero for installations where the sheaths are single-point bonded, and for installations where the sheaths are cross-bonded and each major section is divided into three electrically identical minor sections with keeping the currents flowing in the conductors are balanced.

4.3.2 Sheath eddy current and its loss

4.3.2.1 Introduction

In the development of equations for the sheath losses in the preceding section, it has been assumed that the sheath current density is uniform. In reality the current density is not uniform and the divergence from uniformity increases as the cables are brought closer together. Any lack of uniformity of current density will increase the ohmic losses, and the increased loss due to a non-uniform distribution will be referred to sheath eddy losses [20].

The eddy current losses occur in both 3-core and single-core cables, irrespective of the method of bonding [21]. Arnold [20], who is the author of previous equations which have been listed in IEC-287, has proved that the total loss in the sheath at any instant equals to the sum of the losses caused by the main circulating current and the eddy current, if considered separately, he also has developed an approximate formulas that give the sheath losses due to eddy currents for single-core cable in trefoil and flat formations with sheaths bonded at a single-point or two-points. While IEC-287 introduced formula for calculating eddy sheath

losses for single-core cable with sheaths cross-bonded and at the same time it is used for sheaths bonded at one end only. In this book, Arnold's formulas have been used for calculating eddy sheath losses for single-point bonding and two-points bonding, while IEC-287 formula has been used for calculating eddy sheath losses for cross-bonding.

4.3.2.2 Three phase trefoil arrangement of cables with sheaths bonded at a single-point or two-points [20]

In this case the sheath eddy loss factor and sheath eddy current will be calculated by equations (4-32) and (4-33).

$$\lambda_{SE} = \frac{3\omega^2}{R_S R}\left(\frac{r_{sh}}{S}\right)^2 10^{-14} \qquad (4\text{-}32)$$

$$I_{SE} = \sqrt{\frac{3I^2\omega^2}{R_S^2}\left(\frac{r_{sh}}{S}\right)^2 10^{-14}} \quad \text{A} \qquad (4\text{-}33)$$

Where

λ_{SE} : Sheath eddy-current loss factor

I_{SE} : Sheath eddy-current in A

4.3.2.3 Three phase flat arrangement of cables with sheaths bonded at a single-point or two-points [20]

In this case the sheath eddy loss factor and sheath eddy current will be calculated by equations (4-34) to (4-37).

$$\lambda_{SE1} = \lambda_{SE3} = \frac{3\omega^2}{2R_S R}\left(\frac{r_{sh}}{S}\right)^2 10^{-14} \quad (4\text{-}34)$$

$$I_{SE1} = I_{SE3} = \sqrt{\frac{3I^2\omega^2}{2R_S^2}\left(\frac{r_{sh}}{S}\right)^2 10^{-14}} \quad \text{A} \quad (4\text{-}35)$$

$$\lambda_{SE2} = \frac{6\omega^2}{R_S R}\left(\frac{r_{sh}}{S}\right)^2 10^{-14} \quad (4\text{-}36)$$

$$I_{SE2} = \sqrt{\frac{6I^2\omega^2}{R_S^2}\left(\frac{r_{sh}}{S}\right)^2 10^{-14}} \quad \text{A} \quad (4\text{-}37)$$

Where

λ_{SE1}, λ_{SE3} : Sheath eddy-current loss factor in two outer cables

λ_{SE2} : Sheath eddy-current loss factor in middle cable

I_{SE1}, I_{SE3} : Sheath eddy-current in two outer cables in A

I_{SE2} : Sheath eddy-current in middle cable in A

4.3.2.4 Three phase arrangement with sheaths cross-bonded [19]

In this case the sheath eddy loss factor and sheath eddy current will be calculated by equations (4-38) and (4-39).

$$\lambda_{SE} = \frac{R_S}{R}\left[g_S\lambda_0\left(1+\Delta_1+\Delta_2\right)+\left(\frac{(\beta_1 t_S)^4}{12x10^{12}}\right)\right] \quad (4\text{-}38)$$

$$I_{SE} = \sqrt{\frac{I^2 R\lambda_{SE}}{R_S}} \quad (4\text{-}39)$$

Where

$$g_S = 1 + \left(\frac{t_S}{D_S}\right)^{1.74} \left(\beta_1 D_S 10^{-3} - 1.6\right)$$

$$\beta_1 = \sqrt{\frac{4\pi\omega}{10^7 \rho_S}}$$

ρ_S : The electrical resistivity of sheath material at operating temperature $(\Omega.\mathrm{m})$

D_S : The external diameter of cable sheath (mm)

t_S : The thickness of sheath (mm)

Δ_1 and Δ_2 are factors which their values depend on the types of cable layouts formation.

g_S and β_1 are factors which their values depend on the cable parameters.

For lead-sheathed cables, g_S can be taken as unity and $\frac{(\beta_1 t_S)^4}{12x10^{12}}$ can be neglected.

For aluminum-sheathed cables both terms may need to be evaluated when sheath diameter is greater than about 70 mm or the sheath is thicker than usual.

Formulae for $\lambda 0$, Δ_1 and Δ_2 are given below:

(In which: $m = \frac{\omega}{R_S} 10^{-7}$, for $m \leq 0.1$, Δ_1 and Δ_2 can be neglected)

Where: ‘m’ is a factor depends on power frequency and metallic sheath resistance.

4.3.2.4.1 Three phase trefoil arrangement of cables [19]

In this case the sheath eddy loss factor and sheath eddy current will be calculated by substituting the following parameters in equations (4-38) and (4-39).

$$\lambda_0 = 3\left(\frac{m^2}{1+m^2}\right)\left(\frac{d}{2S}\right)^2$$

$$\Delta_1 = \left(1.14m^{2.45} + 0.33\right)\left(\frac{d}{2S}\right)^{(0.92+1.66)}$$

$$\Delta_2 = 0$$

4.3.2.4.2 Three phase arrangement in a flat

4.3.2.4.2.1 Center cable [19]

In this case the sheath eddy loss factor and sheath eddy current will be calculated by substituting the following parameters in equations (4-38) and (4-39).

$$\lambda_0 = 6\left(\frac{m^2}{1+m^2}\right)\left(\frac{d}{2S}\right)^2$$

$$\Delta_1 = 0.86m^{3.08}\left(\frac{d}{2S}\right)^{(1.4m+0.7)}$$

$$\Delta_2 = 0$$

4.3.2.4.2.2 Outer cable leading phase [19]

In this case the sheath eddy loss factor and sheath eddy current will be calculated by substituting the following parameters in equations (4-38) and (4-39).

$$\lambda_0 = 1.5\left(\frac{m^2}{1+m^2}\right)\left(\frac{d}{2S}\right)^2$$

$$\Delta_1 = 4.7m^{0.7}\left(\frac{d}{2S}\right)^{(0.16m+2)}$$

$$\Delta_2 = 21m^{3.3}\left(\frac{d}{2S}\right)^{(1.47m+5.06)}$$

4.3.2.4.2.3 Outer cable lagging phase [19]

In this case the sheath eddy loss factor and sheath eddy current will be calculated by substituting the following parameters in equations (4-38) and (4-39).

$$\lambda_0 = 1.5\left(\frac{m^2}{1+m^2}\right)\left(\frac{d}{2S}\right)^2$$

$$\Delta_1 = -\frac{0.74(m+2)m^{0.5}}{2+(m-0.3)^2}\left(\frac{d}{2S}\right)^{(m+1)}$$

$$\Delta_2 = 0.92m^{3.7}\left(\frac{d}{2S}\right)^{(m+2)}$$

4.3.3 A.C resistance of conductor

In order to calculate the conductor losses, a number of factors have to be calculated. The A.C resistance, R, of a cable is given by equation

$R = R_{dc}(1 + y_s + y_p)$ [19] (4-39)

$$R_{dc} = R_{20}\left[1+\alpha_{20}(\theta_C - 20)\right] \quad \text{(40-4) [19]}$$

y_s and y_p can be calculated as in [19]

Where

R_{dc} : The d.c. resistance of the conductor at 90 ^oC Ω/m

R_{20} : The d.c. resistance of the conductor at 20 ^oC Ω/m

y_s : The skin effect factor

y_p : The proximity effect factor

α_{20} : The constant mass temperature coefficient of conductor at 20 ^oC per Kelvin

θ_c : Conductor temperature

4.3.4 Sheath resistance

The sheath resistance depends on whether the sheath is a concentric neutral, a tape shield, or tubular configuration.

The ohmic resistance of the metallic sheath at a sheath temperature (θ_s) above 20°C is obtained by using the following formula:

$$R_S = \frac{\rho_{S20}}{A_S}\left[1+\alpha_{S20}(\theta_s - 20)\right] \quad (4\text{-}41)$$

Where

A_S : The sheath cross-sectional area mm^2

ρ_{S20} : The electrical resistivity of sheath material at 20 °C

α_{S20} : The constant mass temperature coefficient at 20 °C per Kelvin

4.3.4.1 Tubular metallic sheath

$A_S = \pi\, d_S\, t_S$ [19] (4-42)

In case of tubular metallic sheath:

$d_S = D_{Se} - t_S$ [19] (4-43)

Where

d_S: The mean diameter of the sheath (mm)

t_S : The thickness of sheath (mm)

D_{Se}: The external diameter of the sheath (mm)

4.3.4.2 Helically metallic sheath

In case of a helically metallic sheath (tape or wires):

The sheath resistance is obtained taking into account that the length of lay of the tape or wires [21].

$$R_S = \frac{\rho_{S20}}{A_S}\left[\sqrt{\left(\frac{\pi d_S}{\ell}\right)^2 + 1}\right]\left[1 + \alpha_{S20}(\theta_s - 20)\right] \qquad (4\text{-}44)$$

Where:

The length of lay of the tape or wire ℓ :

The distance that it takes for one strand of the conductor to make one complete revolution of the layer called the length of lay[22].

In case of a tape sheath, A_s will be calculated as tubular sheath.

In case of a wire sheath, A_s will be calculated per one strand and multiplied by the number of strands [22].

i.e.

$$R_{dc} = \frac{R_{strand}}{n} \qquad (4\text{-}45) \qquad [6]$$

Where

R_{strand}: Resistance of one strand, in Ω/m

n : Number of strands

The above algorithm has been used through MATLAB program and the flowchart of the computation steps is shown in figures (4-3(a)) and (4-3(b)).

Flowchart is given in Fig. (4-3-a) to show the computation steps of sheath currents, their losses and induced sheath voltages for single-core cable in trefoil layout with single-point bonding, two-points bonding and cross-bonding.

Flowchart is given in Fig. (4-3-b) to show the computation steps of sheath currents, their losses and induced sheath voltages for single-

core cable in flat layout with single-point bonding, two-points bonding and cross-bonding.

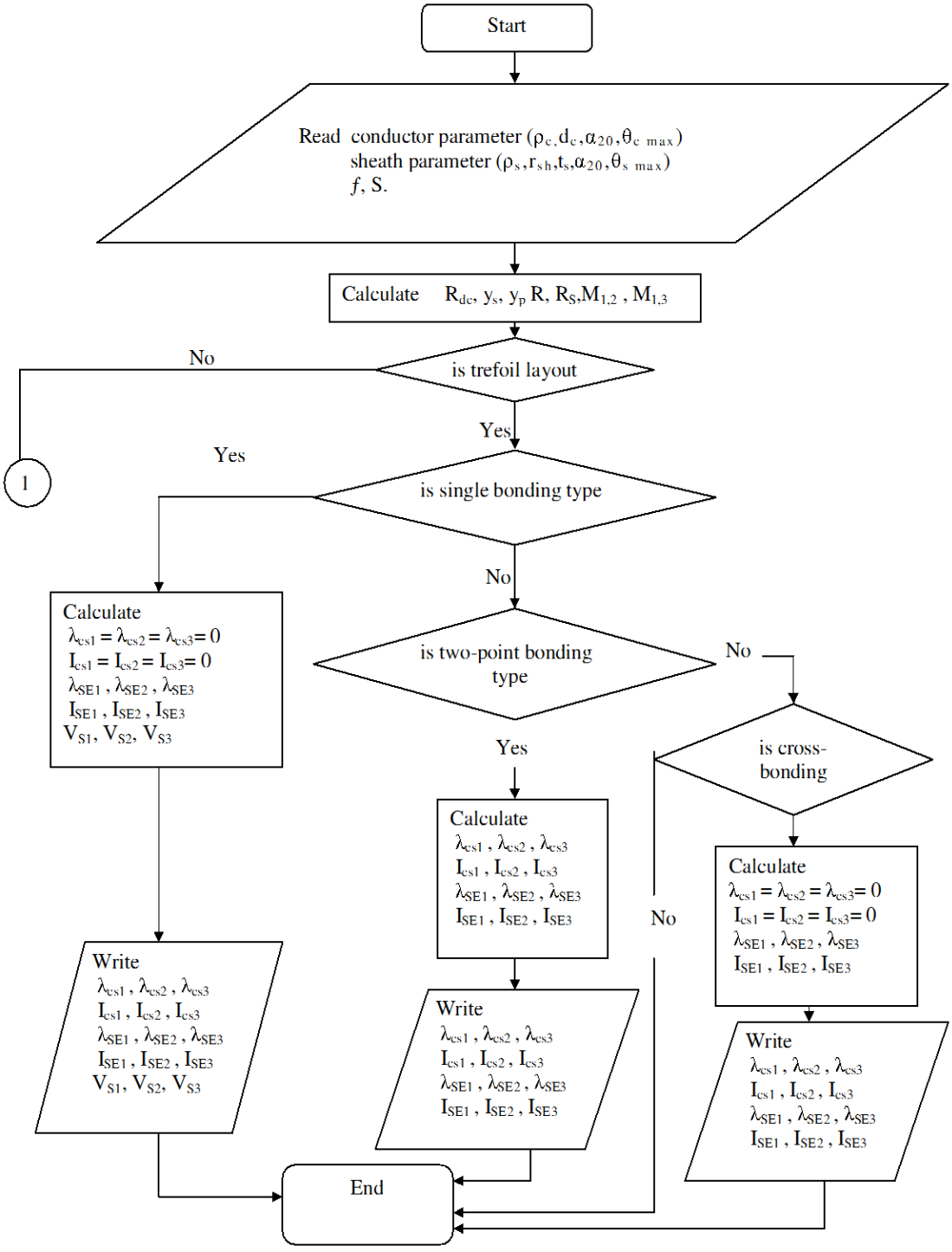


Fig.(4-3-a): Flowchart of the computation steps for trefoil layout

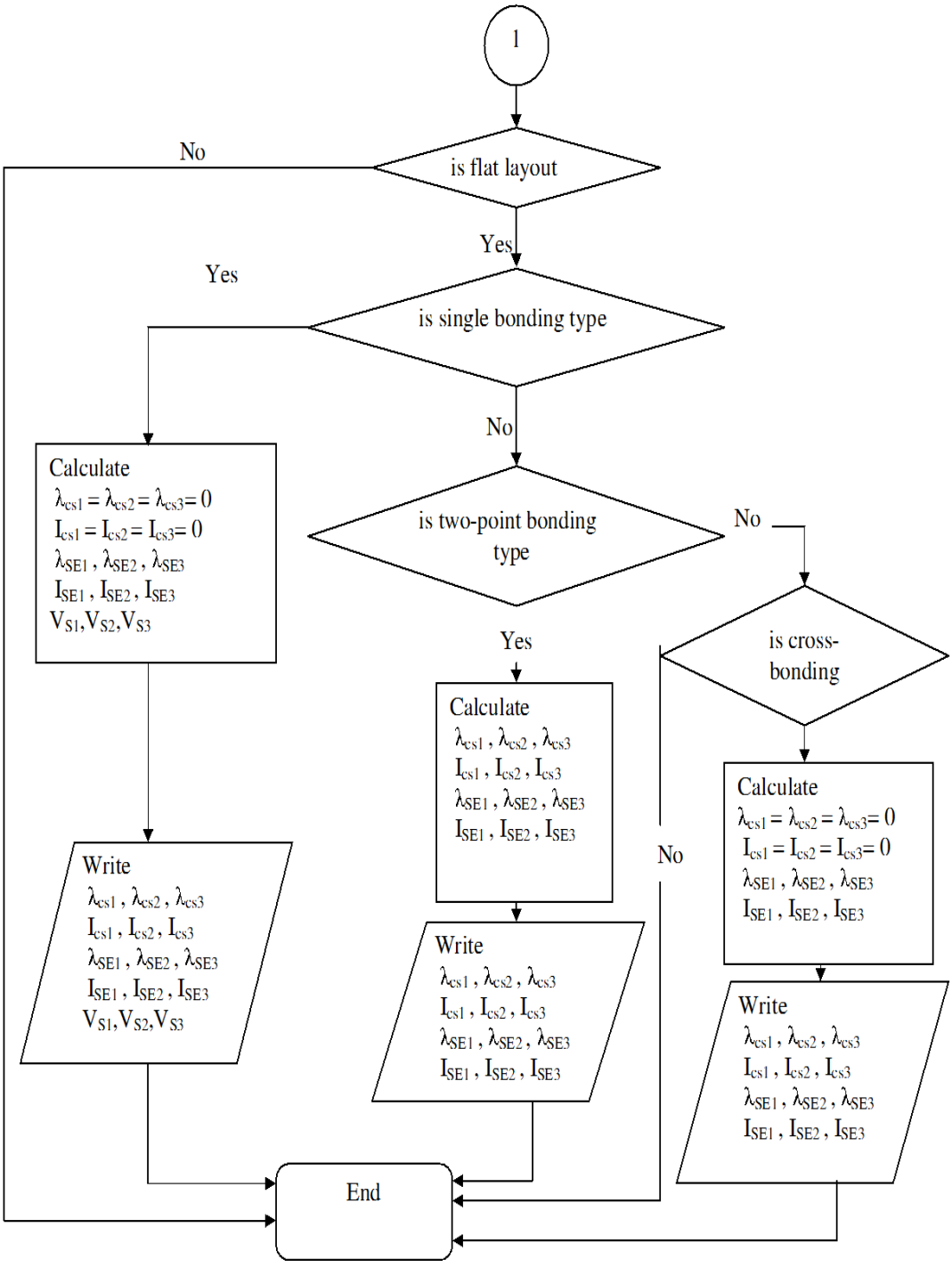


Fig.(4-3-b): Flowchart of the computation steps for flat layout

Where:

ρ_{S20} , ρ_{C20} : The electrical resistivity of sheath & conductor material at 20 ^{o}C respectively.

α_{S20} , α_{C20} : The constant mass temperature coefficient at 20 ^{o}C per Kelvin for sheath & conductor respectively.

d_C : Diameter of conductor

$\theta_{C\ max}$, $\theta_{S\ max}$: maximum operating temperature of conductor & sheath respectively.

ρ_S : The electrical resistivity of sheath material at operating temperature

t_S : The thickness of sheath .

S : Spacing between axes of adjacent conductors

r_{sh} : Mean of outer and inner radii of sheath

R_{dc} : The d.c. resistance of the conductor at 90 ^{o}C

y_s : The skin effect factor

y_p : The proximity effect factor

R : The resistance of sheath at its maximum operating temperature

R_S : The resistance of sheath at its maximum operating temperature.

$M_{1,2}$: The mutual inductance between core (1) and sheath (2).

$M_{1,3}$: The mutual inductance between core (1) and sheath (3).

f : power frequency.

λ_{CS1}, λ_{CS2}, λ_{CS3} : The circulating sheath loss factor for sheaths (1), (2) and (3) respectively.

λ_{SE1} , λ_{SE2} , λ_{SE3} : The eddy sheath loss factor for sheaths (1), (2) and (3) respectively.

I_{SE1} , I_{SE2} , I_{SE3} : The eddy currents in sheaths of phases (1), (2) and (3) respectively.

V_{S1}, V_{S2}, V_{S3} : Induced voltage in sheaths (1), (2) and (3) respectively.

I_{CS1}, I_{CS2}, I_{CS3} : The circulating currents in sheaths of phases (1), (2) and (3) respectively.

4.4 Factors Affecting the Sheath Losses in Single-Core Underground Power Cables

4.4.1 Effect of sheath bonding and cable layout formation on sheath losses

4.4.1.1 Introduction

Sheath circulating currents, sheath eddy currents and their corresponding loss factors for single-point bonding, two-points bonding and cross-bonding and also sheath induced voltages for single-point bonding have been calculated for single-core cable in touch trefoil and touch flat formations with using mathematical algorithm which is explained above to investigate the effect of sheath bonding methods and cable layouts formations on the sheath losses.

4.4.1.2 Cases study

The study is carried out by using single-core cables made of a stranded copper conductor with 800 mm^2 insulated by XLPE and covered by a lead screens, f = 50 Hz, with various voltage levels, to get a wide range of values of these variables, which their parameters [23] are listed in table (4-1).

Table (4-1): Single-core cables 800 mm^2 CU with lead screen parameters

Cable parameters	Voltage level (kV)					
	11	22	66	132	220	500
Conductor size (mm^2)	800	800	800	800	800	800

Diameter of the conductor (mm)	34	34	34	34	34	34
Mean sheath diameter (mm)	46	50.4	62.6	74.5	89.1	115.3
Outer diameter of cable (mm)	53	58	80	93	108	136
DC Resistance of the copper conductor at 20 °C Ω/km	0.0221					
Lead electrical resistivity at 20°C Ω.m	21.4×10^{-8}					
Copper electrical resistivity at 20°C Ω.m	1.7241×10^{-8}					
Temperature coefficient of copper per K at 20 °C	3.93×10^{-3}					
Temperature coefficient of lead per K at 20 °C	4×10^{-3}					

Ground temperature 20°C

Laying depth 1.0 m

Distance “S” between cable axes laid in flat formation De (De: the external diameter of the cable)

Ground thermal resistivity 1.0 Km/W

Assuming the sheath temperature equals to 70°C

Current rating (A) for copper conductor 995 A

4.4.1.3 Obtained results

The outputs of the program which represents the results for unarmored single-core cables are given in tables (4-2-a), (4-2-b) and (4-2-c).

Table (4-2-a) gives the values of sheath currents and their loss factors and induced voltages in the metallic sheaths in case of single-point bonding for touch trefoil and touch flat.

Table (4-2-a): Sheath currents, their loss factors and sheath induced voltages in case of single-point bonding method with lead screens

Voltage levels	Sheath bonding arrangement										Phase no.
	Single-point bonding-touch trefoil					Single-point bonding-touch flat (1 2 3)					
	λ_{CS} %	I_{CS}(A) A	λ_{SE} %	I_{SE} A	V_S (V/km)	λ_{CS} %	I_{CS} A	λ_{SE} %	I_{SE} A	V_S (V/km)	
11 kV	0	0	0.90	14.5 A	61.8	0	0	0.45	10.2 A	91.5	1
	0	0	0.90	14.5 A	61.8	0	0	1.80	20.5 A	61.8	2
	0	0	0.90	14.5 A	61.8	0	0	0.45	10.2 A	91.5	3
22 kV	0	0	1.07	17.1 A	60.9	0	0	0.54	12 A	90.7	1
	0	0	1.07	17.1 A	60.9	0	0	2.15	24.1 A	60.9	2
	0	0	1.07	17.1 A	60.9	0	0	0.54	12 A	90.7	3
66 kV	0	0	2.82	42.1 A	58.7	0	0	1.41	29.8 A	88.7	1
	0	0	2.82	42.1 A	58.7	0	0	5.64	59.6 A	58.7	2
	0	0	2.82	42.1 A	58.7	0	0	1.41	29.8 A	88.7	3
132 kV	0	0	3.96	57.4	57.2	0	0	1.98	40.5	87.3	1

				A					A		
	0	0	3.96	57.4 A	57.2	0	0	7.91	81 A	57.2	2
	0	0	3.96	57.4 A	57.2	0	0	1.98	40.5 A	87.3	3
220 kV	0	0	6.10	85.2 A	55.4	0	0	3.05	60.2 A	85.7	1
	0	0	6.10	85.2 A	55.4	0	0	12.20	120.4 A	55.4	2
	0	0	6.10	85.2 A	55.4	0	0	3.05	60.2 A	85.7	3
500 kV	0	0	10.32	139.2 A	53.7	0	0	5.16	98.4 A	84.2	1
	0	0	10.32	139.2 A	53.7	0	0	20.64	196.8 A	53.7	2
	0	0	10.32	139.2 A	53.7	0	0	5.16	98.4 A	84.2	3

Where:

λ_{CS} : The circulating sheath loss factor percentage of conductor loss

I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

V_S : The induced voltage in the sheath per km

Table (4-2-b) shows the values of sheath currents and their loss factors in case of two-points bonding for touch trefoil and touch flat.

Table (4-2-b): Sheath currents and their loss factors in case of two-points bonding method with lead screens

Voltage levels	**Sheath bonding arrangement**								
	Two-points bonding-touch trefoil				**Two-points bonding-touch flat** (1 2 3)				**Phase no.**
	λ_{CS}	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	λ_{CS} %	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	
11 kV	8.46	44.5 A	0.90	14.5 A	19.14	67 A	0.45	10.2 A	1
	8.46	44.5 A	0.90	14.5 A	4.97	34.1 A	1.80	20.5 A	2
	8.46	44.5 A	0.90	14.5 A	19.94	68.4 A	0.45	10.2 A	3
22 kV	9.51	50.8 A	1.07	17.1 A	21.65	76.8 A	0.54	12 A	1
	9.51	50.8 A	1.07	17.1 A	5.54	38.8 A	2.15	24.1 A	2
	9.51	50.8 A	1.07	17.1 A	22.69	78.6 A	0.54	12 A	3
66 kV	21.32	116 A	2.82	42.1 A	47.38	172.9 A	1.41	29.8 A	1
	21.32	116 A	2.82	42.1 A	12.18	87.7A	5.64	59.6 A	2
	21.32	116 A	2.82	42.1 A	52.79	182.5 A	1.41	29.8 A	3
132 kV	26.91	149.4 A	3.96	57.4 A	58.91	237.1 A	1.98	40.5 A	1

	26.91	149.4 A	3.96	57.4 A	15.19	112.2 A	7.91	81 A	2
	26.91	149.4 A	3.96	57.4 A	67.77	237.1 A	1.98	40.5 A	3
220 kV	35.87	206.6 A	6.10	85.2 A	75.07	298.8 A	3.05	60.2 A	1
	35.87	206.6 A	6.10	85.2 A	19.99	154.2 A	12.20	120.4 A	2
	35.87	206.6 A	6.10	85.2 A	91.25	329.5 A	3.05	60.2 A	3
500 kV	50.95	309.3 A	10.32	139.2 A	93.62	419.2 A	5.16	98.4 A	1
	50.95	309.3 A	10.32	139.2 A	28.50	231.3 A	20.64	196.8 A	2
	50.95	309.3 A	10.32	139.2 A	125.48	485.4 A	5.16	98.4 A	3

Where:

λ_{CS} : The circulating sheath loss factor percentage of conductor loss

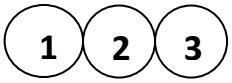
I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

Table (4-2-c) shows the values of sheath currents and their loss factors in case of cross bonding for touch trefoil and touch flat.

Table (4-2-c): Sheath currents and their loss factors in case of cross-bonding method with lead screens

Voltage levels	**Sheath bonding arrangement**								
	Cross bonding-touch trefoil				**Cross bonding-touch flat** 				**Phase no.**
	λ_{CS}	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	λ_{CS} %	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	
11 kV	0	0	0.90	14.5 A	0	0	0.45	10.2 A	1
	0	0	0.90	14.5 A	0	0	1.80	20.5 A	2
	0	0	0.90	14.5 A	0	0	0.45	10.2 A	3
22 kV	0	0	1.07	17 A	0	0	0.54	12 A	1
	0	0	1.07	17 A	0	0	2.15	24.1 A	2
	0	0	1.07	17 A	0	0	0.54	12 A	3
66 kV	0	0	2.81	42 A	0	0	1.4	29.7 A	1
	0	0	2.81	42 A	0	0	5.61	59.5 A	2
	0	0	2.81	42 A	0	0	1.4	29.7 A	3
132 kV	0	0	3.93	57.1 A	0	0	1.97	40.3 A	1
	0	0	3.93	57.1	0	0	7.86	80.7	2

				A				A	
	0	0	3.93	57.1 A	0	0	1.97	40.3 A	3
220 kV	0	0	6.22	86 A	0	0	3.54	64.9 A	1
	0	0	6.22	86 A	0	0	12.04	119.6 A	2
	0	0	6.22	86 A	0	0	2.71	56.7 A	3
500 kV	0	0	10.34	139.3 A	0	0	6.27	108.4 A	1
	0	0	10.34	139.3 A	0	0	19.96	193.6 A	2
	0	0	10.34	139.3 A	0	0	4.35	90.3 A	3

Where:

λ_{CS} : The circulating sheath loss factor percentage of conductor loss

I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

4.4.1.4 Results discussion

From the previous calculations for single-point bonding in table (4-2-a), it is noticed that:

- In case of single-point bonding there is no circulating current as no closed circuit for the sheath, hence there is no sheath circulating losses but sheath eddy losses are still present and they can be neglected as their values are small, also there is induced voltage at the open end.

- For trefoil layout the eddy losses are equal, while for flat layout the eddy losses in the outer cable sheaths are equal and usually smaller than the value of the middle cable sheath, for example, in case of 66 kV single-core cable:

For trefoil: $\lambda_{SE1}=\lambda_{SE2}=\lambda_{SE3}=2.82$ %.

For flat : $\lambda_{SE1}= \lambda_{SE3}=1.41$ % & $\lambda_{SE2} =5.64$ %.

But it must be noticed that, the total sheath eddy losses per circuit in trefoil are equal that in flat, i.e.

For trefoil: $\lambda_{SE1}+\lambda_{SE2}+\lambda_{SE3}=3*2.82 = 8.46$ %.

For flat : $\lambda_{SE1}+ \lambda_{SE3}+ \lambda_{SE2} =2*1.41 + 5.64 = 8.46$ %.

- For trefoil layout the induced sheath voltages are equal, while for flat layout the voltages induced in the outer cable sheaths are equal and usually larger than the voltage induced in the middle cable sheath, the values of induced sheath voltages in trefoil are equal to the value of induced sheath voltage in the middle cable sheath in flat formation, for example, in case of 66 kV single-core cable:

For trefoil: $V_{S1}=V_{S2}=V_{S3}=58.7$ V/km.

For flat : $V_{S1}=V_{S3}=88.7$ V/km & $V_{S2}=58.7$ V/km.

Or this also may be clearly appearing in Fig.(4-4).

- If the cables are laid in trefoil formation instead of flat arrangement, the induced voltages in the screens can be minimized.
- The sheath induced voltages for single-core cables with single-point bonding may be reached to hazard values in normal operations, so the length of cables must be limited to keep them within permissible limits, so it is expected the cable length in case of trefoil is longer than flat layout.
- The sheath induced voltages reduce with increasing the system voltages due to reducing (S/r_{sh}) ratio, factory cable design, as S in that case equals D_e.
- Addition to conductor current and cable length, the induced sheath voltage depends mainly on the spacing between phases as shown in Fig.(4-4)

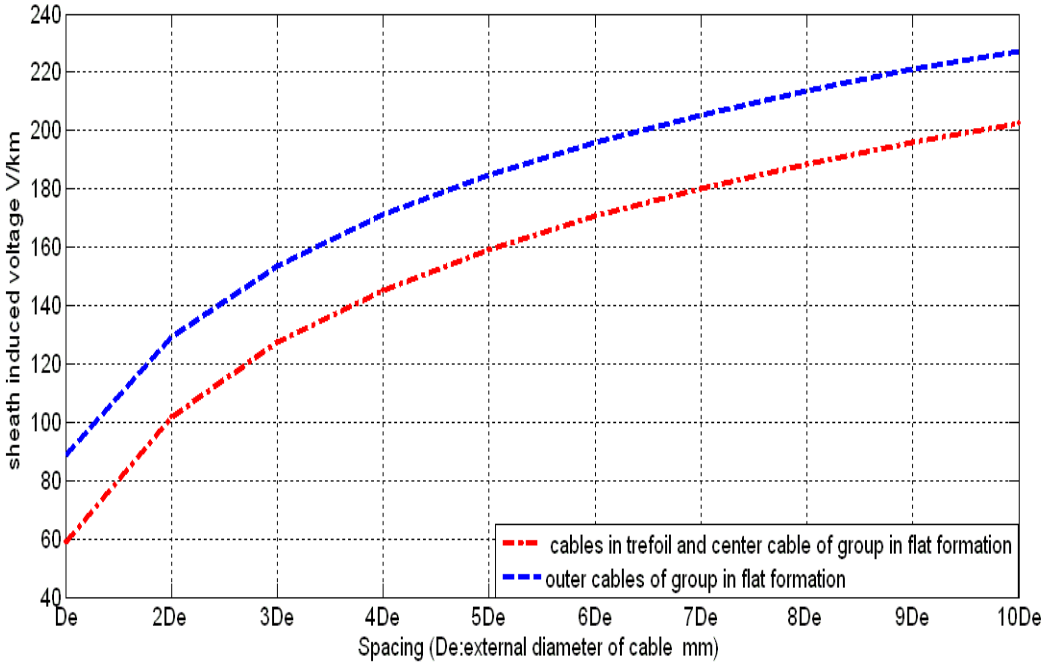


Fig.(4-4): Sheath induced voltage vs. cable spacing for single-core cable 66 kV in trefoil and flat formations with single-point bonding

From the previous calculations for two-points bonding in table (4-2-b), it is noticed that:

- Eddy loss could be disregarded with comparing to circulating loss but it must be noticed that the eddy loss value of middle conductor in flat formation with close spacing between phases especially for extra high system voltages cables must be taken into consideration as its value approaches to the value of sheath circulating loss for the same conductor, for example, in case of 500 kV single-core cable in table (4-2-b):

For flat : λ_{SE2} =20.64 % & λ_{CS2} =28.50 %.

- The trefoil configuration has lower total sheath losses than flat formation when sheaths are bonded at both ends; also it introduces symmetrical results for all calculations.
- For cables in flat configuration when sheaths are bonded at both ends, the sheath circulating losses have unequal magnitude; the least value occurs in the sheath of the middle cable, values in sheaths of outer cables are of unequal magnitude too. Thereby, the cable sheath of the lag phase has a higher value.

- The sheath circulating losses in two-points bonding method could be reached to more than the conductor losses, as in case of 500 kV single-core cable in table (4-2-b) for two-points arrangement with flat formation λ_{CS3}=125.48 %, this cause the insulation of the conductor to be subjected to temperatures may be excess of the insulation ratings, so the cable ampacity must be de-rated.

From the previous calculations for cross bonding in table (4-2-c), it is noticed that:

- According to IEC 60287, the eddy loss only exists where the sheaths are cross bonded and each major section is divided into three identical minor sections.
- In cross-bonding arrangement, the total sheaths losses per circuit in trefoil formation are approximately equal the total sheath losses in flat formation.

From the previous calculations in tables (4-2-a), (4-2-b) and (4-2-c), it is noticed that:

- Both single-point bonding and cross-bonding have sheath losses lower than two-points bonding arrangement.

- The sheath losses in single-point bonding are approximately the same as in cross bonding as shown in tables (4-2-b) and (4-2-c). In single-point bonding, the induced voltage limits the cable length, while cross bonding is preferred in long run lengths due to sheath voltages cancelation in each major section.

- The sheath eddy losses in outer two phases in flat formation are equal for single-point bonding and two-points bonding while for cross-bonding this is true only for m factor ≤ 0.1 ($m = \frac{\omega}{R_S} 10^{-7}$), refer to clause 4.3.2.4 . For example, in case of 500 kV single-core cable:

  In table (4-2-a) for single-point bonding:

For flat : $\lambda_{SE1} = \lambda_{SE3}$=5.16 % .

In table (4-2-b) for two-points bonding:

For flat : $\lambda_{SE1} = \lambda_{SE3}$=5.16 %.

In table (4-2-c) for cross bonding:

For flat : λ_{SE1} =6.27 % & λ_{SE3}=4.35 % ($\lambda_{SE1} \neq \lambda_{SE3}$).

While in case of 66 kV single-core cable:

In table (4-2-c) for cross bonding:

For flat : $\lambda_{SE1} = \lambda_{SE3}$=1.4 %.

For the same layouts formations, Arnold equations for calculating eddy losses (tables (4-2-a) & (4-2-b)) give approximately the same values which have been given in case of using IEC-287 equation (table (4-2-c)), less divergence occurs with increasing m factor over 0.1, so one of equations could be used for calculating eddy losses to any sheath bonding method. The maximum values of the total sheath currents per phase (i.e. sheath eddy current + sheath circulating current) with sheaths bonded at two-points in touch trefoil reached to 45 % of line current (in table (4-2-b) in case of 500 kV $I_{SE3} + I_{CS3}$= 45 % of rating current) while they reached to 58.7 % in touch flat formations (in table (4-2-b) in case of 500 kV $I_{SE3} + I_{CS3}$= 58.7 % of rating current) and they reached to 14 % and 19.5 % respectively with sheaths cross bonded (in table (4-2-c) in case of 500 kV). The total sheath loss factors per circuit in touch trefoil and flat formations (i.e. $\lambda_{CS1}+ \lambda_{CS2}+\lambda_{CS3} +\lambda_{SE1}+ \lambda_{SE2} +\lambda_{SE3}$) with sheaths bonded at two-points reached to 183.81 % and 278.56 % of conductor copper losses respectively (in table (4-2-c) in case of 500 kV) while they reached to 31.02 % and 30.58 % of conductor copper losses respectively with sheaths cross bonded (in table (4-2-c) in case of 500 kV).

4.4.2 Effect of cable parameters (conductor size & its resistivity) on the sheath losses in single-core cables

4.4.2.1 Introduction

The purpose of the core conductor is to transmit the required current with low losses. copper and aluminum of the metals are commonly used for

conductors with various conductor sizes, so the conductor material resistivity, which is determined by the material the conductor is made from, and the conductor sizes have been examined by using the mathematical algorithm, which is explained in clause 4.3, by calculating the sheath losses for aluminum and copper conductor with the same dimensions to examine the effect of conductor resistivity on the sheath losses, and also using single-core cables with various conductor sizes with the same voltage to examine the effect of conductor sizes on the sheath losses.

4.4.2.2 Cases study

1- 66 kV single-core cable, made of a stranded copper conductor with 800 mm^2 insulated by XLPE and covered by a lead screen, f = 50 Hz, which its parameters have been listed in table (4-1), and another single-core cable made of a stranded aluminum conductor with the same dimensions, which their electrical properties have been listed in tables (4-3), are used to examine the effect of conductor material resistivity on the sheath losses.

2- 66 kV Single-core cables, made of a stranded copper conductor with various cross sectionals insulated by XLPE and covered by a lead screens ,f = 50 Hz, which their parameters have been listed in table (4-4), are used to examine the effect of conductor's size on the sheath losses.

Table (4-3): Electrical dc resistances and temperature coefficients for 800 mm^2 copper and aluminum conductors

Conductor material	DC Resistance of the conductor at 20°C, Ω/km	Temperature coefficient per K at 20°C
CU (I=995 A)	0.0221	$3.93x10^{-3}$
AL (I=795 A)	0.0367	$4.03x10^{-3}$

Table (4- 4): Single-core cables 66 kV-CU with lead screens parameters

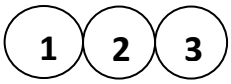
Cable parameters	conductor size mm^2					
	240	300	400	500	630	800
Current rating (A)	530.1	599.9	683.6	776.6	889	995
Outer diameter of cable (mm)	62	64	67	72	76	80
Mean sheath diameter (mm)	46.2	50.8	53.8	57.1	61	62.6
DC Resistance of the copper conductor (20°C), Ω/km	0.0754	0.0601	0.0470	0.0366	0.0283	0.0221
diameter of the conductor (mm)	18.1	20.4	23.2	26.3	30.2	34

4.4.2.3 Obtained results

4.4.2.3.1 Effect of conductor material resistivity on the sheath losses

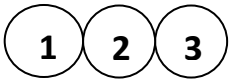
Tables (4-5-a) and (4-5-b) show the values of sheath currents and their loss factors for touch trefoil and touch flat layouts in two single-core cables, one of them is made of a stranded copper conductor and the other is made of a stranded aluminum conductor in case of two-points bonding and cross bonding respectively to indicate the effect of conductor material resistivity on the sheath losses.

Table (4- 5-a): Sheath currents and their loss factors in case of two-points bonding method for copper and aluminum conductors

Conductor material	**Sheath bonding arrangement**		
	Two-points bonding-touch trefoil	**Two-points bonding-touch flat** 	**Phase no.**

	λ_{CS}	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	λ_{CS} %	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	
CU	21.32	116 A	2.82	42.1 A	47.38	172.9 A	1.41	29.8 A	1
	21.32	116 A	2.82	42.1 A	12.18	87.7A	5.64	59.6 A	2
	21.32	116 A	2.82	42.1 A	52.79	182.5 A	1.41	29.8 A	3
AL	13.78	92.6 A	1.82	33.7 A	30.62	138.1 A	0.91	23.8 A	1
	13.78	92.6 A	1.82	33.7 A	7.87	70 A	3.64	47.6 A	2
	13.78	92.6 A	1.82	33.7 A	34.11	145.8 A	0.91	23.8 A	3

Table (4-5-b): Sheath currents and their loss factors in case of cross-bonding method for copper and aluminum conductors

Conductor material	**Sheath bonding arrangement**				**Phase no.**
	Cross bonding-touch trefoil		**Cross bonding-touch flat** 		
	λ_{SE} %	I_{SE} (A)	λ_{SE} %	I_{SE} (A)	
CU	2.81	42 A	1.4	29.7 A	1
	2.81	42 A	5.61	59.5 A	2
	2.81	42 A	1.4	29.7 A	3
AL	1.81	33.6 A	0.91	23.7 A	1

	1.81	33.6 A	3.63	47.5 A	2
	1.81	33.6 A	0.91	23.7 A	3

Where:

λ_{CS} : The circulating sheath loss factor percentage of conductor loss

I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

4.4.2.3.2 Effect of conductor sizes on the sheath losses

Tables (4-6-a) and (4-6-b) show the values of sheath currents and their loss factors for touch trefoil and touch flat layouts in single-core cables with various sizes in case of two-points bonding and cross bonding to indicate the effect of conductor sizes on the sheath losses.

Table (4-6-a): Sheath currents and their loss factors for various sizes in case of two-points bonding method

Cross section mm^2	**Sheath bonding arrangement**								**Phase no.**
	Two-points bonding-touch trefoil				**Two-points bonding-touch flat** (1 2 3)				
	λ_{CS}	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	λ_{CS} %	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	
240	4.67	38.9 A	0.5	12.7 A	10.38	58 A	0.25	9 A	1
	4.67	38.9 A	0.5	12.7 A	2.75	29.8 A	1	18 A	2
	4.67	38.9	0.5	12.7	11.10	59.9	0.25	9 A	3

		A		A		A			
300	6.17	47.5 A	0.72	16.2 A	13.83	71.2 A	0.36	11.4 A	1
	6.17	47.5 A	0.72	16.2 A	3.59	36.2 A	1.43	22.9 A	2
	6.17	47.5 A	0.72	16.2 A	14.88	73.9 A	0.36	11.4 A	3
400	8.41	59.2 A	1.03	20.7 A	18.92	88.9 A	0.51	14.6 A	1
	8.41	59.2 A	1.03	20.7 A	4.85	45 A	2.05	29.2 A	2
	8.41	59.2 A	1.03	20.7 A	20.50	92.5 A	0.51	14.6 A	3
500	12.01	75.4 A	1.40	25.7 A	26.57	112.2 A	0.70	18.2 A	1
	12.01	75.4 A	1.40	25.7 A	6.99	57.5 A	2.80	36.4 A	2
	12.01	75.4 A	1.40	25.7 A	29.04	117.3 A	0.70	18.2 A	3
630	15.61	91.1 A	1.94	32 A	34.74	135.9 A	0.97	22.6 A	1
	15.61	91.1 A	1.94	32 A	9	69.1 A	3.87	45.3 A	2
	15.61	91.1 A	1.94	32 A	38.18	142.5 A	0.97	22.6 A	3
800	21.32	116 A	2.82	42.1 A	47.38	172.9 A	1.41	29.8 A	1
	21.32	116 A	2.82	42.1 A	12.18	87.7A	5.64	59.6 A	2

	21.32	116 A	2.82	42.1 A	52.79	182.5 A	1.41	29.8 A	3

Where:

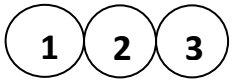
λ_{CS} : The circulating sheath loss factor percentage of conductor loss

I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

Table (4-6-b): Sheath currents and their loss factors for various sizes in case of cross-bonding method

Cross section mm^2	**Sheath bonding arrangement**				
	Cross bonding-touch trefoil		**Cross bonding-touch flat** 		**Phase no.**
	λ_{SE} %	I_{SE}(A)	λ_{SE} %	I_{SE}(A)	
240	0.5	12.7 A	0.25	9 A	1
	0.5	12.7 A	1	18 A	2
	0.5	12.7 A	0.25	9 A	3
300	0.71	16.1 A	0.36	11.4 A	1
	0.71	16.1 A	1.43	22.9 A	2
	0.71	16.1 A	0.36	11.4 A	3
400	1.02	20.6 A	0.51	14.6 A	1
	1.02	20.6 A	2.05	29.2 A	2

	1.02	20.6 A	0.51	14.6 A	3
500	1.40	25.7 A	0.70	18.2 A	1
	1.40	25.7 A	2.79	36.4 A	2
	1.40	25.7 A	0.70	18.2 A	3
630	1.93	32 A	0.96	22.6 A	1
	1.93	32 A	3.86	45.3 A	2
	1.93	32 A	0.96	22.6 A	3
800	2.81	42 A	1.4	29.7 A	1
	2.81	42 A	5.61	59.5 A	2
	2.81	42 A	1.4	29.7 A	3

Where:

λ_{CS} : The circulating sheath loss factor percentage of conductor loss

I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

4.4.2.4 Discussion of the obtained results

From tables (4-5-a) and (4-5-b) when the conductor materials are copper and aluminum It is noticed that:

- Both sheath circulating loss factor and sheath eddy loss factor decrease as the conductor resistivity increase, i.e. the sheath loss factors (λ_{SE} & λ_{CS}) are inversely proportional to the conductor resistivity.
- The sheath losses in flat formation with two-points bonding have more sensitivity to conductor material resistivity than other types of bonding, as the total sheath losses factor per circuit in touch flat and

touch trefoil formations increased by 42.75 % and 25.62% of conductor losses respectively in case of using copper conductor instead of using aluminum conductor, as shown in table (4-5-a), i.e.((λ_{CS1}+ λ_{CS2}+λ_{CS3} +λ_{SE1}+ λ_{SE2} +λ_{SE3}) with CU)-((λ_{CS1}+ λ_{CS2}+λ_{CS3} +λ_{SE1}+ λ_{SE2} +λ_{SE3}) with AL) = +42.75 in touch flat & +25.62 in touch trefoil), while they increased by 2.96 % and 3.62 % of conductor losses respectively with sheaths cross bonded, as shown in table (4-5-b), i.e. ((λ_{CS1}+ λ_{CS2}+λ_{CS3} +λ_{SE1}+ λ_{SE2} +λ_{SE3}) with CU)-((λ_{CS1}+ λ_{CS2}+λ_{CS3} +λ_{SE1}+ λ_{SE2} +λ_{SE3}) with AL) = +2.96 in touch flat & +3.62 in touch trefoil, i.e. conductor resistivity is one of the method for controlling the sheath losses in two-points bonding arrangement.

From tables (4-6-a) and (4-6-b) with changing the conductor sizes, it is noticed that:

- Both sheath circulating loss factor and sheath eddy loss factor increase with increasing the conductor sizes.
- The cross-bonding has very low sensitivity to the changing of small conductor sizes, while flat formation with two-points bonding has more sensitivity to the changing of small conductor sizes.
- In lower conductor sizes, both sheath circulating loss factor and sheath eddy loss factor can be neglected.
- Trefoil layout introduces a good solution to overcome the problems of high sheath circulating losses values in two-points bonding method rather than flat layout.

4.4.3 Effect of cable spacing on the sheath losses

4.4.3.1 Introduction

The previous mathematical algorithm, explained in clause 4.2, is used to investigate the effect of varying the spacing between cables on the sheath losses by varying the spacing between cables from D_e to $2D_e$. (D_e: The external diameter of the cable)

4.4.3.2 Cases study

66 kV single-core cable, made of a stranded copper conductor with 800 mm^2 insulated by XLPE and covered by a lead screen, f = 50 Hz, which its parameters have been listed in table (4-1), is used in this case study.

4.4.3.3 Obtained results by using IEC 60287

The obtained results using IEC 60287 have been shown in tables (4-7-a) and (4-7-b). In these tables sheath currents and their losses are calculated with changing the axial spacing between the cables from D_e to 2 D_e in case of two-points bonding (trefoil & flat) and cross bonding (trefoil & flat) respectively.

Table (4-7-a): Sheath currents and their loss factor in case of two-points bonding methods with D_e and $2D_e$ spacing between cables

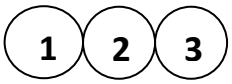
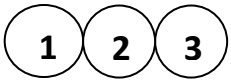
Spacing	**Sheath bonding arrangement**								
	Two-points bonding-touch trefoil				**Two-points bonding-touch flat** 				**Phase no.**
	λ_{CS}	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	λ_{CS} %	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	
D_e mm	21.32	116 A	2.82	42.1 A	47.38	172.9 A	1.41	29.8 A	1
	21.32	116 A	2.82	42.1 A	12.18	87.7A	5.64	59.6 A	2
	21.32	116 A	2.82	42.1 A	52.79	182.5 A	1.41	29.8 A	3
$2D_e$ mm	64.47	198.9 A	0.72	21 A	98.92	246.4 A	0.36	14.9 A	1
	64.47	198.9 A	0.72	21 A	48.01	171.1 A	1.45	29.8 A	2
	64.47	198.9 A	0.72	21 A	113.63	264.1 A	0.36	14.9 A	3

Table (4-7-b): Sheath currents and their loss factor in case of cross bonding methods with D_e and $2D_e$ spacing between cables

Spacing	**Sheath bonding arrangement**				
	Cross bonding-touch trefoil		**Cross bonding-touch flat** 		**Phase no.**
	λ_{SE} %	I_{SE}(A)	λ_{SE} %	I_{SE}(A)	
D_e mm	2.81	42 A	1.4	29.7 A	1
	2.81	42 A	5.61	59.5 A	2
	2.81	42 A	1.4	29.7 A	3
$2D_e$ mm	0.72	21 A	0.36	14.8 A	1
	0.72	21 A	1.44	29.7 A	2
	0.72	21 A	0.36	14.8 A	3

Where:

λ_{CS} : The circulating sheath loss factor percentage of conductor loss

I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

4.4.3.4 Discussion of the obtained results

From tables (4-7-a) and (4-7-b), it is seen that:

- In case of trefoil and flat formation when sheaths are bonded at both ends, the sheath circulating losses increase with increasing the cable spacing.
- The sheath circulating losses could be reached to more than its double values with duplicating the spacing between phases.

- The sheath eddy losses decrease with increasing the cable spacing, so it can be deduced that for larger cables the effect of spacing on total sheath losses is much lesser than that on the sheath circulating losses alone.

In general, the effect of spacing on the sheath circulating losses and sheath eddy losses for single-core cable can be shown in figures (4-5),(4-6), (4-7) and (4-8).

Figure (4-5) shows the values of sheath circulating loss factor with varying the axial spacing between the conductors for single-core cable in case of its sheaths bonded at two-points with trefoil formation.

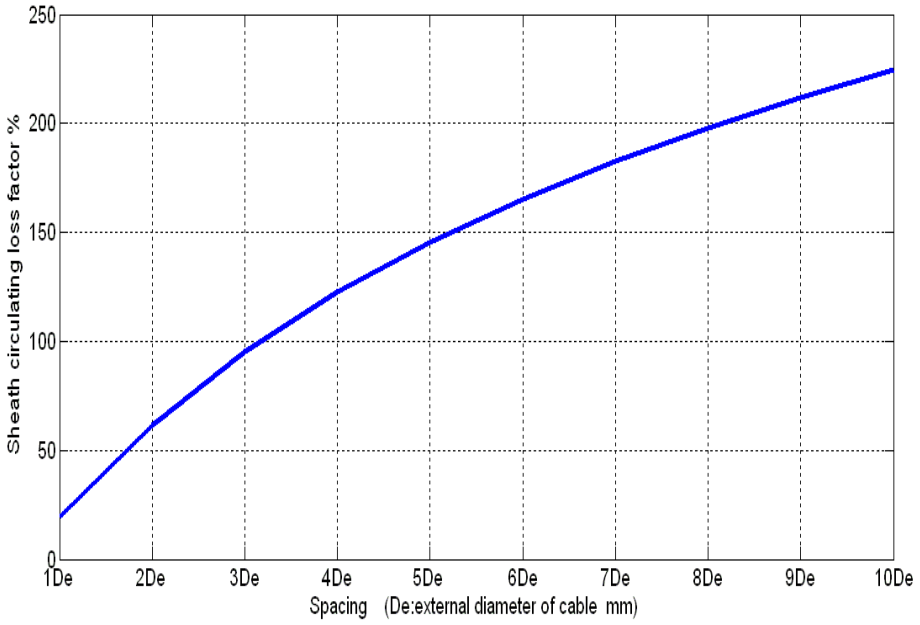


Fig. (4-5): Sheath circulating loss factor vs. spacing for 66 kV single-core cable trefoil formation with two-points bonding

Figure (4-6) shows the values of sheath circulating loss factor with varying the axial spacing between the conductors for single-core cable in case of its sheaths bonded at two-points with flat formation.

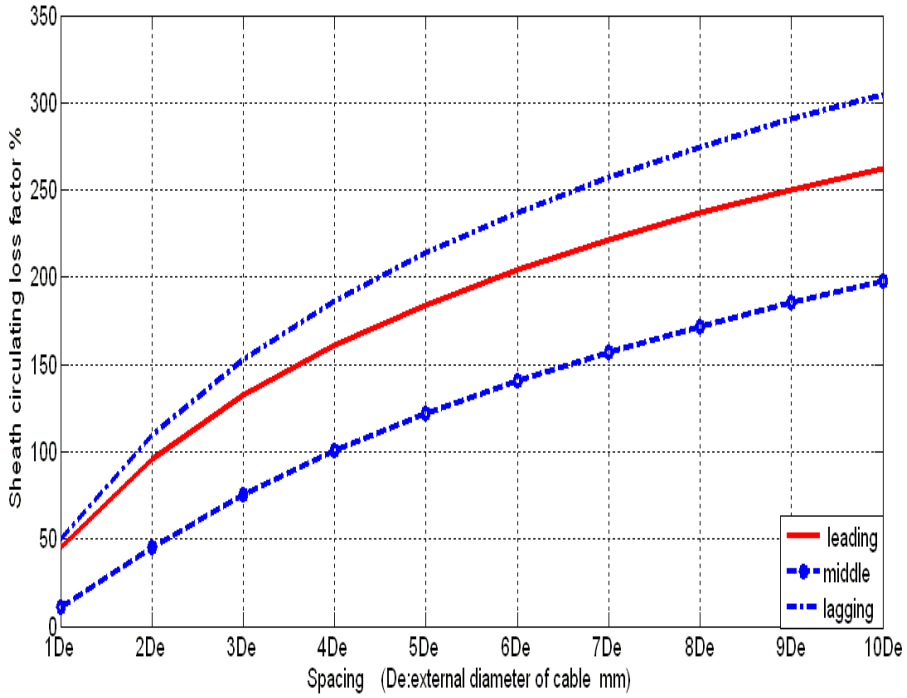


Fig. (4-6): Sheath circulating loss factor vs. spacing for 66 kV single-core cable flat formation with two-points bonding

Figure (4-7) shows the values of sheath eddy loss factor with varying the axial spacing between the conductors for single-core cable in case of its sheaths bonded at two-points with trefoil formation.

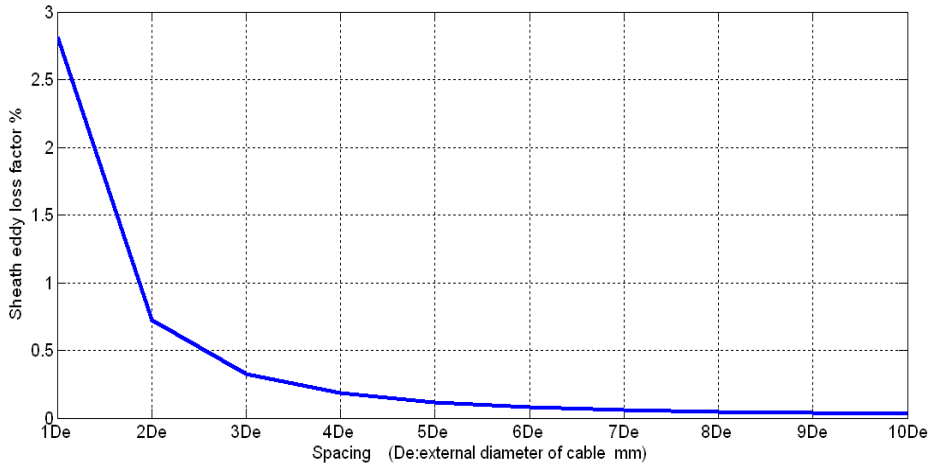


Fig. (4-7): Sheath eddy loss factor vs. spacing for 66 kV single-core cable trefoil formations with two-points bonding

Figure (4-8) shows the values of sheath eddy loss factor with varying the axial spacing between the conductors for single-core cable in case of its sheaths bonded at two-points with flat formation.

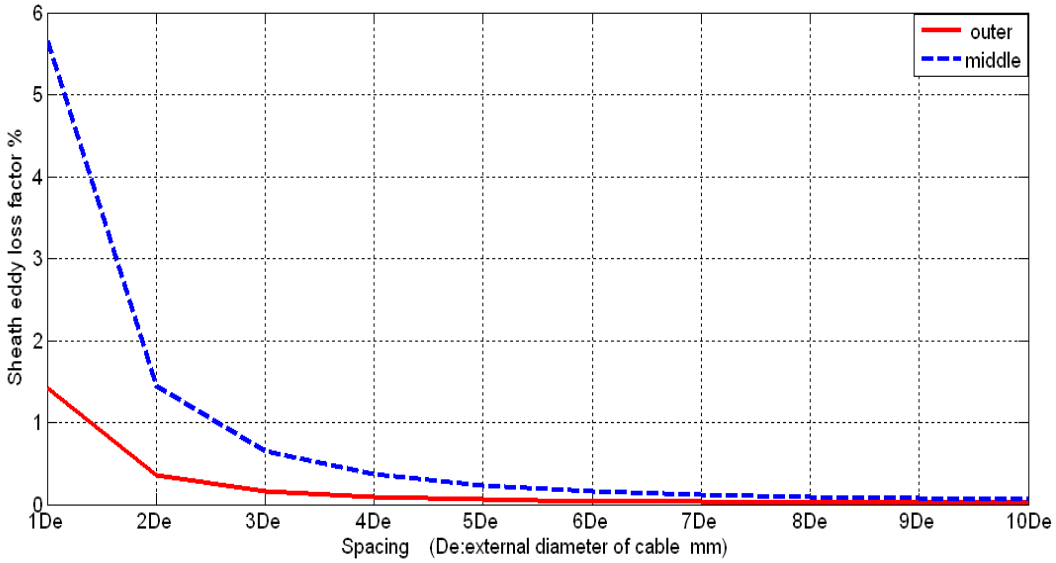


Fig. (4-8): Sheath eddy loss factor vs. spacing factor for 66 kV single-core cable flat formations with two-points bonding

From figures (4-5) and (4-6) which show the effect of cable spacing on sheath circulating losses, it can be seen that:

- The sheath circulating losses are proportional to the spacing between phases.
- The sheath circulating losses could be reached to more than two times the conductor loss depending on the spacing between phases.

From figures (4-7) and (4-8) which show the effect of cable spacing on sheath eddy losses, it is clearly appearing that:

- The sheath eddy losses are inversely proportional to the spacing between phases.
- The sheath eddy losses reduce rapidly at lower spacing, while reduce very slowly at large spacing.
- The sheath eddy losses can be neglected at large spacing.
- Cross-bonding method is more active method with increasing the spacing between phases but in one condition which is keeping the minor section lengths of sheath are equal, because according to IEC 60287 the eddy losses are then only exist which are inversely proportional to spacing between cables.

4.4.4 Effect of sheath resistance on the sheath losses

4.4.4.1 Introduction

Mathematical algorithm, which is explained in clause 4.3, is used to examine more closely the effect of sheath resistance on the sheath losses.

4.4.4.2 Cases study

66 kV single-core cable, made of a stranded copper conductor with 800 mm^2 insulated by XLPE and covered by a lead screen, f = 50 Hz, which its parameters have been listed in table (4-1), is used in this case study. A.C sheath resistance (R_S) at 70 ^oC = 0.5 Ω/km.

4.4.4.3 Obtained results by using IEC 60287

Figures (4-9), (4-10), (4-11), (4-12) and (4-13) show the obtained results.

4.4.4.3.1 Effect of sheath resistance on the sheath circulating losses

Figure (4-9) shows the sheath circulating loss factor with varying A.C sheath resistance of single-core cable in trefoil formation in case of two-points bonding with axial spacing between cables De and 2De.

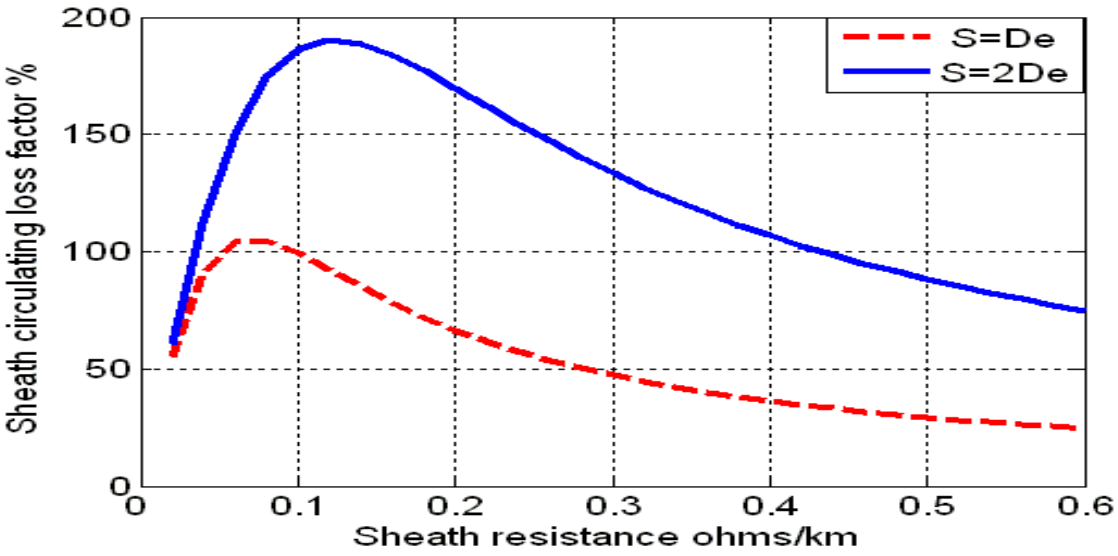


Fig. (4-9): Sheath circulating loss factor vs. sheath resistance in trefoil formation with two-points bonding for D_e and $2D_e$ spacing between cables

Figure (4-10) shows the sheath circulating current with varying A.C sheath resistance of single-core cable in trefoil formation in case of two-points bonding with axial spacing between cables D_e and $2D_e$.

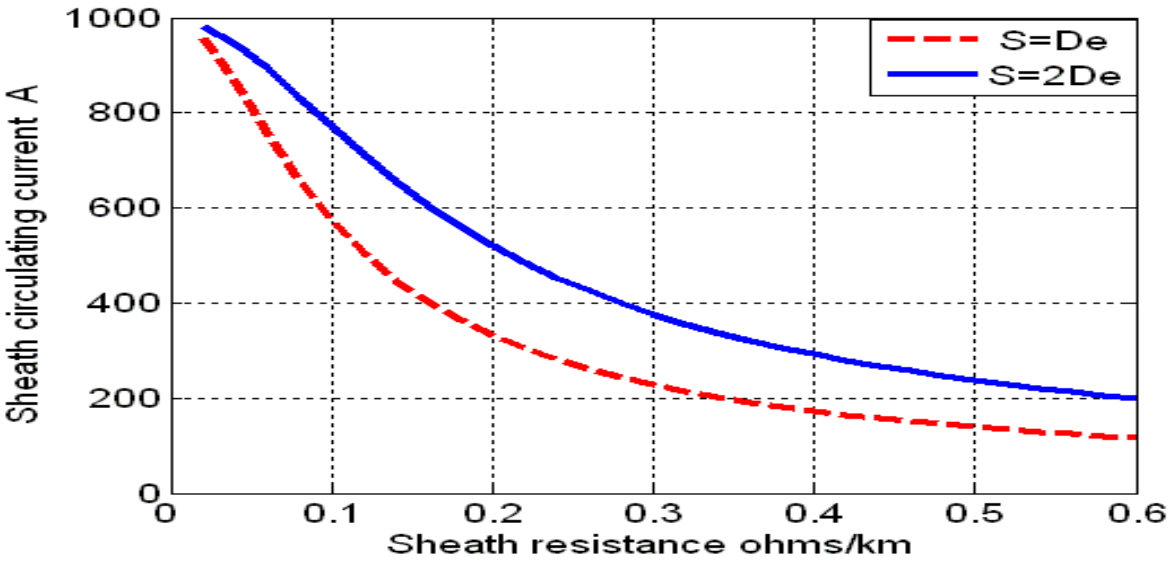


Fig. (4-10): Sheath circulating current vs. sheath resistance in trefoil formation with two-points bonding for D_e and $2D_e$ spacing between cables

Figure (4-11) shows the sheath circulating loss factor with varying A.C sheath resistance of single-core cable in flat formation in case of two-points bonding.

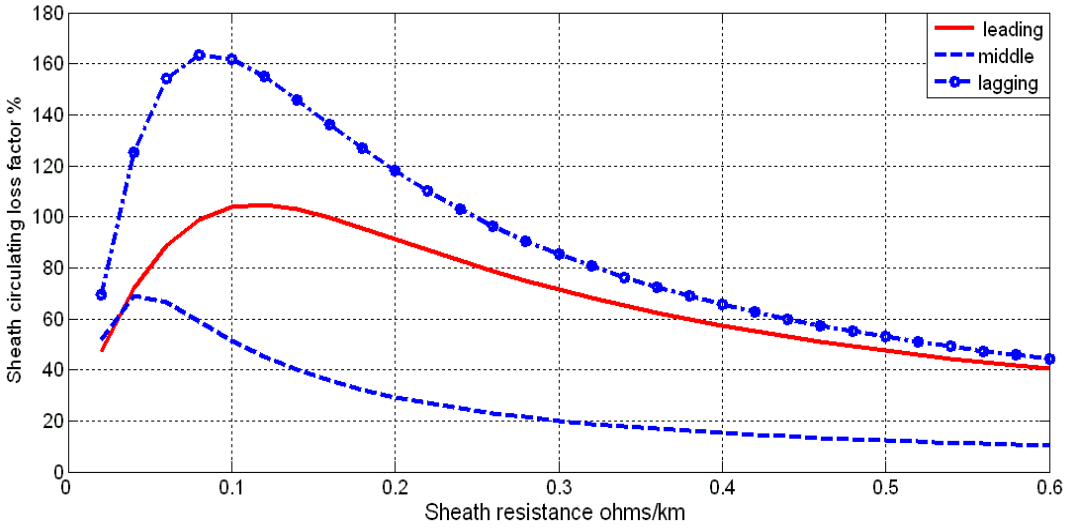


Fig. (4-11): Sheath circulating loss factor vs. sheath resistance in touch flat formation with two-points bonding

4.4.4.3.2 Effect of sheath resistance on the sheath eddy losses

Figure (4-12) shows the sheath eddy loss factor with varying A.C sheath resistance of single-core cable in trefoil formation in case of two-points bonding.

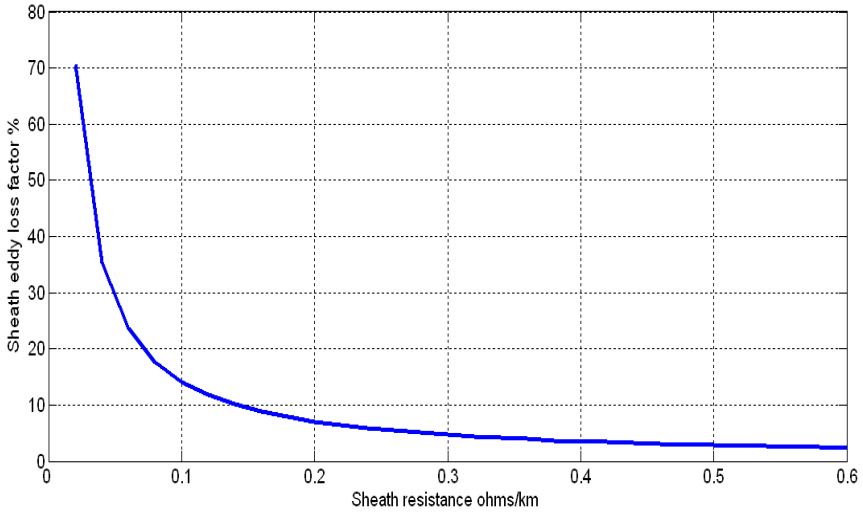


Fig. (4-12): Sheath eddy loss factor vs. sheath resistance for trefoil formation with two-points bonding

Figure (4-13) shows the sheath eddy loss factor with varying A.C sheath resistance of single-core cable in flat formation in case of two-points bonding.

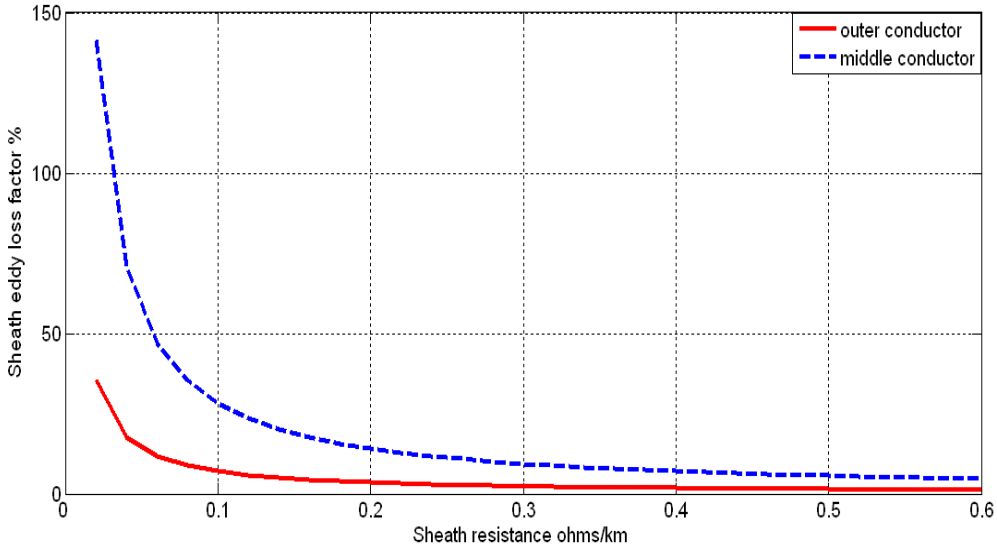


Fig. (4-13): Sheath eddy loss factor vs. sheath resistance for flat formation with two-points bonding

4.4.4.4 Discussion of the obtained results

From Figures (4-9), (4-10) and (4-11) which indicate the effect of sheath resistance on the sheath circulating losses it is noticed that:

- At the maximum sheath current, equal to full conductor current, (i.e., for the case of zero sheath resistance), the circulating-current loss is obviously zero. While the sheath current falls with increasing sheath resistance, i.e. the sheath current is inversely proportional to the sheath resistance, the sheath circulating loss first rises to a maximum, and then falls, again approaching zero at infinite sheath resistance, so the sheath circulating loss would be eliminated when the sheath resistance tends to either zero or infinity.
- The value of sheath resistance which gives maximum-sheath circulating-current loss is called critical sheath resistance, values of sheath resistance higher or lower than this critical value will give lower circulating-current losses than those for the critical sheath resistance, so the cable designer must be aware to avoid this value.
- Attention is also called to the fact, indicated in Fig. (4-9), that the critical sheath resistance for a given cable is diminished when the spacing between phases is reduced.
- The critical value of sheath resistance in flat formation differs from conductor to other in flat formation as shown in Fig. (4-11).

From Figures (4-12) and (4-13) which indicate the effect of sheath resistance on the sheath eddy losses it can be seen that:

- The sheath eddy losses are inversely proportional to the sheath resistance.
- The sheath eddy losses can be neglected at large values of sheath resistances.

4.4.4.5 Factors affecting the sheath resistance

4.4.4.5.1 Introduction

By referring to equation (4-41), it can be noticed that the sheath resistance is affecting by:

1-Sheath material resistivity

2- Temperature of the sheath material

The effect of each factor on the sheath losses is studied by using the mathematical algorithm which is explained in clause 4.3.

4.4.4.5.2 Cases study

1- Single-core cables, made of a stranded copper conductor with 800 mm^2 insulated by XLPE, f = 50 Hz, with various voltages levels and screens (copper wire, copper tape, stainless steel and aluminum) , which their parameters have been listed in tables (4-8), (4-9), (4-10), (4-11) and (4-12), are used to investigate the effect of sheath material resistivities of different metallic screens on the sheath losses.

2- 66 kV single-core cable, made of a stranded copper conductor with 800 mm^2 insulated by XLPE and covered by a lead screen and laid in touch trefoil with sheaths bonded at two-points, f = 50 Hz, which its parameters have been listed in table (4-1), is used to see the effect of temperature of the sheath material on the sheath losses.

Table (4-8): Electrical resistivities and temperature coefficients for different metallic sheaths materials

Metal	Resistivity (Ω m) at 20 ^oC	Temperature coefficient per K at 20°C
Copper	1.7241×10^{-8}	3.93×10^{-3}
Lead and its alloy	21.4×10^{-8}	4×10^{-3}
Stainless steel	70×10^{-8}	negligible
Aluminum	2.84×10^{-8}	4.03×10^{-3}

Table (4- 9): Single-core cable 800 mm^2 CU, with copper tape screen parameters

Cable parameters	Voltage level (kV)					
	11	22	66	132	220	500
Outer diameter of cable (mm)	51	55	64	74	90	112
Mean sheath diameter (mm)	45	49.2	58.2	68.2	84.2	106.2
DC Resistance of the copper conductor (20°C), Ω/km	0.0221	0.0221	0.0221	0.0221	0.0221	0.0221
Diameter of the conductor (mm)	33.7	33.7	33.7	33.7	33.7	33.6
Diameter over insulation (mm)	42.7	46.9	55.9	65.9	81.9	103.9
sheath resistance at 70°C Ω	1.4	1.2	1.1	0.90	0.73	0.58

Ground temperature 20°C

Laying depth 1.0 m

Distance “S” between cable axes laid in flat formation D_e

Ground thermal resistivity 1.0 Km/W

- Assuming the sheath temperature equals to 70°C
- Current rating (A) for copper conductor 995 A
- Thickness of copper tape 2x0.15 mm

Table (4-10): Single-core cable 800 mm^2 CU with copper wire screen parameters

Cable parameters	Voltage level (kV)					
	11	22	66	132	220	500
Cross section of screen (mm^2)	35	35	35	95	185	185
Outer diameter of cable (mm)	53	58	67	79	97	120
Mean sheath diameter (mm)	49	54	63	75	93	116
DC Resistance of the copper conductor (20°C), Ω/km	0.0221	0.0221	0.0221	0.0221	0.0221	0.0221
DC Resistance of the copper sheath (20°C), Ω/km	0.524	0.524	0.524	0.193	0.0991	0.0991
diameter of the conductor (mm)	33.7	33.7	33.7	33.7	33.7	33.6
sheath resistance at 70°C Ω/km	1.99	1.9	1.8	0.65	0.47	0.33

Ground temperature 20°C

Laying depth 1.0 m

Distance "S" between cable axes laid in flat formation D_e

Ground thermal resistivity 1.0 Km/W

- Assuming the sheath temperature equals to 70°C
- Current rating (A) for copper conductor 995 A

Table (4-11): Single-core cable 800 mm^2 CU with stainless steel screen parameters

Cable parameters	Voltage level (kV)					
	11	22	66	132	220	500
Outer diameter of cable (mm)	53	58	80.2	93.2	108.2	136.2

Mean sheath diameter (mm)	46	50.4	62.7	74.6	89.2	115.4
DC Resistance of the copper conductor (20°C), Ω/km	0.0221	0.0221	0.0221	0.0221	0.0221	0.0221
diameter of the conductor (mm)	34	34	34	34	34	34
sheath resistance at (70°C) Ω/km	3.7	2.9	1.3	1	0.69	0.44

Ground temperature 20°C

Laying depth 1.0 m

Distance "S" between cable axes laid in flat formation D_e

Ground thermal resistivity 1.0 Km/W

- Assuming the sheath temperature equals to 70°C
- Current rating (A) for copper conductor 995 A

Table (4-12): Single-core cable 800 mm^2 CU with aluminum screen parameters

Cable parameters	Voltage level (kV)					
	11	22	66	132	220	500
Outer diameter of cable (mm)	53	58	87	101	116	148
Mean sheath diameter (mm)	46	50.4	61.9	73.8	88	114
DC Resistance of the copper conductor (20°C), Ω/km	0.0221	0.0221	0.0221	0.0221	0.0221	0.0221
diameter of the conductor (mm)	34	34	34	34	34	34
sheath resistance at 70°C Ω/km	0.18	0.14	0.092	0.067	0.051	0.032

Ground temperature 20°C

Laying depth 1.0 m

Distance “S” between cable axes laid in flat formation D_e

Ground thermal resistivity 1.0 Km/W

- Assuming the sheath temperature equals to 70°C
- Current rating (A) for copper conductor 995 A

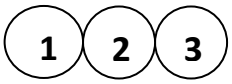
4.4.4.5.3 Obtained results

4.4.4.5.3.1 Results of the effect of sheath material resistivity on the sheath losses

Tables (4-13-a), (4-13-b), (4-14-a), (4-14-b), (4-15-a) (4-15-b), (4-16-a), (4-16-b) show the values of sheath currents and their loss factors for single-core cables with various voltage levels and metallic sheaths materials (copper wire, copper tape, stainless steel and aluminum) to investigate the effect of sheath material resistivities on the sheath losses in case of two-points bonding for touch trefoil and touch flat and also in case of cross bonding touch trefoil and touch flat.

Table (4-13-a) shows the values of sheath currents and their loss factors for single-core cables with various voltage levels and their metallic sheaths materials made of copper tape in case of two-points bonding for touch trefoil and touch flat.

Table (4-13-a): Sheath currents and their loss factors in case of two-points bonding method for copper tape screens

Voltage levels	**Sheath bonding arrangement**								
	Two-points bonding-touch trefoil				**Two-points bonding-touch flat** 				**Phase no.**
	λ_{CS}	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	λ_{CS} %	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	
11 kV	5.74	37.5 A	1.25	17.5 A	15.10	60.8 A	0.63	12.3 A	1
	5.74	37.5 A	1.25	17.5 A	2.96	26.9 A	2.51	24.7 A	2

	5.74	37.5 A	1.25	17.5 A	15.66	61.9 A	0.63	12.3 A	3
22 kV	6.14	40.3 A	1.43	19.4 A	16.35	65.7 A	0.71	13.7 A	1
	6.14	40.3 A	1.43	19.4 A	3.12	28.7 A	2.85	27.4 A	2
	6.14	40.3 A	1.43	19.4 A	17.01	67 A	0.71	13.7 A	3
66 kV	7.11	46.6 A	1.78	23.3 A	19.17	76.6 A	0.89	16.5 A	1
	7.11	46.6 A	1.78	23.3 A	3.56	33 A	3.56	33 A	2
	7.11	46.6 A	1.78	23.3 A	20.06	78.4 A	0.89	16.5 A	3
132 kV	8.16	53.7 A	2.17	27.7 A	22.20	88.7 A	1.09	19.6 A	1
	8.16	53.7 A	2.17	27.7 A	4.02	37.7 A	4.34	39.2 A	2
	8.16	53.7 A	2.17	27.7 A	23.41	91 A	1.09	19.6 A	3
220 kV	9.80	65 A	2.80	34.7 A	26.89	107.7 A	1.40	24.5 A	1
	9.80	65 A	2.80	34.7 A	4.76	45.3 A	5.6	49.1 A	2
	9.80	65 A	2.80	34.7 A	28.67	111.2 A	1.40	24.5 A	3
500 kV	12.03	80.5 A	3.66	44.4 A	33.04	133.4 A	1.83	31.4 A	1
	12.03	80.5	3.66	44.4	5.75	55.6	7.33	62.8	2

		A		A		A		A	
	12.03	80.5 A	3.66	44.4 A	35.77	138.8 A	1.83	31.4 A	3

Where: λ_{CS} : The circulating sheath loss factor percentage of conductor loss

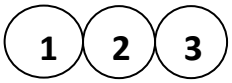
I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

Table (4-13-b) shows the values of sheath currents and their loss factors for single-core cables with various voltage levels and their metallic sheaths materials made of copper tape in case of cross bonding for touch trefoil and touch flat.

Table (4-13-b): Sheath currents and their loss factors in case of cross-bonding methods for copper tape screens

Voltage levels	**Sheath bonding arrangement**				
	Cross bonding-touch trefoil		**Cross bonding-touch flat** 		**Phase no.**
	λ_{SE} %	I_{SE}(A)	λ_{SE} %	I_{SE}(A)	
11 kV	1.26	17.5 A	0.63	12.4 A	1
	1.26	17.5 A	2.51	24.7 A	2
	1.26	17.5 A	0.63	12.4 A	3
22 kV	1.43	19.4 A	0.71	13.7 A	1
	1.43	19.4 A	2.85	27.4 A	2
	1.43	19.4 A	0.71	13.7 A	3

66 kV	1.78	23.3 A	0.89	16.5 A	1
	1.78	23.3 A	3.56	33 A	2
	1.78	23.3 A	0.89	16.5 A	3
132 kV	2.17	27.7 A	1.09	19.6 A	1
	2.17	27.7 A	4.34	39.2 A	2
	2.17	27.7 A	1.09	19.6 A	3
220 kV	2.80	34.7 A	1.40	24.5 A	1
	2.80	34.7 A	5.59	49.1 A	2
	2.80	34.7 A	1.40	24.5 A	3
500 kV	3.66	44.3 A	1.83	31.3 A	1
	3.66	44.3 A	7.31	62.7 A	2
	3.66	44.3 A	1.83	31.3 A	3

Where:

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

Table (4-14-a) shows the values of sheath currents and their loss factors for single-core cables with various voltage levels and their metallic sheaths materials made of copper wire in case of two-points bonding for touch trefoil and touch flat.

Table (4-14-a): Sheath currents and their loss factors in case of two-points bonding method for copper wire screens

Voltage levels	**Sheath bonding arrangement**								
	Two-points bonding-touch trefoil				**Two-points bonding-touch flat** (1 2 3)				**Phase no.**
	λ_{CS}	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	λ_{CS} %	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	
11 kV	3.99	27.4 A	1.07	14.2 A	11.10	45.7 A	0.54	10 A	1
	3.99	27.4 A	1.07	14.2 A	1.96	19.2 A	2.15	20.1 A	2
	3.99	27.4 A	1.07	14.2 A	11.40	46.3 A	0.54	10 A	3
22 kV	4	27.5 A	1.10	14.3 A	11.13	45.6 A	0.55	10.1 A	1
	4	27.5 A	1.10	14.3 A	1.97	19.3 A	2.21	20.2 A	2
	4	27.5 A	1.10	14.3 A	11.44	46.4 A	0.55	10.1 A	3
66 kV	4.08	27.6 A	1.15	14.4 A	11.16	45.7 A	0.57	10.2 A	1
	4.08	27.6 A	1.15	14.4 A	2	19.4 A	2.29	20.4 A	2
	4.08	27.6 A	1.15	14.4 A	11.46	46.8 A	0.57	10.2 A	3
132 kV	10.52	71.7 A	3.22	39.7 A	29.16	119.5 A	1.61	28 A	1
	10.52	71.7	3.22	39.7	5.02	49.5	6.44	56.1	2

		A		A		A		A	
	10.52	71.7 A	3.22	39.7 A	31.30	123.8 A	1.61	28 A	3
220 kV	14.36	98.3 A	4.62	55.7 A	39.26	162.5 A	2.31	39.4 A	1
	14.36	98.3 A	4.62	55.7 A	6.79	67.6 A	9.24	78.8 A	2
	14.36	98.3 A	4.62	55.7 A	43.27	170.6 A	2.31	39.4 A	3
500 kV	19.65	135.5 A	6.54	78.7 A	52.01	220.4 A	3.32	55.6 A	1
	19.65	135.5 A	6.54	78.7 A	9.24	92.9 A	13.27	111.3 A	2
	19.65	135.5 A	6.54	78.7 A	59.48	235.7 A	3.32	55.6 A	3

Where:

λ_{CS} : The circulating sheath loss factor percentage of conductor loss

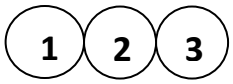
I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

Table (4-14-b) shows the values of sheath currents and their loss factors for single-core cables with various voltage levels and their metallic sheaths materials made of copper wire in case of cross bonding for touch trefoil and touch flat.

Table (4- 14-b): Sheath currents and their loss factors in case of cross-bonding method for copper wire screens

Voltage levels	**Sheath bonding arrangement**				
	Cross bonding-touch trefoil		**Cross bonding-touch flat** 		**Phase no.**
	λ_{SE} %	I_{SE}(A)	λ_{SE} %	I_{SE}(A)	
11 kV	1.07	14.2 A	0.54	10 A	1
	1.07	14.2 A	2.15	20.1 A	2
	1.07	14.2 A	0.54	10 A	3
22 kV	1.10	14.3 A	0.55	10.1 A	1
	1.10	14.3 A	2.21	20.2 A	2
	1.10	14.3 A	0.55	10.1 A	3
66 kV	1.15	14.4 A	0.57	10.2 A	1
	1.15	14.4 A	2.29	20.4 A	2
	1.15	14.4 A	0.57	10.2 A	3
132 kV	3.21	39.6 A	1.61	28 A	1
	3.21	39.6 A	6.42	56 A	2
	3.21	39.6 A	1.61	28 A	3
220 kV	4.60	55.6 A	2.30	39.3 A	1
	4.60	55.6 A	9.20	78.7 A	2
	4.60	55.6 A	2.30	39.3 A	3
500 kV	6.58	78.4 A	3.29	55.4 A	1

	6.58	78.4 A	13.15	110.8A	2
	6.58	78.4 A	3.29	55.4 A	3

Where:

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

Table (4-15-a) shows the values of sheath currents and their loss factors for single-core cables with various voltage levels and their metallic sheaths materials made of stainless steel in case of two-points bonding for touch trefoil and touch flat.

Table (4-15-a): Sheath currents and their loss factors in case of two-points bonding method for stainless steel screens

Voltage levels	**Sheath bonding arrangement**								
	Two-points bonding-touch trefoil				**Two-points bonding-touch flat** (1 2 3)				**Phase no.**
	λ_{CS}	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	λ_{CS} %	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	
11 kV	2.20	14 A	0.45	6.3 A	5.78	22.6 A	0.22	4.4 A	1
	2.20	14 A	0.45	6.3 A	1.15	10.1 A	0.89	8.9 A	2
	2.20	14 A	0.45	6.3 A	5.86	22.8 A	0.22	4.4 A	3
22 kV	2.81	17.6 A	0.57	7.9 A	7.38	28.6 A	0.29	5.6 A	1
	2.81	17.6 A	0.57	7.9 A	1.47	12.7 A	1.15	11.2 A	2

	2.81	17.6 A	0.57	7.9 A	7.51	28.8 A	0.29	5.6 A	3
66 kV	8.25	44.5 A	1.07	16 A	19.37	68.3 A	0.54	11.3 A	1
	8.25	44.5 A	1.07	16 A	4.69	33.6 A	2.15	22.7 A	2
	8.25	44.5 A	1.07	16 A	20.19	69.7 A	0.54	11.3 A	3
132 kV	10.45	57.4 A	1.50	21.7 A	24.80	88.4 A	0.75	15.3 A	1
	10.45	57.4 A	1.50	21.7 A	5.85	42.9 A	3	30.7 A	2
	10.45	57.4 A	1.50	21.7 A	26.17	90.8 A	0.75	15.3 A	3
220 kV	14.10	79.5 A	2.30	32.1 A	33.72	123.1 A	1.15	22.7 A	1
	14.10	79.5 A	2.30	32.1 A	7.73	58.9 A	4.60	45.4 A	2
	14.10	79.5 A	2.30	32.1 A	36.37	127.8 A	1.15	22.7 A	3
500 kV	20.91	121.4 A	3.87	52.2 A	49.31	186.5 A	1.94	36.9 A	1
	20.91	121.4 A	3.87	52.2 A	11.25	89 A	7.75	73.9 A	2
	20.91	121.4 A	3.87	52.2 A	55.38	197.6 A	1.94	36.9 A	3

Where:

λ_{CS} : The circulating sheath loss factor percentage of conductor loss

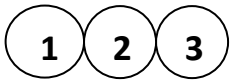
I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

Table (4-15-b) shows the values of sheath currents and their loss factors for single-core cables with various voltage levels and their metallic sheaths materials made of stainless steel in case of cross bonding for touch trefoil and touch flat.

Table (4-15-b): Sheath currents and their loss factors in case of cross-bonding method for stainless steel screens

Voltage levels	**Sheath bonding arrangement**				
	Cross bonding-touch trefoil		**Cross bonding-touch flat** 		**Phase no.**
	λ_{SE} %	I_{SE}(A)	λ_{SE} %	I_{SE}(A)	
11 kV	0.45	6.3 A	0.22	4.4 A	1
	0.45	6.3 A	0.89	8.9 A	2
	0.45	6.3 A	0.22	4.4 A	3
22 kV	0.57	7.9 A	0.29	5.6 A	1
	0.57	7.9 A	1.15	11.2 A	2
	0.57	7.9 A	0.29	5.6 A	3
66 kV	1.08	16.1 A	0.54	11.4 A	1
	1.08	16.1 A	2.15	22.7 A	2
	1.08	16.1 A	0.54	11.4 A	3
132 kV	1.51	21.8 A	0.76	15.4 A	1
	1.51	21.8 A	3.01	30.8 A	2

	1.51	21.8 A	0.76	15.4 A	3
220 kV	2.31	32.2 A	1.16	22.8 A	1
	2.31	32.2 A	4.61	45.5 A	2
	2.31	32.2 A	1.16	22.8 A	3
500 kV	3.88	52.3 A	1.95	37 A	1
	3.88	52.3 A	7.75	73.9 A	2
	3.88	52.3 A	1.95	37 A	3

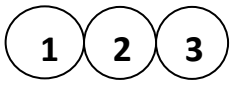
Where:

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

Table (4-16-a) shows the values of sheath currents and their loss factors for single-core cables with various voltage levels and their metallic sheaths materials made of aluminum in case of two-points bonding for touch trefoil and touch flat.

Table (4-16-a): Sheath currents and their loss factors in case of two-points bonding method for aluminum screens

Voltage levels	**Sheath bonding arrangement**								
	Two-points bonding-touch trefoil				**Two-points bonding-touch flat** 				**Phase no.**
	λ_{CS}	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	λ_{CS} %	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	
11 kV	41.73	276 A	9.16	129.3 A	82.03	387 A	4.58	91.4 A	1
	41.73	276	9.16	129.3	22.66	203.4	18.32	182.9	2

		A		A		A		A	
	41.73	276 A	9.16	129.3 A	106.85	441.7 A	4.58	91.4 A	3
22 kV	50.98	340.7 A	11.77	163.7 A	90.24	453.4 A	5.88	115.7 A	1
	50.98	340.7 A	11.77	163.7 A	28.22	253.5 A	23.54	231.5 A	2
	50.98	340.7 A	11.77	163.7 A	124.86	533.3 A	5.88	115.7 A	3
66 kV	95.94	572.3 A	12.74	208.5 A	109.97	612.7 A	6.37	147.4 A	1
	95.94	572.3 A	12.74	208.5 A	66.59	476.8 A	25.48	294.9 A	2
	95.94	572.3 A	12.74	208.5 A	175.53	774.1 A	6.37	147.4 A	3
132 kV	99.97	683.6 A	18.70	295.6 A	98.34	678.1 A	9.35	209 A	1
	99.97	683.6 A	18.70	295.6 A	73.41	585.9 A	37.40	418.1 A	2
	99.97	683.6 A	18.70	295.6 A	166.50	882.3 A	9.35	209 A	3
220 kV	95.56	760.2 A	26.36	399.3 A	85.63	719.6 A	13.18	282.3 A	1
	95.56	760.2 A	26.36	399.3 A	73.44	666.4 A	52.73	564.7 A	2
	95.56	760.2 A	26.36	399.3 A	148.72	948.4 A	13.18	282.3 A	3
500 kV	79.46	879.2 A	44.31	656.5 A	65.63	799 A	22.16	464.2 A	1

	79.46	879.2 A	44.31	656.5 A	68.36	815.5 A	88.62	928.5 A	2
	79.46	879.2 A	44.31	656.5 A	108.07	1025.3	22.16	464.2 A	3

Where:

λ_{CS} : The circulating sheath loss factor percentage of conductor loss

I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

Table (4-16-b) shows the values of sheath currents and their loss factors for single-core cables with various voltage levels and their metallic sheaths materials made of aluminum in case of cross bonding for touch trefoil and touch flat.

Table (4-16-b): Sheath currents and their loss factors in case of cross-bonding method for aluminum screens

Voltage levels	**Sheath bonding arrangement**				
	Cross bonding-touch trefoil		**Cross bonding-touch flat** (1 2 3)		**Phase no.**
	λ_{SE} %	I_{SE}(A)	λ_{SE} %	I_{SE}(A)	
11 kV	9.25	129.9A	5.58	100.9A	1
	9.25	129.9A	17.82	180.4A	2
	9.25	129.9A	3.89	84.3 A	3
22 kV	11.70	163.2A	7.29	128.9A	1
	11.70	163.2A	22.54	226.5A	2

	11.70	163.2A	4.84	105 A	3
66 kV	11.75	200.2A	7.23	157.1A	1
	11.75	200.2A	23.05	280.5A	2
	11.75	200.2A	5	130.6A	3
132 kV	16.23	275.4A	10.57	222.2A	1
	16.23	275.4A	31.86	385.9A	2
	16.23	275.4A	6.80	178.3A	3
220 kV	20.68	353.6A	14.22	293.2A	1
	20.68	353.6A	40.75	496.4A	2
	20.68	353.6A	8.49	226.5A	3
500 kV	25.80	501 A	18.83	427.9A	1
	25.80	501 A	51.06	704.7A	2
	25.80	501 A	10.62	321.4A	3

Where:

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

4.4.4.5.3.2 Obtained results of the effect of temperature of the sheath material on the sheath losses

Figure (4-14) shows values of A.C resistance of lead sheath with varying its temperature for 66 kV single-core cable.

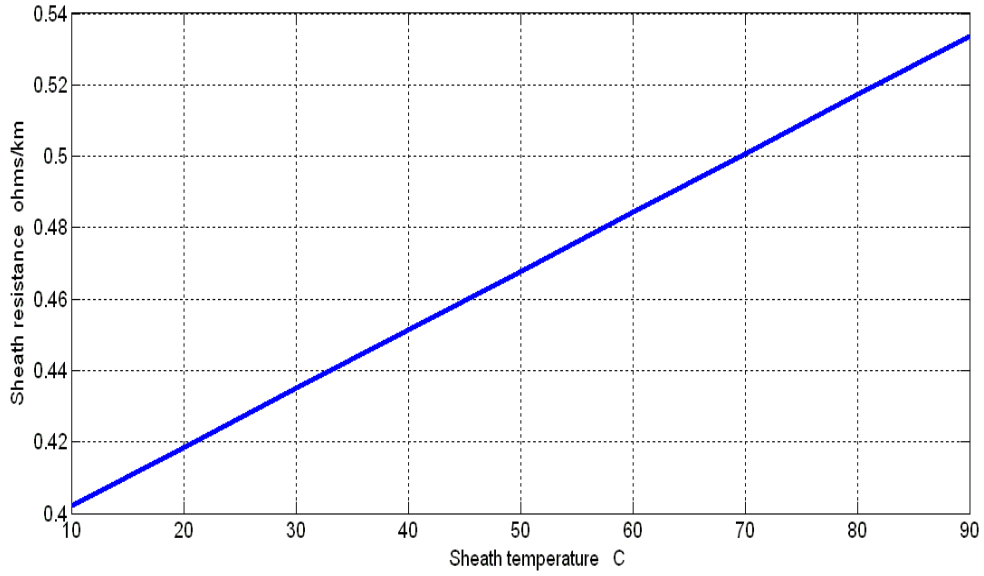


Fig. (4-14): Sheath resistance vs. sheath temperature

Figure (4-15) shows the sheath circulating loss factor with varying A.C resistance of lead sheath for 66 kV single-core cable in touch trefoil with its sheaths bonded at two-points.

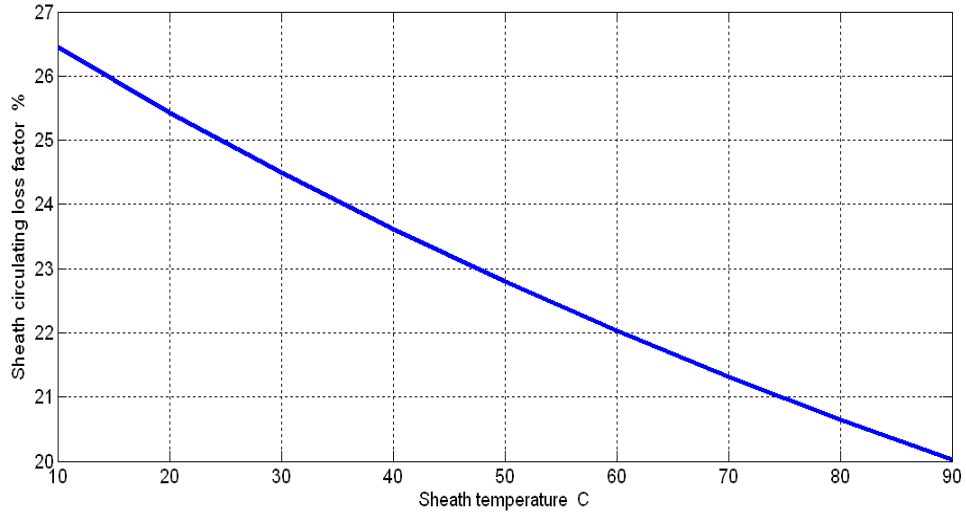


Fig. (4-15): Sheath loss factor vs. sheath temperature

4.4.4.5.4 Discussion of the obtained results

4.4.4.5.4.1 Results discussion of the effect of sheath material resistivity on the sheath losses

From the above calculations according to IEC 60287 are given in tables

(4-2-b), (4-2-c), (4-13-a), (4-13-b), (4-14-a), (4-14-b), (4-15-a), (4-15-b), (4-16-a) and (4-16-b) it is noticed that:

- The sheath material plays a great role in controlling the values of sheath losses, as, for example, the total sheath losses can be reduced to more than 92 % by replacing the sheath material from aluminum to copper tape (in case of 500 kV, two-points bonding, flat formation lagging phase (phase no. 3)).
- Single-core cables covered by copper wire screen, copper tape or stainless steel introduce a best solution to reduce the sheath losses and overcoming the problems of lead sheath especially at higher voltages.
- Eddy losses could be neglected with respect to circulating losses except in aluminum sheath as the eddy losses could be greater than the circulating losses as shown in table (4-16-a) for 500 kV in touch flat with two-points bonding, where λ_{SE2} = 88.62 % and λ_{CS2} = 68.36 % of copper losses.
- From calculations given in tables (4-16-a,b) in case of aluminum screen, it can be noticed that:

1-Single-core cable with aluminum sheath introduces higher sheath losses and currents due to its low resistivity as the total sheath current could be reached to more than 149 % of line current (for 500 kV single-core cable in touch flat with two-points bonding, where $I_{SE3} + I_{CS3}$ = 149.7 % of rated current), so IEEE std 536 recommended to overcome this problem by selecting thinner sheaths and by using special sheath bonding methods (single-point bonding or cross bonding).

2-Single-core cable covered by aluminum sheath introduces irregular behavior towards the values of sheath circulating loss factors in extra high voltages as they are reducing with increasing the system voltage levels, this is due to reducing the sheath resistance to values lower than the critical sheath resistance as explained in clause 3.4.4. This may be become clearly by discussing the sheath circulating loss factors in trefoil formation with sheaths bonded at two-points with referring to Fig. (4-16) where R_S refers to the value of sheath resistance corresponding to each cable of system voltages 66,

132, 220 is corresponding to single-core cable of 500 kV system voltage followed by 220, 66 and 132 kV as shown in table (4-17).

3- By reducing the values of sheath resistance of each cable to values lower than R_S, for example 0.75 R_S , it can be seen clearly that the sheath circulating loss factor is inversely proportional to system voltage level.

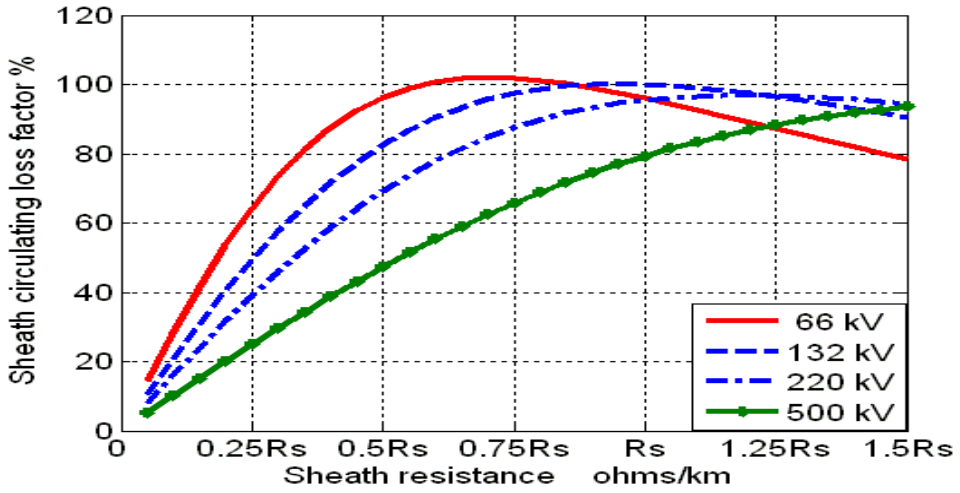


Fig. (4-16): Sheath resistance vs. sheath circulating loss factor with aluminum screen

4.4.4.5.4.2 Results discussion of the effect of sheath temperature on the sheath losses

From figures (4-14) and (4-15) it is noticed that:

With increasing the temperature of the sheath material, the sheath losses reduce due to increasing the sheath resistance.

4.4.5 Effect of phase rotation on the sheath circulating loss factor for two-points bonding – flat arrangements

4.4.5.1 Introduction

The above calculations are carried out on flat arrangement with phase rotation shown in Fig.(4-1(b)), to examine the effect of phase rotation on sheath circulating loss factor for two-points bonding, there are another two configurations must be taken into considerations which are shown in Fig.(4-17).



(a) S-T-R configuration (b) S-R-T configuration

Fig.(4-17) Phase rotation in flat formation

The previous mathematical algorithm, which is explained in 4.3.1.2, is used but with assuming the phase rotation for S-T-R configuration

$$I_2 = I_1\left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right)$$
$$I_3 = I_1\left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)$$

And the phase rotation for S-R-T configuration

$$I_2 = I_1\left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)$$
$$I_3 = I_1\left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right)$$

4.4.5.2 Cases study

66 kV single-core cable, made of a stranded copper conductor with 800 mm^2 insulated by XLPE and covered by a lead screen, f = 50 Hz, which its parameters have been listed in table (4-1), is used as case study.

4.4.5.3 Obtained results by using IEC 60287

The results are shown in table (4-17). In this table the sheath circulating loss factor in each phase of single-core cable in flat formation is calculated with corresponding to three different phase rotation arrangements of the cable.

Table (4-17): Sheath circulating loss factors for different configuration in flat formation

SHEATH CIRCULATING LOSS FACTOR (%)	CABLE CONFIGURATION		
	R S T	S R T	S T R
λ_{CS-R}	47.38	12.18	52.79
λ_{CS-S}	12.18	52.79	47.38
λ_{CS-T}	52.79	47.38	12.18

Where

λ_{CS-R}, λ_{CS-S}, λ_{CS-T} : The sheath circulating loss factors in R, S and T phases respectively.

4.4.5.4 Discussion of the obtained results

From table (4-17), it is noticed that:

- Always the central conductor has the lowest sheath circulating loss value, due to magnetic cancellation.
- The sheath circulating losses of the outer conductors are depending mainly on the phase rotation and its arrangement.

4.4.6 Effect of conductor current on the sheath losses

4.4.6.1 Introduction

The previous mathematical algorithm, which is explained in clause 4.2, has been used to investigate the effect of variations of conductor current on the sheath losses by calculating sheath losses for full and half values of ampacity.

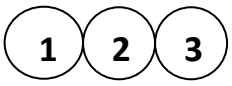
4.4.6.2 Cases study

66 kV single-core cable, made of a stranded copper conductor with 800 mm^2 insulated by XLPE and covered by a lead screen, f = 50 Hz, which its parameters have been listed in table (4-1), is used in this case study.

4.4.6.3 Obtained results by using IEC 60287

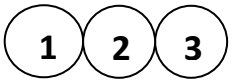
The results are shown in tables (4-18-a) and (4-18-b). In these tables sheath currents and their losses are calculated with changing the conductor current from full rating value to its half in case of two-points bonding (touch trefoil & touch flat) and cross bonding (touch trefoil & touch flat) respectively.

Table (4-18-a): Sheath currents and their loss factors for single-core cables with full rating current and its half value for two-points bonding method

Current	**Sheath bonding arrangement**								
	Two-points bonding-touch trefoil				**Two-points bonding-touch flat** 				**Phase no.**
	λ_{CS}	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	λ_{CS} %	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	
Full rating	21.32	116 A	2.82	42.1 A	47.38	172.9 A	1.41	29.8 A	1
	21.32	116 A	2.82	42.1 A	12.18	87.7A	5.64	59.6 A	2
	21.32	116 A	2.82	42.1 A	52.79	182.5 A	1.41	29.8 A	3
Half rating	21.32	58 A	2.82	21 A	47.38	86.4 A	1.41	14.9 A	1
	21.32	58 A	2.82	21 A	12.18	43.8 A	5.64	29.8 A	2

	21.32	58 A	2.82	21 A	52.79	91.2 A	1.41	14.9 A	3

Table (4-18-b): Sheath currents and their loss factor for single-core cables with full rating current and its half value for cross bonding method

current	**Sheath bonding arrangement**				
	Cross bonding-touch trefoil		**Cross bonding-touch flat** 		**Phase no.**
	λ_{SE} %	I_{SE}(A)	λ_{SE} %	I_{SE}(A)	
Full rating	2.81	42 A	1.4	29.7 A	1
	2.81	42 A	5.61	59.5 A	2
	2.81	42 A	1.4	29.7 A	3
Half rating	2.81	21 A	1.4	14.8 A	1
	2.81	21 A	5.61	29.7 A	2
	2.81	21 A	1.4	14.8 A	3

Where:

λ_{CS} : The circulating sheath loss factor percentage of conductor loss

I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

4.4.6.4 Discussion of the obtained results

From tables (4-18-a) and (4-18-b), it is noticed that:

- The sheath currents (eddy and circulating) duplicate with duplicating the conductor current.
- The sheath losses factors (eddy and circulating) did not changed because the ratio of sheath current and conductor current is fixed.

4.4.7 Effect of power frequency (50 or 60 Hz) on the sheath losses

4.4.7.1 Introduction

Power frequency in Egypt is 50 Hz, but in some other countries is 60 Hz, this difference may be due to economical and other factors which are not suitable to be mentioned here. The previous mathematical algorithm, which is explained in clause 3.2, is used to study the effect of power frequencies on the sheath losses by calculating the sheath losses for f = 50 and 60 Hz.

4.4.7.2 Cases study

66 kV single-core cable, made of a stranded copper conductor with 800 mm^2 insulated by XLPE and covered by a lead screen, 50 Hz, which its parameters have been listed in table (4-1), is used as case study.

4.4.7.3 Obtained results

The results are shown in tables (4-19-a) and (4-19-b). In these tables sheath currents and their losses are calculated for f = 50 and 60 Hz in case of two-points (touch trefoil & touch flat) and cross bonding (touch trefoil & touch flat) respectively.

Table (4-19-a): Sheath currents and their loss factors for single-core cables with two-points bonding method with power frequencies 50 and 60 Hz

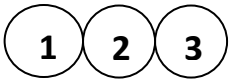
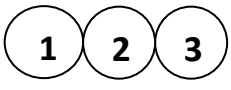
Frequency	**Sheath bonding arrangement**								
	Two-points bonding-touch trefoil				**Two-points bonding-touch flat** 				**Phase no.**
	λ_{CS}	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	λ_{CS} %	I_{CS}(A)	λ_{SE} %	I_{SE}(A)	
50 Hz	21.32	116 A	2.82	42.1 A	47.38	172.9 A	1.41	29.8 A	1
	21.32	116 A	2.82	42.1 A	12.18	87.7A	5.64	59.6 A	2
	21.32	116 A	2.82	42.1 A	52.79	182.5 A	1.41	29.8 A	3
60 Hz	29.22	138.8 A	3.89	50.6 A	63.55	204.6 A	1.94	35.7 A	1
	29.22	138.8 A	3.89	50.6 A	16.75	105 A	7.77	71.5 A	2
	29.22	138.8 A	3.89	50.6 A	72.31	218.3 A	1.94	35.7 A	3

Table (4-19-b): Sheath currents and their loss factors for single-core cables with cross bonding method with power frequencies 50 and 60 Hz

Frequ-ency	**Sheath bonding arrangement**		
	Cross bonding-touch trefoil	**Cross bonding-touch flat** 	**Phase no.**

	λ_{SE} %	I_{SE}(A)	λ_{SE} %	I_{SE}(A)	
50 Hz	2.81	42 A	1.4	29.7 A	1
	2.81	42 A	5.61	59.5 A	2
	2.81	42 A	1.4	29.7 A	3
60 Hz	3.86	50.4 A	1.93	35.6 A	1
	3.86	50.4 A	7.73	71.3 A	2
	3.86	50.4 A	1.93	35.6 A	3

Where:

λ_{CS} : The circulating sheath loss factor percentage of conductor loss

I_{CS} : The circulating current in the sheath

λ_{SE} : The sheath eddy loss factor percentage of conductor loss

I_{SE} : The eddy current in the sheath

4.4.7.4 Discussion of the obtained results

From tables (4-19-a) and (4-19-b) it is noticed that:

- Both sheath eddy losses and sheath circulating losses increase with increasing power frequency.
- The two-points bonding for flat formation has more sensitivity to the changing of power frequency than other type of bonding arrangement.

4.4.8 Effect of the minor section length on the sheath circulating current in cross-bonding arrangement

4.4.8.1 Introduction

When the cables in each minor section have the same length, it is said the cables are balanced and the length imbalance rate is zero [30]. Supposing three single-core cables with the sheath of each single-core cable consists of

three minor sections and cross bonded as shown in Fig. (4-18) and the lengths of the second and third minor section equal 300 meters. With changing the length of the first minor section between 200 and 400 meters and calculating the sheath circulating current to study the effect of minor section length variation on the sheath circulating currents with using the following mathematical algorithm [10] which depends on clause 4.3.From Fig. (4-18), it can be deduced that:

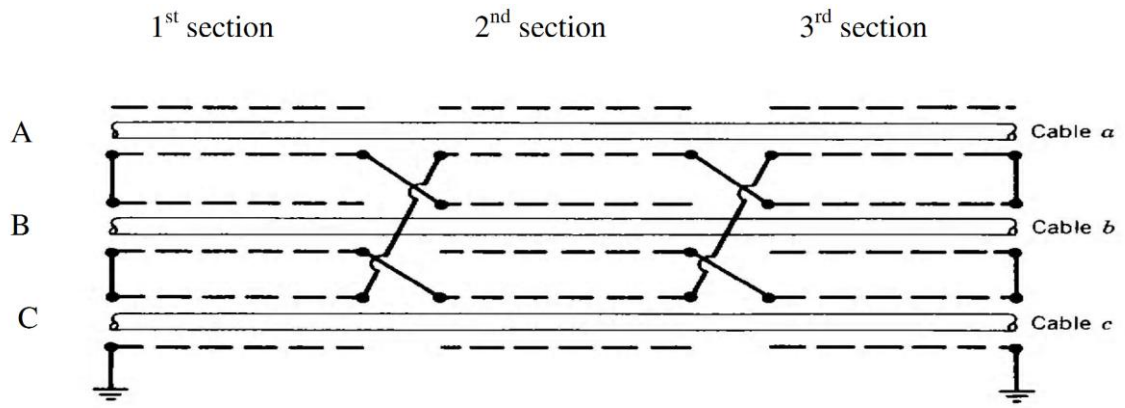


Fig.(4-18): Cross-bonded cables without transposition

Let

$I_A = I$, $I_B = h^2 I$, $I_C = hI$

Where

h is an operator which rotates a phasor 120° counter clock-wise

$$h = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$$

$$h^2 = -\frac{1}{2} - j\frac{\sqrt{3}}{2}$$

Then

Namely

Circuit X: Consisting of A in section 1, B in section 2 and C in section 3;

Circuit Y: Consisting of B in section 1, C in section 2 and A in section 3;

Circuit Z: Consisting of C in section 1, A in section 2 and B in section 3.

The induced sheath voltages of X, Y and Z circuits are given as follows:

$$V_X = jI\left(\ell_1 X_A + h^2 \ell_2 X_B + h\ell_3 X_C\right)$$
$$V_Y = jI\left(h^2 \ell_1 X_B + h\ell_2 X_C + \ell_3 X_A\right) \quad [10]\ (4\text{-}46)$$
$$V_Z = jI\left(h\ell_1 X_C + \ell_2 X_A + h^2 \ell_3 X_B\right)$$

Where ℓ_i is the length of section number i, i=1, 2, 3.

The sheath current in each sheath circuit can be calculated by the following equations:

$$I_{CSX} = \frac{V_X}{Z_X}\ ,\ I_{CSY} = \frac{V_Y}{Z_Y}\ ,\ I_{CSZ} = \frac{V_Z}{Z_Z} \qquad (47\text{-}4)$$

Where I_{CSX}, I_{CSY} and I_{CSZ} are the sheath circulating currents in sheath circuits X, Y and Z respectively and the sheath impedances of the X, Y and Z circuits respectively are:

$$\left.\begin{aligned} Z_X &= \ell_1(R_S + jX_A) + \ell_2(R_S + jX_B) + \ell_3(R_S + jX_C) \\ Z_Y &= \ell_1(R_S + jX_B) + \ell_2(R_S + jX_C) + \ell_3(R_S + jX_A) \\ Z_Z &= \ell_1(R_S + jX_C) + \ell_2(R_S + jX_A) + \ell_3(R_S + jX_B) \end{aligned}\right\} (4\text{-}48)$$

In trefoil formation:

$$X_A = X_B = X_C = X = 2\omega 10^{-7} \ln\left(\frac{S}{r_{sh}}\right) \qquad (4\text{-}49)$$

So $Z_X = Z_Y = Z_Z$, $I_{CSX} = I_{CSY} = I_{CSZ} = I_{CS}$ (4-50)

So for $\ell_2 = \ell_3 = 300$ m and varying ℓ_1 with calculating the sheath circulating current.

So from equations (4-49) and (4-50) by substituting in equations (4-46), (4-47) and (4-48)

$$Z_X = Z_Y = Z_Z = Z = \ell_1(R_S + jX) + 300(R_S + jX) + 300(R_S + jX) \quad (4\text{-}51)$$

$$V_X = V_Y = V_Z = V_S = jI(\ell_1 X + 300h^2 X + 300hX) \quad (4\text{-}52)$$

$$I_{CS} = \frac{V_S}{Z} \quad (4\text{-}53)$$

4.4.8.2 Cases study

66 kV single-core cable, made of a stranded copper conductor with 800 mm^2 insulated by XLPE and covered by a lead screen, 50 Hz, which its parameters have been listed in table (4-1), is used as case study.

4.4.8.3 Obtained results

The result is shown in Fig. (4-19). This figure shows the values of sheath circulating current with varying the length of first minor section from 200 to 400 meters.

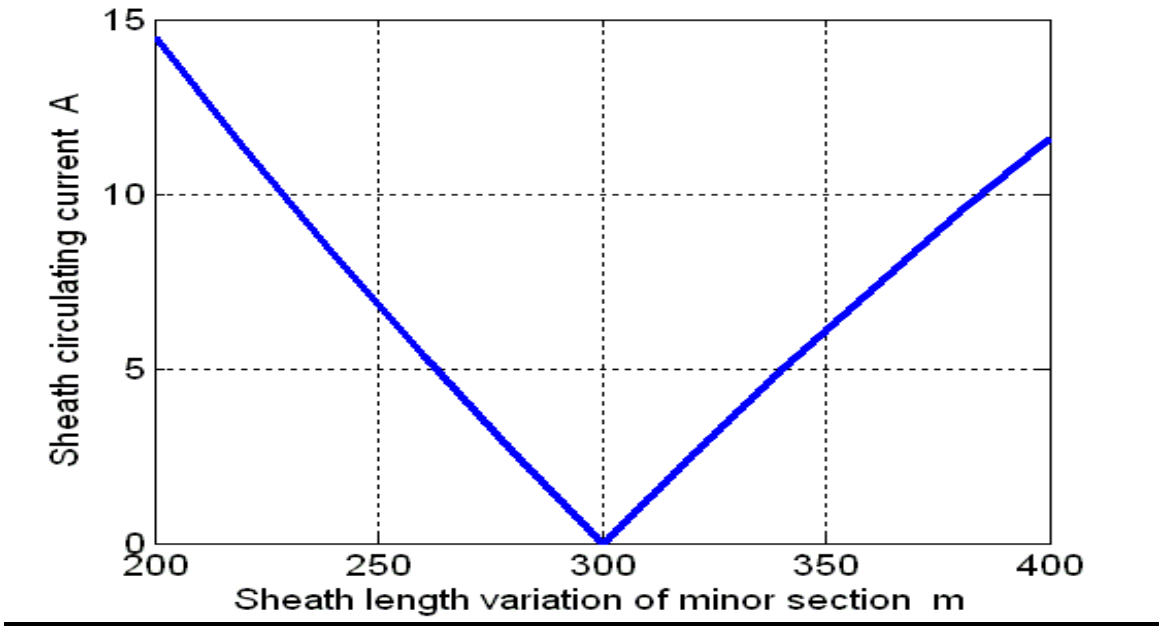


Fig. (4-19): Sheath current vs. sheath length of minor section for trefoil formation.

4.4.8.4 Discussion of the obtained results

From Fig. (4-19) it can be seen that:

- When the minor sections have the same length (300 m), the sheath circulating current reaches zero because the vectorial summations of induced voltages in the three minor sections of metallic sheath equal zero as shown in Fig. (4-20).

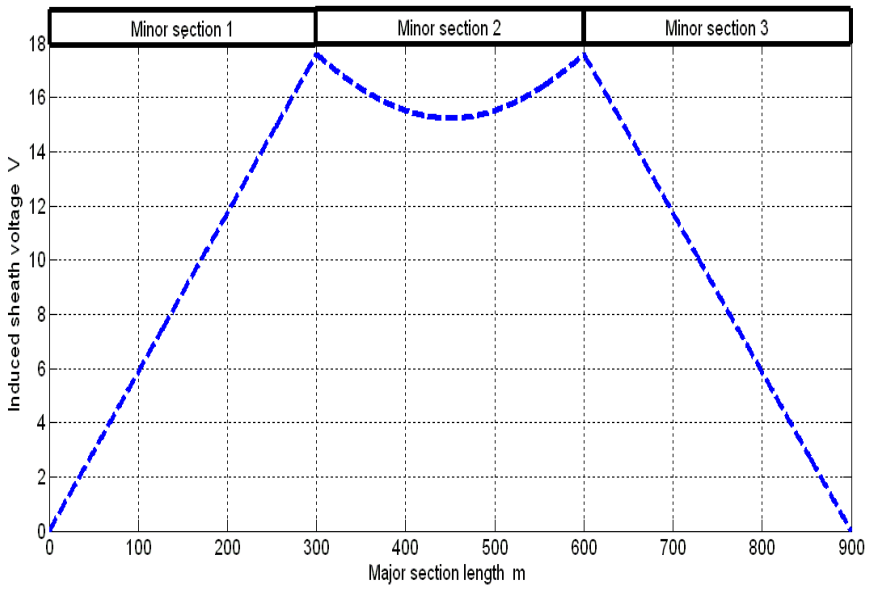


Fig. (4-20): Sheath induced voltage vs. total sheath length for trefoil formation.

- Any unbalance in the length of the minor sections of the cross bonded systems will result in circulating currents in the cable screens even when the currents in the phase conductors are symmetric.

4.4.9 Effect of cable armoring on the sheath losses

4.4.9.1 Introduction

In order to protect the cables from mechanical damage such as pick or spade blows, ground subsidence or excessive vibrations cable armoring is employed [24].

Armored single-core cables for general use in A.C systems usually have nonmagnetic armor. This is because of the very high losses that would occur in closely spaced single-core cables with magnetic armor. On the other hand, when magnetic armor is used, losses due to eddy currents and hysteresis in the steel must be considered. [25].

To calculate the sheath and armour losses for single-core cables with nonmagnetic armor according to IEC 60287 [19], mathematical algorithm in clause 4.3 is used, but with using the parallel combination of sheath and armour resistance in place of single sheath resistance, and the root mean square value of the sheath and armour diameter replaces the mean sheath diameter, i.e.

$$R_e = \frac{R_S R_A}{R_S + R_A} \qquad (4\text{-}54)$$

$$d = \sqrt{\frac{d_S^2 + d_A^2}{2}} \qquad (4\text{-}55)$$

So

$$I_S = (R_e/R_S)\, I_{SA} \qquad (4\text{-}56)$$

$$I_A = (R_e/R_A)\, I_{SA} \qquad (4\text{-}57)$$

Where

R_e: The equivalent resistance of sheath and armour in parallel (Ω/m)

R_A: The resistance of armour per unit length of cable at its maximum operating temperature (Ω/m)

R_S : The resistance of sheath at its maximum operating temperature (Ω/m).

d: The mean diameter of sheath and armour (mm)

d_S: The mean diameter of sheath (mm)

d_A: The mean diameter of armour (mm)

I_S: Sheath current (circulating or eddy) in A

I_A: Armour current (circulating or eddy) in A

I_{SA}: Sheath-armour combination current (circulating or eddy) in A

Thus the addition of the armour is at least equivalent to lowering of the sheath resistance, so from discussion in clause 4.4.4, if R_e is lower than the critical value of sheath resistance, the addition of the armour may be tends to reduce or increase the combined sheath-armour circulating losses, if R_e is higher than the critical value of sheath resistance, the addition of the armour, no doubt in that case, tends to increase the combined sheath-armour circulating losses, while for combined sheath-armour eddy loss as well as combined sheath-armour current (circulating or eddy) it is expected increasing them because they are inversely proportional to sheath resistance.

It is of interest to show the effect of armour resistance on the sheath and armour currents. Fig. (4-21) is prepared for this purpose, so if the armour resistance equals the sheath resistance, I_{SA} is equally divided between sheath and armour resistance i.e. the armour current will be equal the sheath current (intersection point in Fig. (4-21)), and if the armour resistance is lower than the sheath resistance, the armour current will be higher than the sheath current and vice versa.

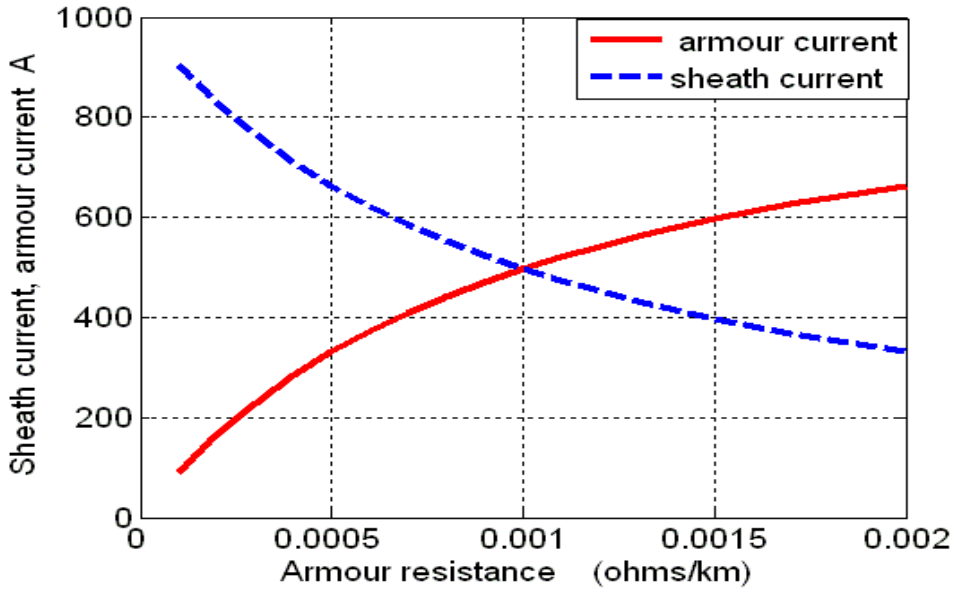


Fig. (4-21): Sheath, armour current vs. armour resistance

4.4.9.2 Cases study

66 kV single-core cable, made of a stranded copper conductor with 800 mm^2 insulated by XLPE and covered by a lead screen and armored with aluminum wire, 50 Hz which its parameters have been listed in table (4-20), is used as a case study.

Table (4-20): Armored Single-core cable 800 mm^2, 66 kV CU with lead covered and aluminum wire armored parameters

Outer diameter of cable (mm)	93
Mean armour diameter (mm)	82.5
Mean sheath diameter (mm)	62.6
DC Resistance of the copper conductor at 20°C ohm/km	0.0221
diameter of the conductor (mm)	34
Thickness of lead (mm)	2.6
No. of armour wires	50

$R_S = 0.5$ Ω /km , $R_A = 0.39$ Ω/km and $R_e = 0.22$ Ω/km

4.4.9.3 Obtained results by using IEC 60287

The results are shown in table (4-21). This table shows the values of sheath currents and armor currents with their corresponding losses for armored single-core cable in case of two-points bonding method and cross bonding method for touch trefoil and touch flat.

.Table (4-21): Sheath, armour currents and their loss factors for nonmagnetic armored single-core cable with two-points bonding and cross-bonding methods

Parameters	Two-bonding point		Cross-bonding	
	Touch trefoil	Touch flat	Touch trefoil	Touch flat
$\lambda_{CS1} + \lambda_{AC1}$	46.01	87.35	0	0
$\lambda_{CS2} + \lambda_{AC2}$	46.01	26.8	0	0
$\lambda_{CS3} + \lambda_{AC3}$	46.01	110.76	0	0

$\lambda_{SE1}+\lambda_{AE1}$	6.59	3.30	6.66	3.82
$\lambda_{SE2}+\lambda_{AE2}$	6.59	13.19	6.66	12.93
$\lambda_{SE3}+\lambda_{AE3}$	6.59	3.30	6.66	2.9
I_{CS1}	112.1 A	154.4 A	0	0
I_{CS2}	112.1 A	85.5 A	0	0
I_{CS3}	112.1 A	173.9 A	0	0
λ_{CS1}	20.10	38.16	0	0
λ_{CS2}	20.10	11.71	0	0
λ_{CS3}	20.10	48.39	0	0
I_{SE1}	42.4 A	30 A	42.6 A	32.3 A
I_{SE2}	42.4 A	60 A	42.6 A	59.4 A
I_{SE3}	42.4 A	30 A	42.6 A	29.9 A
λ_{SE1}	2.88	1.44	2.91	1.67
λ_{SE2}	2.88	5.76	2.91	5.65
λ_{SE3}	2.88	1.44	2.91	1.47
I_{AC1}	144.5 A	199.1 A	0	0
I_{AC2}	144.5 A	110.2 A	0	0
I_{AC3}	144.5 A	224.2 A	0	0
λ_{AC1}	25.91	49.19	0	0
λ_{AC2}	25.91	15.09	0	0
λ_{AC3}	25.91	62.37	0	0
I_{AE1}	54.7 A	38.6 A	54.9 A	41.6 A
I_{AE2}	54.7 A	77.3 A	54.9 A	76.6 A
I_{AE3}	54.7 A	38.6 A	54.9 A	36.2 A
λ_{AE1}	3.71	1.86	3.75	2.15
λ_{AE2}	3.71	7.43	3.75	7.28

λ_{AE3}	3.71	1.86	3.75	1.63

Where:

I_{CS1}, I_{CS2}, I_{CS3} : Circulating current in sheath of phase no. 1,2 and 3 respectively

λ_{CS1}, λ_{CS2}, λ_{CS3} : Circulating loss factor in sheath of phase no. 1,2 and 3 respectively

I_{SE1}, I_{SE2}, I_{SE3} : Eddy current in sheath of phase no. 1,2 and 3 respectively

λ_{SE1}, λ_{SE2}, λ_{SE3} : Eddy loss factor in sheath of phase no. 1,2 and 3 respectively

I_{AC1}, I_{AC2}, I_{AC3} : Circulating current in armour of phase no. 1,2 and 3 respectively

λ_{AC1}, λ_{AC2}, λ_{AC3} : Circulating loss factor in armour of phase no. 1,2 and 3 respectively

I_{AE1}, I_{AE2}, I_{AE3} : Eddy current in armour of phase no. 1,2 and 3 respectively

λ_{AE1}, λ_{AE2}, λ_{AE3} : Eddy loss factor in armour of phase no. 1,2 and 3 respectively

4.4.9.4 Discussion of the obtained results

From results in table (4-21) with using armored single-core cable instead of unarmored single-core cable which its results are listed in tables (4-2-a) and (4-2-b) it can be seen that:

- The combined sheath-armour circulating losses ($\lambda_{CS} + \lambda_{AC}$) and the combined sheath-armour eddy losses ($\lambda_{SE} + \lambda_{AE}$) increased due to R_e is higher than the critical value of sheath resistance which can be seen in figures (4-9) and (4-11).
- The sheath circulating losses and the sheath eddy losses are lower than the armour circulating losses and the armour eddy losses respectively because

the armour resistance (R_A = 0.39Ω/km) is lower than the sheath resistance (R_S = 0.5Ω /km).

- The sheath current value in armored single-core cable is depending mainly on the (R_e/R_S) ratio.

CHAPTER (5)

SHEATH OVERVOLTAGES DUE TO EXTERNAL FAULTS IN SPECIALLY BONDED CABLE SYSTEM

5.1 Introduction

In chapter 4, It is shown that the types of the bonding are one of the important factors which effect on the sheath losses in single-core cables, and it is concluded that both single-point bonding and cross bonding, which are known as special bonding, introduce the lowest losses in the metallic sheath of the cable.

To take the advantages of the specially bonded cable systems it is necessary to insulate the cable sheath from earth to avoid corrosion. This is achieved by having an extruded serving of PVC or PE on the cables and housing the joints in compound filled fiberglass boxes to insulate them from the surrounding soil [26].

The use of special bonding gives rise to sheath over-voltages at sheath sectionalizing insulators in cross bonded cable system and insulators in a single-point bonded cable system due to lightning, switching surges or faults [6,27].

One of the factors affecting the sheath losses in single-core underground power cables in case of special bonding types is the sheath overvoltage. Those over-voltages may cause the sheath multi-points break-down which result in a large sheath currents and losses and hence may cause overheating of the cables and finally leading to operation faults [6, 28].

As mentioned before, faults are one of reasons which cause sheath over-voltages. System faults may be divided into internal faults occurring within the cables themselves and external faults for which the cables carry some or all of the fault current. The sheath voltages resulting from internal faults may greatly exceed those caused by external faults [27].

A fault in the cables themselves inevitably involves repair work and hence it is not so important if the sheath insulation adjacent to the fault is also damaged. The sheath bonding design should preclude the damage cascading to other parts of the cable system i.e. the cable installation must clearly be capable of safely withstanding the effects of any fault in the system external to the cables [6,27,29]. So it is important to consider the performance of special sheath bonding methods in relation to power frequency external fault currents. Three types of external faults are considered:

1- Three-phase symmetrical fault

2- Phase-to-phase fault

3- Single-phase ground fault

These three types represent extreme cases and, hence, may be expected to show maximum values of sheath voltage [27]. Transient voltages induced in the cable sheaths are particularly important because of the possibility of excessive voltages that can cause harm to personnel, the cable or equipment connected to the cable. Also the level of transient voltages induced in the sheaths will have a direct bearing on measurement actuators and sensors used in any cable monitoring system [8].Consideration must be given to assess the magnitude of those over-voltages, so in this chapter over-voltages will be calculated for single-point bonding and cross bonding under three types of external faults which are listed above for systems having solidly earthed neutral, with introducing a suitable method to protect the outer jacket sheath of the cable.

5.2 Mathematical Algorithm

In deriving the equations that give the sheath voltage gradients due to the external faults types which listed above for special sheath bonding methods, the following assumptions are made [6, 27,29 ,30 , 31]:

1- The short circuit current is known and is unaffected in value by the characteristics of the cable system.

2- Symmetric currents flow during three-phase faults.

3- No currents flow other than currents in phase conductors for phase to phase fault and three-phase fault, i.e. no induced circulating currents in screens, or any other parallel conductors are considered when calculating the induced voltages. Parallel conductors which are connected to earth at both ends will generally act as screening conductors reducing the induced voltages. So this assumption will give results on the safe side.

4- The cables, for cross bonded systems, be laid with constant spacing and equal lengths.

These assumptions have to be set up in a way that most of the practical cases are covered and the deviations from the exact values will be on the safe side. It must be refer here that, the studies support the use of the following equations to within good accuracy and with the benefit of being simple to apply [31].

5.2.1 Single-point bonding cables:

As mentioned in chapter 3, clause 3.2.2, the sheaths of single-point bonded cables provide no path for returning fault current; hence, an additional earth conductor is normally laid with such cables. To avoid circulating currents in this earth conductor, it is laid where possible at spacing from the center cable of 0.7 times the spacing of the main cables and transposed at the center of the route when the power cables are not transposed as shown in Fig. (5-1).

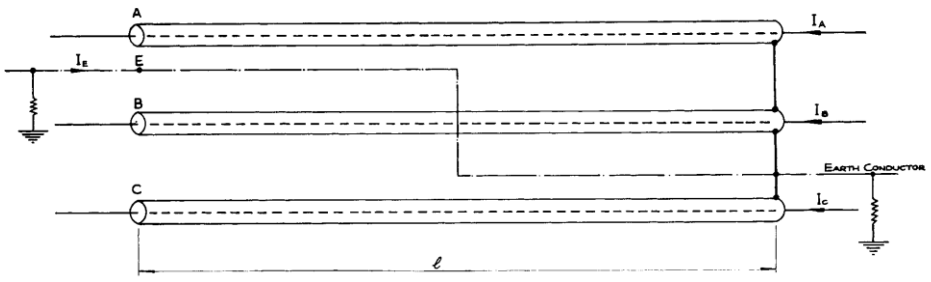


Fig. (5-1): Arrangement of single-point bonded cables

5.2.1.1 Three phase symmetrical fault

For a symmetrical three-phase fault, the equations are the same as for normal balanced load currents and are given as following:

5.2.1.1.1 Trefoil formation [6, 29, 30, 31]**:**

For cables in trefoil formation the induced voltages between sheath and local earth reference are given by the formulae shown below:

$$E_{AE} = j\omega I_F \left(2.10^{-7}\right)\left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)\ln\left(\frac{2S}{d}\right) \text{ V/m} \qquad (5\text{-}1)$$

$$E_{BE} = j\omega I_F \left(2.10^{-7}\right)\ln\left(\frac{2S}{d}\right) \text{ V/m} \qquad (5\text{-}2)$$

$$E_{CE} = j\omega I_F \left(2.10^{-7}\right)\left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right)\ln\left(\frac{2S}{d}\right) \text{ V/m} \qquad (5\text{-}3)$$

From equations (5-1), (5-2) and (5-3) it can be said that, the magnitudes of the voltages between sheath and local earth reference in trefoil formation are equal and are given by:

$$E = \omega I_F \left(2.10^{-7}\right)\ln\left(\frac{2S}{d}\right) \text{ V/m} \qquad (5\text{-}4)$$

5.2.1.1.2 Flat formation [6, 27, 29, 30, and 31]**:**

For cables in flat formation the induced voltages between sheath and earth conductor are given by the formulae shown below:

$$E_{AE} = j\omega I_F \left(2.10^{-7}\right)\left(-\frac{1}{2}\ln\left(\frac{S}{d}\right) + j\frac{\sqrt{3}}{2}\ln\left(\frac{4S}{d}\right)\right) \text{ V/m} \qquad (5\text{-}5)$$

$$E_{BE} = j\omega I_F \left(2.10^{-7}\right)\ln\left(\frac{2S}{d}\right) \text{ V/m} \qquad (5\text{-}6)$$

$$E_{CE} = j\omega I_F \left(2.10^{-7}\right)\left(-\frac{1}{2}\ln\left(\frac{S}{d}\right) - j\frac{\sqrt{3}}{2}\ln\left(\frac{4S}{d}\right)\right) \quad \text{V/m} \qquad (5\text{-}7)$$

Where:

E_{AE},E_{BE},E_{CE} : Voltages between sheaths of phases A,B and C respectively and the earth conductor

I_F : Short-circuit current in cable conductor (rms) in A

S : Spacing between axes of adjacent conductors in m

d : Mean of outer and inner diameter of sheath in m

ω : 2π x frequency (in cycles per second).

5.2.1.2 Phase-to-phase fault

If the fault current is asymmetrical in relation to the earth conductor, a current will flow in this conductor of value depending on its resistance and the earth resistances at its terminals and it will act as a screening conductor. This small effect is ignored in the following equations so that they represent the worst case. For a phase-to-phase fault, the equations are given as following:

5.2.1.2.1 Trefoil formation [6, 29, 30, 31]**:**

Assuming a fault is carried out between phases A and B.

$$E_{AE} = j\omega I_F \left(2.10^{-7}\right)\left(\ln\left(\frac{2S}{d}\right)\right) \quad \text{V/m} \qquad (5\text{-}8)$$

$$E_{BE} = -j\omega I_F \left(2.10^{-7}\right)\left(\ln\left(\frac{2S}{d}\right)\right) \quad \text{V/m} \qquad (5\text{-}9)$$

$$E_{CE} = 0 \quad \text{V/m} \qquad (5\text{-}10)$$

5.2.1.2.2 Flat formation:

For a phase-to-phase fault, two cases are possible; fault current in one outer cable with return in either the other outer or the center cable, the formulae of each case are shown below:

5.2.1.2.2.1 Fault between two outers cables [27]

$$E_{AE} = j\omega I_F \left(2.10^{-7}\right)\left(\ln\left(\frac{4S}{d}\right)\right) \quad \text{V/m} \quad (5\text{-}11)$$

$$E_{BE} = 0 \quad \text{V/m} \quad (5\text{-}12)$$

$$E_{CE} = -j\omega I_F \left(2.10^{-7}\right)\left(\ln\left(\frac{4S}{d}\right)\right) \quad \text{V/m} \quad (5\text{-}13)$$

5.2.1.2.2.2 Fault between inner and outer cables (phase A & phase B) [6, 29, 30, 31]

$$E_{AE} = j\omega I_F \left(2.10^{-7}\right)\left(\ln\left(\frac{2S}{d}\right)\right) \quad \text{V/m} \quad (5\text{-}14)$$

$$E_{BE} = -j\omega I_F \left(2.10^{-7}\right)\left(\ln\left(\frac{2S}{d}\right)\right) \quad \text{V/m} \quad (5\text{-}15)$$

$$E_{CE} = -j\omega I_F \left(2.10^{-7}\right)(\ln 2) \quad \text{V/m} \quad (16\text{-}5)$$

5.2.1.3 Single-phase ground fault (solidly earthed neutral)

Under earth fault conditions the return current will flow through the mass of the earth and through the earth continuity conductor. Calculating the division of current between the mass of earth and the earth continuity conductor depends on a number of factors that are not often known. Because of this, it is assumed in this case that all fault current returns in the earth conductor and none returns in the ground, this results in the highest values of sheath voltage. If an earth fault is in phase A, the sheaths to earth conductor voltages are:

5.2.1.3.1 **Trefoil formation** [6, 29, 30, 31]

V/m (5-17) $$E_{AE} = I_F\left(R_C + j\omega.2.10^{-7}.\ln\left(\frac{2S_{AE}^2}{d.r_c}\right)\right)$$

V/m (5-18) $$E_{BE} = I_F\left(R_C + j\omega.2.10^{-7}.\ln\left(\frac{S_{AE}.S_{BE}}{S.r_c}\right)\right)$$

V/m (5-19) $$E_{CE} = I_F\left(R_C + j\omega.2.10^{-7}.\ln\left(\frac{S_{AE}.S_{CE}}{S.r_c}\right)\right)$$

5.2.1.3.2 **Flat formation** [6, 29, 30, and 31]

$$E_{AE} = I_F\left(R_C + j\omega.2.10^{-7}.\ln\left(\frac{2S_{AE}^2}{dr_c}\right)\right) \quad \text{V/m} \quad (5\text{-}18)$$

$$E_{BE} = I_F\left(R_C + j\omega.2.10^{-7}.\ln\left(\frac{S_{AE}.S_{BE}}{S.r_c}\right)\right) \quad \text{V/m} \quad (5\text{-}19)$$

$$E_{CE} = I_F\left(R_C + j\omega.2.10^{-7}.\ln\left(\frac{S_{AE}.S_{CE}}{2.S.r_c}\right)\right) \quad \text{V/m} \quad (5\text{-}20)$$

Where:

S_{AE},S_{BE},S_{CE}: The geometric mean spacing between cables A, B and C respectively and the earth conductor (S_{AE},S_{BE},S_{CE} = 0.7S)

R_C : Resistance of earth conductor, ohm/m

r_c : Geometric mean radius of earth conductor (for stranded conductors take 0.75 overall radius)

From the above equations, it is appearing that the magnitude of the induced voltage due to earth fault current rather than other external faults is characterized by that is a function of the spacing between the earth continuity conductor and the line conductors.

5.2.2 Cross bonding cables:

5.2.2.1 Three-phase symmetrical fault [6, 29, 30, 31]

The sheath voltage gradients are given by the same equations as those are given in clause (5.2.1.1) of this chapter for single-point bonded systems with using the longest minor section length in case of minor sections unbalance as a worst case.

5.2.2.2 Phase-to-phase fault [6, 29, 30, 31]

The sheath voltage gradients are given by the same equations as those are given in clause (5.2.1.2) of this chapter for single-point bonded systems with using the longest minor section length in case of minor sections unbalance as a worst case.

5.2.2.3 Single-phase ground fault (solidly earthed neutral)

Under single phase to earth fault conditions the return current divides between the three sheaths in parallel and the earth. The proportion of current returning via the earth depends on the sheath resistance and the earthing resistances at the ends of the circuit. Equations can be given for the voltages between sheaths but the voltages from sheath to ground will depend strongly on the earthing resistances at the ends of the circuit and they can not be simply calculated. The voltages between sheaths are given by the following equations for earth fault in phase (A) by using the simple assumption that sheaths are earthed at one point only and that the whole of the returning current divides between the three sheaths:

5.2.2.3.1 Trefoil formation [6, 29, 30, 31]**:**

$$E_{AB} = j\omega I_F \left(2.10^{-7}\right)\left(\ln\left(\frac{2S}{d}\right)\right) \quad \text{V/m} \quad (5\text{-}23)$$

$$E_{BC} = 0 \quad \text{V/m} \quad (5\text{-}24)$$

$$E_{CA} = -j\omega I_F \left(2.10^{-7}\right)\left(\ln\left(\frac{2S}{d}\right)\right) \quad \text{V/m} \quad (5\text{-}25)$$

5.2.2.3.2 Flat formation [6, 29, 30, and 31]**:**

V/m (5-26) $E_{AB} = I_F \left(j\omega 2.10^{-7} \right) \left(\ln \left(\frac{2(2)^{1/3}.S}{d} \right) \right)$

V/m (5-27) $E_{BC} = I_F \left(j\omega 2.10^{-7} \right) \left(\ln \left(2^{2/3} \right) \right)$

V/m (5-28) $E_{CA} = I_F \left(j\omega 2.10^{-7} \right) \left(\ln \left(\frac{4S}{d} \right) \right)$

Where:

E_{AB},E_{BC},E_{CA} : Voltages between sheaths of phases A&B, B&C and C&A respectively

The above algorithm has been used through MATLAB program and the flowcharts of the computation steps are shown in Figures (5-2-a), (5-2-b), (5-2-c) and (5-2-d). Flowcharts in Figs (5-2-a) & (5-2-b) show the computation steps of induced sheath voltages due to three phase symmetrical fault, phase to phase fault and single phase ground fault for single-core cable in trefoil layout with both single-point bonding & cross bonding methods respectively. Flowcharts in Figs (5-2-c) & (5-2-d) show the computation steps of induced sheath voltages due to three phase symmetrical fault, phase to phase fault and single phase ground fault for single-core cable in flat layout with both single-point bonding & cross bonding methods respectively.

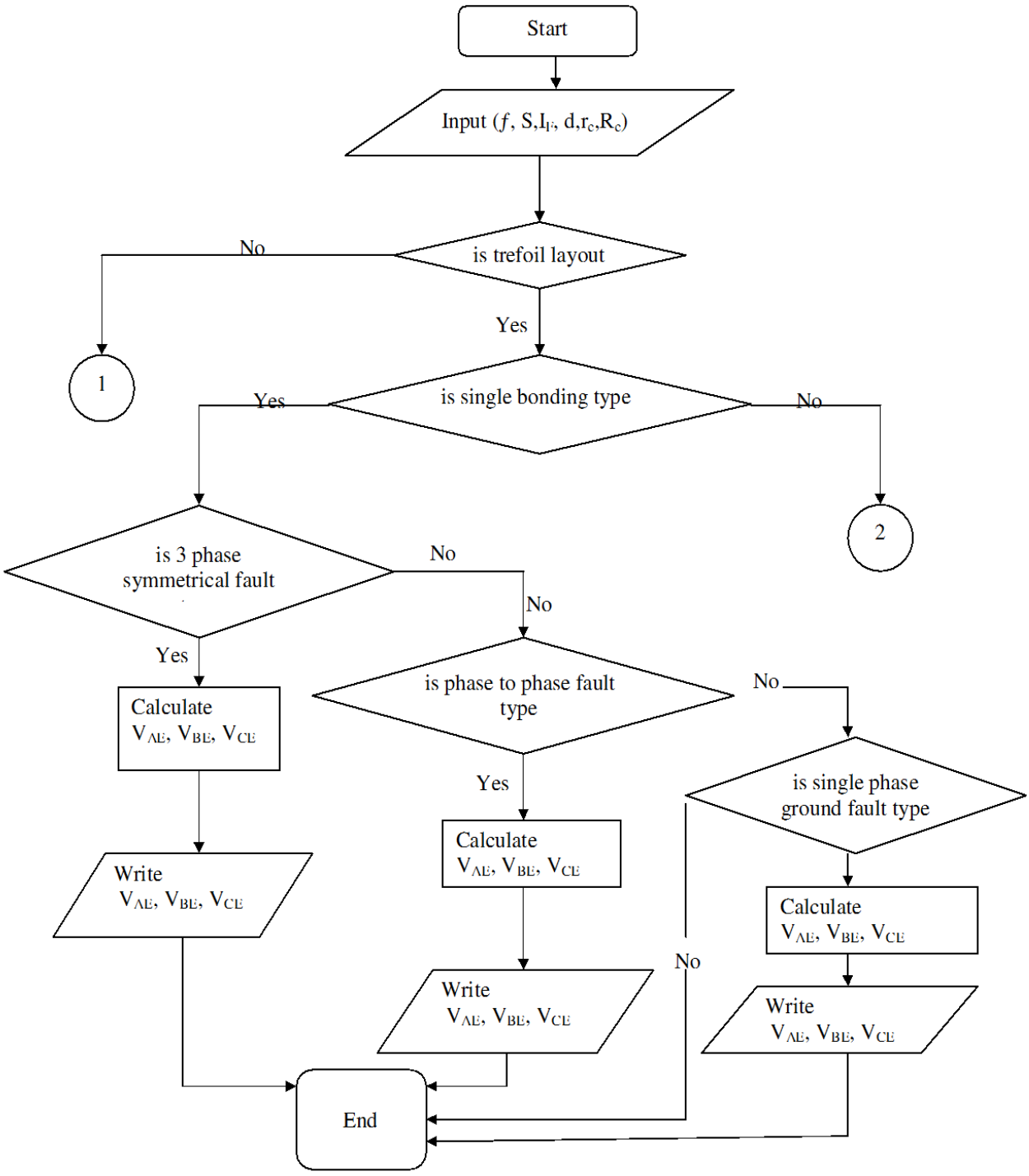


Fig.(5-2-a): Flowchart of the computation steps of sheath induced overvoltage for trefoil layout with single-point bonding

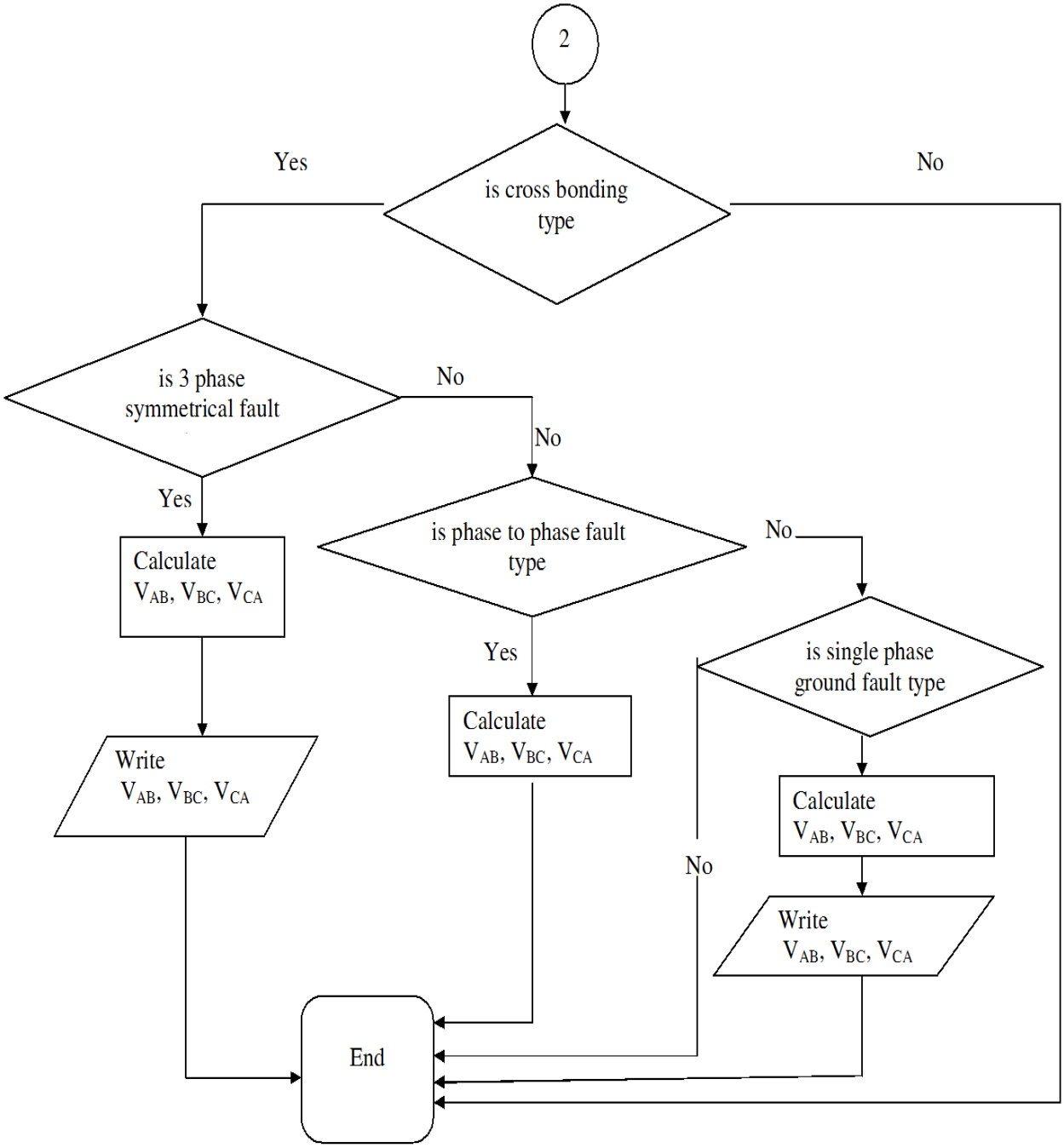


Fig.(5-2-b): Flowchart of the computation steps of sheath induced overvoltage for trefoil layout with cross bonding

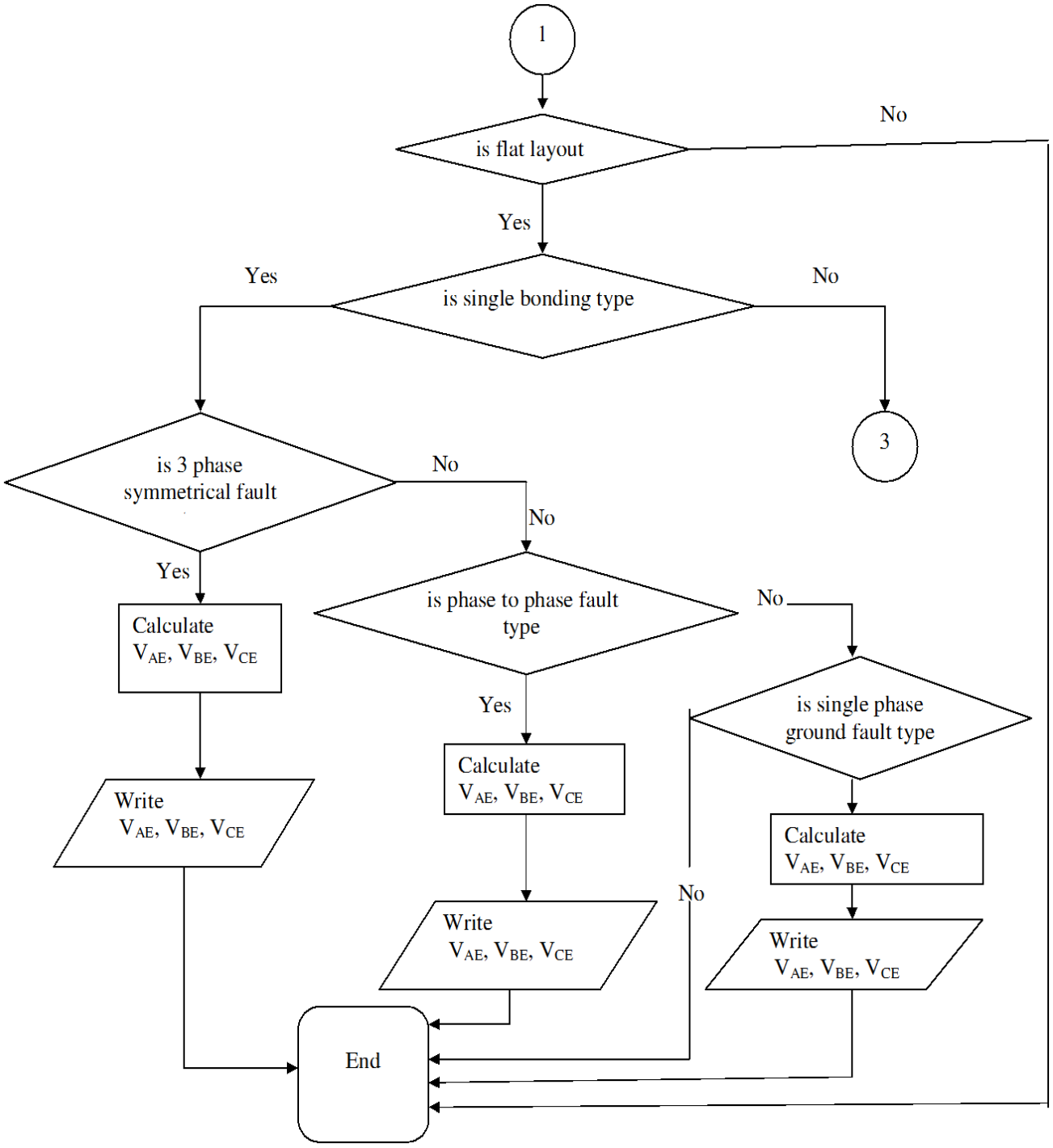


Fig.(5-2-c): Flowchart of the computation steps of sheath induced overvoltage for flat layout with single-point bonding

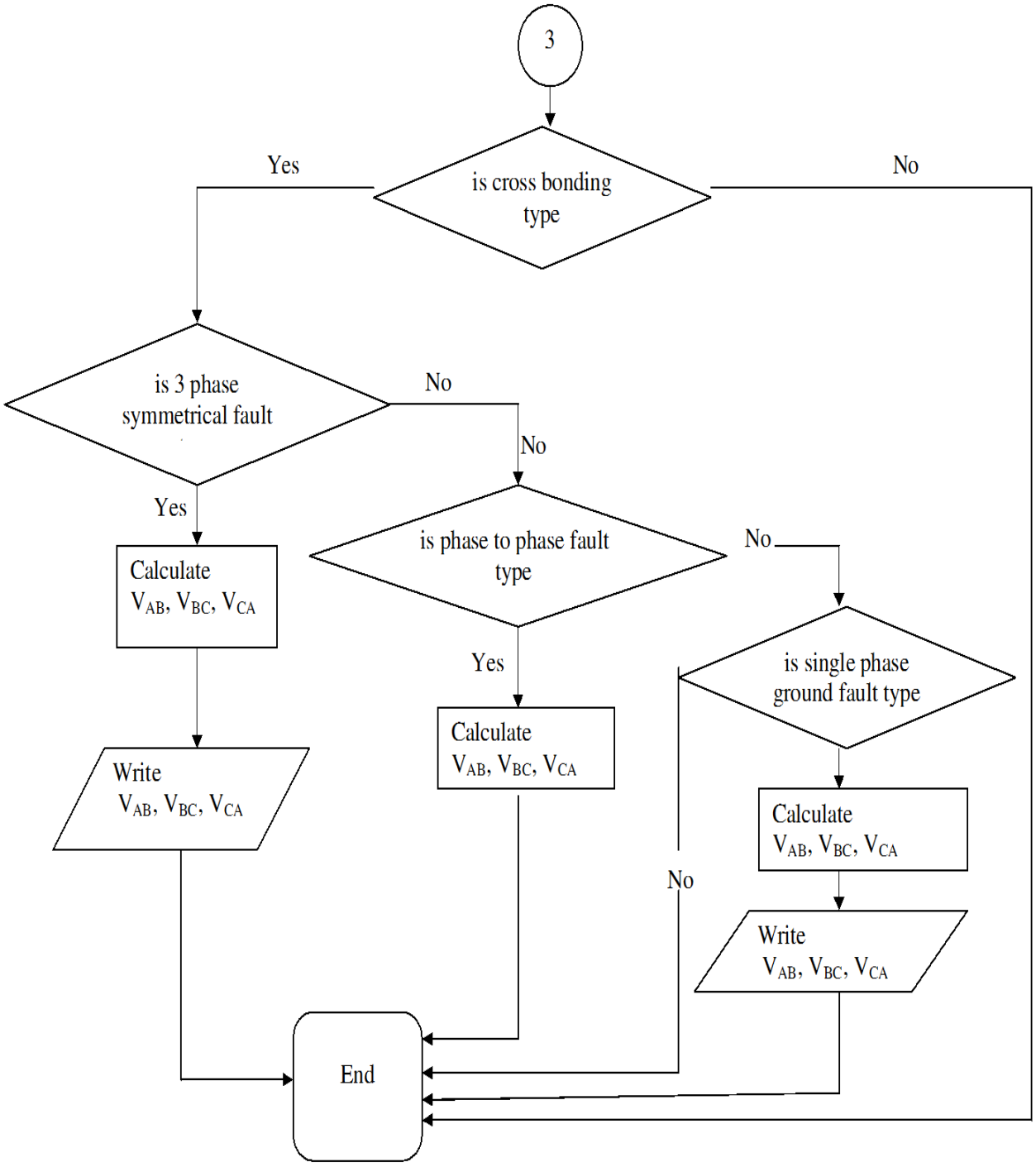


Fig.(5-2-d): Flowchart of the computation steps of sheath induced overvoltage for flat layout with cross bonding

Where:

I_F : Short-circuit current in cable conductor (rms) in A

S : Spacing between axes of adjacent conductors in m

d : Mean of outer and inner diameter of sheath in m

f: power frequency (50 Hz)

R_C : Resistance of earth conductor, ohm/m

r_c : Geometric mean radius of earth conductor (for stranded conductors take 0.75 overall radius)

E_{AE},E_{BE},E_{CE} : Voltages between sheaths of phases A,B and C respectively and the earth conductor

E_{AB},E_{BC},E_{CA} : Voltages between sheaths of phases A&B, B&C and C&A respectively

5.3 Case Study

66 kV single-core cable made of a stranded copper conductor with 800 mm^2 insulated by XLPE and covered by a lead screen, f = 50 Hz, which its parameters have been listed in table (4-1) with taking the distance between axial cable spacing (S) equals [2De], an earth continuity conductor (ecc) size 240 mm^2 (r_c = 13.5 mm & R_C = 0.076 Ω/km), is used in this case study to calculate the induced sheath voltages due to different types of external faults for single-point bonding and cross bonding methods.

To compare between values of induced sheath voltages it is preferred to calculate those values as between sheath and earth continuity conductor in case of single-point bonding, while they are calculated as between sheaths in case of cross bonding because it is not easy to calculate those values as a sheath to ground in case of single ground fault as has mentioned before.

5.4 Obtained Results

The outputs of the program that represents the results for unarmored single-core cables are shown in tables (5-1) and (5-2) for single-point bonding and cross bonding methods respectively. Table (5-1) shows the values of the voltages between sheaths and local earthing system in single-core cable due

to different types of external faults in case of single-point bonding for trefoil and flat layouts with S = 2De. Table (5-2) shows the values of the sheath to sheath voltages in single-core cable due to different types of external faults in case of cross bonding for trefoil and flat layouts with S = 2De.

Table (5-1): Voltages between sheaths and local earthing system due to different external faults in single-core cables with single-point bonding

<table>
<tr><td rowspan="3">Fault type</td><td colspan="6">Sheath voltage to local earth V/(km.kA)
(Single-point bonding)</td></tr>
<tr><td colspan="3">Trefoil-formation</td><td colspan="3">Flat-formation
A B C</td></tr>
<tr><td>E_{AE}</td><td>E_{BE}</td><td>E_{CE}</td><td>E_{AE}</td><td>E_{BE}</td><td>E_{CE}</td></tr>
<tr><td>3 phase sym. fault</td><td>102.5</td><td>102.5</td><td>102.5</td><td>129.9</td><td>102.5</td><td>129.9</td></tr>
<tr><td rowspan="4">Phase to phase fault</td><td colspan="3">Fault between phases A&B</td><td colspan="3">Fault between two outers cables</td></tr>
<tr><td rowspan="3">102.5</td><td rowspan="3">102.5</td><td rowspan="3">0</td><td>146</td><td>0</td><td>146</td></tr>
<tr><td colspan="3">Fault between inner & outer (A&B)</td></tr>
<tr><td>102.5</td><td>102.5</td><td>43.6</td></tr>
</table>

Single phase ground fault	**226.4**	**134.4**	**134.4**	**226.4**	**134.4**	**101.7**

E_{AE},E_{BE},E_{CE} : Voltages between sheaths of phases A,B and C respectively and the earth conductor

Table (5-2): Sheath to sheath voltages due to different external faults in single-core cables with cross bonding method for trefoil & flat layouts

Fault type	**Sheath to sheath voltage V/(km.kA) (Cross bonding)**					
	Trefoil-formation			**Flat-formation** (A B C)		
	E_{AB}	**E_{BC}**	**E_{CA}**	**E_{AB}**	**E_{BC}**	**E_{CA}**
3 phase sym. fault	**177.6**	**177.6**	**177.6**	**182.8**	**182.8**	**253**
Phase to phase fault	**Fault between phases A&B**			**Fault between two outers cables**		
	205	**102.5**	**102.5**	**146**	**146**	**292.1**
				Fault between inner & outer (A&B)		

				205	**59**	**146**
Single phase ground fault	**102.5**	**0**	**102.5**	**117**	**29**	**146**

E_{AB},E_{BC},E_{CA} : Voltages between sheaths of phases A&B, B&C and C&A respectively

5.5 Discussion of the Obtained Results:

From calculations in table (5-1) for single-point bonding method, it is noticed that:

- In case of 3-phase symmetrical fault the values of voltages between the sheath and earth continuity conductor (ecc) for trefoil layout are equal, while for flat layout the voltages induced in the outer cable sheaths are equal and usually larger than the voltage induced in the middle cable sheath. The voltage induced in the middle cable sheath is the same as in trefoil layout. This comment is the same as for the normal case but with taking the fault current value into consideration (refer to clause 4.4.1.4, Fig. (4-4)).
- For the phase to phase fault, in case of trefoil layout, the sheath voltage in the healthy phase will be zero due to symmetrical arrangement of phases, while it will be equal in the two faulty phases and their values are the same as three-phase symmetrical fault for the same fault current. In case of flat formation, the sheath voltages in the two faulty phases are equal and the highest sheath voltages result when the fault is between the two outer cables, it is also noticed that the voltage of the healthy phase in case of fault between two outer cables equals zero due to symmetrical case rather than in case of fault between inner and outer cables.
- For an earth fault, for a fault in phase (A), the highest sheath voltage is in the faulty phase for trefoil and flat formations and they have the same value in case of the distance between the faulty phase and the earth continuity conductor is equal for each, the effect of R_c can generally be neglected, so in

flat formation the equation (5-17) – which gives the maximum sheath voltage – can be expressed as:

$$E_{AE} = I_F \left(j\omega.2.10^{-7}.\ln\left(\left(\frac{S}{d}\right)^2.\frac{d}{r_c}\right)\right) \text{ [6]} \qquad \text{V/m} \qquad (5\text{-}29)$$

- Maximum values of voltages between the sheath and earth continuity conductor (ecc) under different faults in flat formation could be clearly appearing in Fig. (5-3) as following:

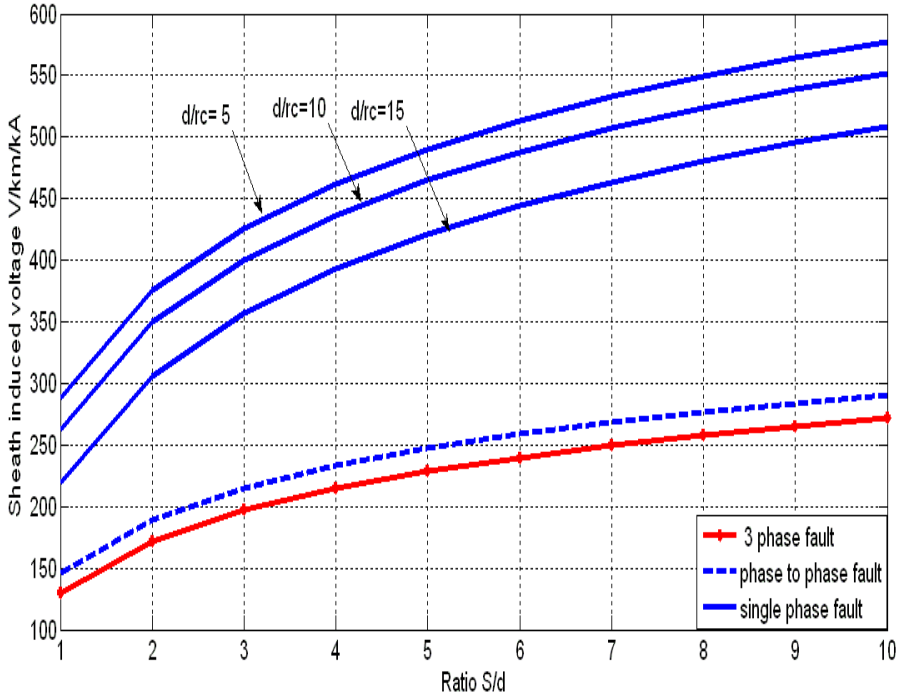


Fig. (5-3): Maximum induced sheath voltage gradients (sheath to earth) for various faults in single-point bonded cable system-flat

From Fig. (5-3) it can be seen that the sheath overvoltage due to the single phase fault are much more important than with respect to the other types of fault for systems having solidly earthed neutral and it also indicates the effect of (d/r_c), the ratio between mean of outer and inner diameter of metallic sheath and geometric mean radius of earth conductor (ecc), on the sheath induced voltage in case of single phase fault, as sheath induced voltage is inversely proportional to that ratio. From calculations in table (5-2) for cross bonding method, it is noticed that:

- In case of 3-phase symmetrical fault the values of voltages between the sheaths for trefoil layout are symmetrical, while for flat layout the maximum voltage is reached between the two outer cables.
- For the phase to phase fault, in case of trefoil layout, the highest sheath voltage is between the sheaths of two faulty phases. In flat formation, the highest sheath voltage is between the two outer cables in two cases which are studied (fault between two outer cables & fault between inner and outer cables).
- For an earth fault, for a fault in phase (A), the highest sheath to sheath voltage is between the two outer cables in case of flat layout.
- In all cases, the maximum induced voltages between sheaths in flat layouts are higher than the maximum induced voltages between sheaths in trefoil layouts.

- Maximum values of voltages between the sheaths at the cross bond position per unit length of km of the minor section length under different faults in flat formation could be clearly appearing in Fig. (5-4).

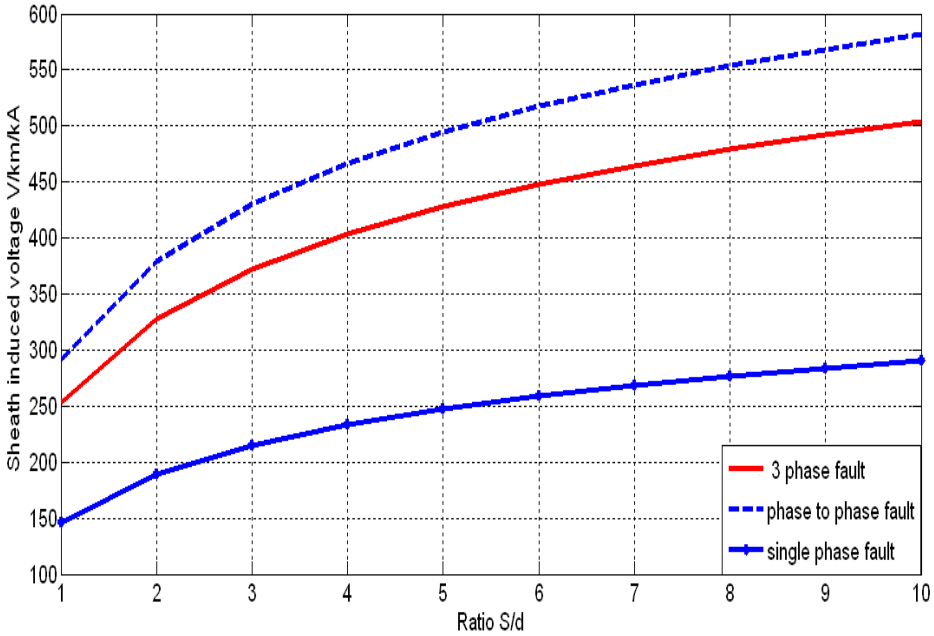


Fig. (5-4): Maximum induced sheath voltage gradients (sheath to sheath) for various faults in cross bonded cable system-flat

From Fig. (5-4) it can be seen that the sheath overvoltage due to the phase to phase fault is much more important than other types of faults for systems having solidly earthed neutral.

CHAPTER 6
CONCLUSION

From this study, some important conclusions are summarized as follows:

- Arnold equations for calculating eddy losses give approximately the same values which have been given in case of using IEC-287 equation, so one of equations could be used for calculating eddy losses to any sheath bonding method.
- Trefoil layout introduces symmetrical values of voltages and currents in its three metallic sheaths than flat layout. As for flat layout : the voltages induced in the outer cable sheaths are equal and usually larger than the voltage induced in the middle cable sheath in case of single-point bonding, the eddy currents in the outer cable sheaths are equal and usually smaller than the value of the middle cable sheath, the sheath circulating currents have unequal magnitude; the least value occurs in the sheath of the middle cable, values in sheaths of outer cables are of unequal magnitude too in case of two-points bonding.
- The sheath circulating losses could be reached to more than the conductor losses, this causes the insulation of the conductor to be subjected to temperatures may be excess of the insulation ratings, so the cable ampacity must be de-rated.
- Eddy loss could be disregarded with comparing to circulating loss but it must be noticed that the eddy loss value of middle conductor in flat formation with close spacing between phases especially for extra high system voltages cables must be taken into consideration as its value approaches to the value of sheath circulating loss for the same conductor.
- Single-point bonding and cross bonding methods introduce a solution to overcome the problems of sheath losses in case of two-points bonding method.
- The sheath loss factors (eddy & circulating) are inversely proportional to the conductor resistivity while they are proportional to the conductor sizes.
- The sheath circulating losses are proportional to the spacing between phases, while the sheath eddy losses are inversely proportional to it so they can be neglected at large spacing.

- The sheath eddy currents, eddy losses and circulating currents are inversely proportional to the spacing between phases.
- The sheath circulating losses could be reduced by large increase in sheath resistance or large reduce in the sheath resistance.
- Single-core cables covered by copper wire screen, copper tape or stainless steel introduce a best solution to reduce the sheath losses and overcoming the problems of lead sheath especially at higher voltages.
- Eddy losses could be neglected with respect to circulating losses except in aluminum sheath as the eddy losses could be greater than the circulating losses.
- Single-core cable with aluminum sheath introduces higher sheath losses and currents due to its low resistivity; it also introduces irregular behavior towards the values of sheath circulating loss factors in extra high voltages as they are reducing with increasing the system voltage levels.
- In flat formation the central conductor always has the lowest sheath circulating current value, while the values of two outer conductors are depending on the phase rotation and its arrangement.
- The sheath current duplicates with duplicating the conductor current.
- The sheath loss factor increases with increasing power frequency.
- Two-point bonding for flat formation has more sensitivity to the changing of power frequency than other bonding types.
- When the minor sections have the same length, the sheath current reaches zero because the vectorial summation of induced voltages in the three minor sections of metallic sheath equals zero.
- Any unbalance in the length of the minor sections of the cross bonded systems will result in circulating currents in the cable screens even when the currents in the phase conductors are symmetric.
- In case of armoring single-core cables, the combined sheath and armor circulating losses could be lower or greater than the sheath circulating losses without armoring depending mainly on the equivalent resistance of sheath and armour in parallel (R_e).
- The values of sheath current and armor current are depending mainly on the armour resistance (R_A) and sheath resistance (R_s).

- For systems having solidly earthed neutral, the overvoltage due to the single-phase fault are much more important than the other types of fault, while the overvoltage due to phase to phase fault are much more important than the other types of fault in case single-point bonding and cross bonding respectively.

Finally it can be said that “The studying of the factors affecting the sheath losses in single-core underground cables helps engineers who dealing with high voltage single-core cables to be more active by introducing a suitable solutions to overcome the sheath losses problems".

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